

Quantum Theory Through Physical Roles

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Quantum theories differ in what they treat as state, dynamics, constraint, measurement, protocol, or external realization. Predictive closure connects these choices and identifies the physical structure missing from an incomplete description.

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When External Conditions Become Quantum Physics

Quantum theory changes when a quantity once treated as fixed becomes part of the physical state or its dynamics. A prescribed electromagnetic potential becomes a quantum field. A classical measuring device becomes an interacting quantum subsystem. A passive environment becomes a source of memory when its correlations influence later motion. Quantum gravity asks the same question of geometry itself. In each case, the variables retained by the theory must determine future observable probabilities, and equivalent physical transformations must compose to the same result. We call this condition predictive closure.

At a chosen resolution, a physical theory is closed when its declared state fixes future observable probabilities and its allowed transformations compose consistently. If either condition fails, the smallest missing field, state coordinate, operator, closure condition, observable, or protocol must enter the theory.

$$q(h_1) = q(h_2) \implies p(y, t \mid h_1) = p(y, t \mid h_2), \quad T_{\gamma_1} = T_{\gamma_2} \text{ for physically equivalent paths}$$

The first relation compares two histories that produce the same declared state. Their later probabilities must agree. If they do not, the state has discarded a physically active correlation; an internal coordinate or memory kernel restores the missing dependence. The second relation compares two physically equivalent paths. Their transformations must agree when applied to the same state. A non-neutral loop records curvature, frustration, or another obstruction to global composition. Predictive closure also determines the boundary between a mechanism and its realization. A parameter remains external while it selects a member of a fixed theory. It enters the mechanism when it changes the state space, operator domain, dynamical map, admissibility condition, observable, or operation order.

$$a \in A \quad \text{while} \quad (\Xi, \Omega, C, R, P)_a = (\Xi, \Omega, C, R, P), \\ a \longrightarrow X \in \{\Xi, \Omega, C, R, P\} \quad \text{when the physical object } X \text{ changes.}$$

A wall illustrates the distinction. Its position can parameterize one boundary-value problem, while its boundary condition selects the domain of the Hamiltonian. The first quantity belongs to a realization. The second fixes the spectrum and therefore belongs to the closure of the mechanism.

$$(\Omega, \Xi) \longrightarrow M \longrightarrow I_{\text{op}} = (M; C, R, P) \longrightarrow I_{\text{real}} = (I_{\text{op}}; A) \\ q \in \Xi, \quad \dot{q} = \Omega[q; C, P], \quad y = R(q)$$

A physical state q contains the information required to determine later observables at the chosen resolution. The admissible states form Ξ , and Ω acts on them as a generator, channel, projection, constraint, or observable. Their pair $M = (\Omega, \Xi)$ fixes which operation acts on which physical degrees of freedom. Closure C fixes domains and admissibility, R maps states to observable predictions, and P fixes the order of preparation, control, and measurement. The

realization A supplies the fields, material, parameters, geometry, initial data, and apparatus of a particular experiment. Each clause prevents a specific ambiguity. Without the state space, the operator has no domain. Without closure, its solutions or probabilities need not be admissible. Without an observable, the formal evolution has no predicted measurement. Without a protocol, noncommuting operations have no defined order. For a quantum state, this identity reduces to a state space, an evolution map, and a positive measurement law.

$$\begin{aligned}\Xi_Q &= (\mathcal{H}, \mathcal{D}, \mathcal{S}), & \rho_0 &\in \mathcal{S}(\mathcal{H}), \\ \rho_P &= \mathcal{E}_P(\rho_0), & \mathcal{E}_P &\text{ completely positive and trace preserving,} \\ p(y \mid P) &= \text{Tr}(E_y \rho_P), & E_y &\geq 0, \quad \sum_y E_y = I.\end{aligned}$$

The Hilbert space, state class, and operator domain define the admissible states. The channel describes their evolution under protocol P . The positive operators E_y define the possible outcomes, and the trace rule assigns their probabilities. Unitary dynamics, open-system evolution, measurement, and feedback differ in which physical structures enter these three lines. These relations order the following 146 topics by the physical role each one fixes or changes. Entanglement belongs to composite state structure. Commutators belong to observable algebra. Bell experiments join the two through local measurements. Gauge constraints select physical states, while renormalization changes the effective law with scale. This ordering exposes transitions between theories that conventional subject headings place in separate fields.

Part I

Predictive Closure And Physical Role

Chapter 1

The Physical Identity Of A Quantum Mechanism

Quantum predictions join an admissible state space to an operation, the conditions under which that operation is defined, and an observable consequence. The same equation can play different roles when its domain, state, or measurement changes.

1.1 Nested Construction

$$(\Omega, \Xi) \longrightarrow M \longrightarrow I_{\text{op}} = (M; C, R, P) \longrightarrow I_{\text{real}} = (I_{\text{op}}; A)$$

A physical state q belongs to Ξ . The operation Ω acts on q , while $M = (\Omega, \Xi)$ identifies the law together with the state space on which it is meaningful. Closure, an observable map, and protocol give the operational identity. The realization layer fixes the material or field content, scales, parameter values, initial and boundary data, geometry, drives, and apparatus needed for a particular calculation or experiment.

1.2 State Space And Operation Must Agree

$$\begin{aligned}\Xi_Q &= (\mathcal{H}, \mathcal{D}, \mathcal{S}, \mathcal{A}_{\text{obs}}), \\ q &= \rho \in \mathcal{S}(\mathcal{H}), \\ \Omega_Q &= \{G, \mathcal{E}, \{E_y\}, \mathcal{U}, \dots\}, \\ M_Q &= (\Omega_Q, \Xi_Q).\end{aligned}$$

The state space Ξ_Q contains the Hilbert or Fock space, admissible density operators, operator domains, and, when needed, a subsystem or observable algebra. The operation Ω_Q contains generators, channels, observables, symmetry actions, projections, or their compositions. Neither factor can be interpreted without the other: the state space fixes what operations are defined, and the operation selects which structure of the state enters a prediction.

1.3 Completion And Realization

Normalization, positivity, self-adjointness, domains, and gauge constraints form the closure C_Q . Spectral measures, positive operator-valued measures, correlators, currents, and detector outcomes form the observable map R_Q . Preparation, evolution, control, and measurement order

form the protocol P_Q . The realization A_Q then fixes the particles, fields, material, geometry, parameter values, drives, and apparatus.

$$\begin{aligned} I_{\text{op},Q} &= ((\Omega_Q, \Xi_Q); C_Q, R_Q, P_Q), \\ I_{\text{real},Q} &= (I_{\text{op},Q}; A_Q), \\ I_{\text{real},Q} &\longmapsto p(y), \langle O \rangle, S(\omega), J, \text{ or another stated consequence.} \end{aligned}$$

Changing the state space can destroy self-adjointness, locality, or normalization. Changing the protocol can change the channel itself. The closure and observable must therefore be derived again whenever one of these physical roles changes.

1.4 Transformations And Compatibility

$$I_i = ((\Omega_i, \Xi_i); C_i, R_i, P_i) \xrightarrow{\tau} I_j = ((\Omega_j, \Xi_j); C_j, R_j, P_j).$$

The transformation τ relates two descriptions or realizations. It may change representation, complete a partial model, project onto a reduced description, transfer a law to another state space, compose operations, or take a controlled limit. Its physical content is the relation retained between the two constructions: an amplitude, expectation value, operator algebra, balance law, probability distribution, or approximation with a stated error. The retained relation is tested by maps that carry states and outputs between the two descriptions.

$$\begin{aligned} \alpha : \Xi_A &\longrightarrow \Xi_B, \\ \beta : \Omega_A(\Xi_A) &\longrightarrow \Omega_B(\Xi_B), \\ \Delta_{\alpha,\beta} &= \Omega_B\alpha - \beta\Omega_A. \end{aligned}$$

The map α carries physical states from the source description to the target, while β carries the corresponding outputs. Exact compatibility means that acting first and then mapping gives the same result as mapping first and acting in the target. The residual $\Delta_{\alpha,\beta}$ records the part of the source dynamics that the proposed target description cannot represent.

$$\begin{cases} \text{transfer,} & \Delta_{\alpha,\beta} = 0, \\ \text{reject,} & \Delta_{\alpha,\beta} \neq 0 \text{ without stable closure,} \\ \text{promote}(\Delta), & \Delta_{\alpha,\beta} \neq 0 \text{ and closes} \end{cases}$$

A vanishing residual, together with agreement of the stated observables, identifies another realization of the same mechanism. A residual with no stable form rejects the proposed transfer. When $\Delta_{\alpha,\beta}$ recurs, obeys its own closure relation, and changes a measurable consequence, it becomes a candidate correction operator, interaction, boundary term, or hidden state variable.

Chapter 2

When External Structure Becomes Dynamical

A background field, geometry, environment, boundary, constraint, or control sequence changes the theory when it alters the state space, dynamics, operator domain, observable, or operation order.

2.1 Backgrounds, Geometry, And Environment

A background field supplies prescribed coefficients to a Hamiltonian. Quantizing that field enlarges the Hilbert space and gives the field its own conjugate variables, fluctuations, and quanta. Quantum electrodynamics makes this change for the electromagnetic field. Quantum gravity asks whether distances and causal relations must likewise participate in the quantum state. The same change occurs in reduced dynamics. Tracing over an environment produces closed Markovian dynamics only when discarded correlations do not influence the future. When they return, the reduced state is incomplete. Retaining environmental coordinates or a memory kernel restores predictive closure. Decoherence, non-Markovian dynamics, and measurement back-action therefore differ by which environmental correlations remain physically active.

2.2 Domains, Constraints, And Controlled Dynamics

A boundary condition enters the physical law when it defines the operator domain. The same differential expression can then possess different self-adjoint extensions and different spectra. Confinement, tunnelling, scattering, and topological edge modes follow from this dependence on the domain. Constraints alter the state space rather than the operator domain. Gauge redundancy and exchange symmetry begin as restrictions on allowed descriptions. Imposing them selects the physical Hilbert-space sectors. Gauss constraints remove gauge-equivalent vectors, while Bose or Fermi symmetry fixes the admissible many-particle states. Charges, particle statistics, and observable algebras follow from the surviving sectors. A sequence of gates, measurements, and conditional controls becomes a single physical map after composition. Its order can change the resulting unitary or channel. In quantum computing, feedback, error correction, and Floquet control, the experimental sequence is therefore part of the dynamics implemented during the experiment.

2.3 One Principle Across The Frontiers

Each enlargement repairs the same failure. A smaller theory omitted information required for future probabilities or for consistent composition. The minimal enlargement identifies the missing physical object: a field, an environmental coordinate, an operator domain, a constrained state sector, or a composed channel. Quantum theories can therefore be compared by the physical role of their variables rather than by the historical field in which those variables were introduced.

Chapter 3

Global Composition: Curvature, Frustration, And Memory

Local quantum relations define one global physical state only when their ordered composition is independent of the path used to connect the descriptions. Curvature, frustration, and memory arise when this composition retains information that is absent from the endpoint variables.

3.1 From Local Transformations To A Global State

Let T_{ij} carry a state, basis, or constraint from one local description to the next. A path is an ordered composition of these maps. It can describe transport through parameter space, constraints around an interaction loop, or a sequence of interventions followed by relaxation. For a closed path γ , the composite map is

$$\mathfrak{H}_\gamma = T_{0n}T_{n,n-1} \cdots T_{21}T_{10}.$$

For changes of description, $\mathfrak{H}_\gamma = I$ means that the local identifications fit one path-independent global description. A physical protocol need not return the state when its external controls close a loop. The corresponding completeness test compares histories that end at the same retained state. If their later observables differ, the retained endpoint variables have omitted physically active path information.

3.2 Curvature Records Transport Around A Loop

A quantum state transported around a loop can return with a phase or an internal rotation. In gauge theory the path-ordered exponential of the connection gives the loop operator; for a small loop its departure from the identity is set by the field strength. Berry curvature gives the corresponding relation in parameter space.

$$W(\gamma) = \text{Tr } \mathcal{P} \exp \left(i \oint_\gamma A_\mu dx^\mu \right),$$
$$\mathfrak{H}_\gamma = I + iF_{\mu\nu}S^{\mu\nu} + O(S^2).$$

The connection specifies how neighboring descriptions are compared; the curvature measures the infinitesimal failure of those comparisons to be path independent. The loop, rather than any one local coordinate choice, carries the observable geometric information.

3.3 Frustration Records Incompatible Local Preferences

In a frustrated quantum magnet, each bond can impose a well-defined local preference while no spin configuration satisfies every bond around a loop. For an Ising loop, a negative product of bond signs is the simplest obstruction:

$$\prod_{\langle ij \rangle \in \gamma} \text{sign } J_{ij} = -1.$$

The mismatch survives every local reassignment of spin signs and is therefore a property of the loop. In quantum magnets the same logic is carried by loop operators, flux sectors, or non-commuting bond terms. The loop flux or frustration sector becomes an additional physical label that local bond variables alone cannot determine.

3.4 Memory Records Hidden Paths Through State Space

A reduced quantum state is complete only if equal present states give equal future observable statistics. Let h_1 and h_2 be two histories that produce the same retained state q . The state is incomplete when hidden system–environment correlations make a later observable distinguish the histories:

$$q(h_1) = q(h_2), \quad R\Phi_t(h_1) \neq R\Phi_t(h_2), \\ V_{t+s} \neq V_t V_s.$$

The second relation is the failure of the reduced propagator to compose as a Markov semigroup. A memory kernel keeps the missing history explicit; an auxiliary mode or correlation coordinate enlarges the state and can restore a local evolution law. For order memory, two write operations followed by the same relaxation are distinguishable precisely when their ordered products leave different retained states.

$$RR_t W_B W_A(q_0) \neq RR_t W_A W_B(q_0).$$

3.5 The Shared Physical Content

Gauge curvature, frustrated loops, and reduced-state memory attach different physical meanings to $\mathfrak{H}_\gamma$. Their common content is failure of global composition: locally valid relations do not assemble into a path-independent state. The obstruction identifies what the smaller description has omitted. A connection and field strength complete local gauge comparisons; a flux or defect sector completes an incompatible interaction network; a memory kernel or hidden coordinate completes reduced dynamics. In each case the enlarged state distinguishes paths that the original endpoint variables identified as the same.

Chapter 4

Equivalence Across Quantum Descriptions

Two quantum descriptions represent the same physical mechanism when a map between their states preserves the dynamics and the stated observable consequences.

Quantum theory uses several forms of this equivalence. A change of representation carries states and observables together. A change of carrier embeds the same operator algebra in another Hilbert space. A reduced description projects onto fewer degrees of freedom, and a measurement or boundary condition completes the relation by fixing its observable content. In each case the equivalence is decided by the quantity that remains unchanged.

4.1 One operator, two roles: generator and observable

This relation connects Hamiltonian Quantum Mechanics, Observable, Spectral Theory. The exponential of the Hamiltonian generates time evolution, whereas its spectral measure defines energy probabilities. The retained physical relation is the self-adjoint operator and its domain.

$$U(t) = e^{-iHt/\hbar}$$
$$H = \int_{\sigma(H)} E dP_H(E)$$

Equivalence requires a common operator domain and self-adjointness; the generated dynamics must be unitary and the spectral measure must reproduce the same energy statistics.

4.2 Schrodinger, Heisenberg, and path-integral representations

This relation connects Schrödinger Picture, Heisenberg Picture, Path Integral Formulation. Time dependence may be carried by states or operators, and the same propagator may be represented by an action-weighted path integral. The retained physical relation is transition amplitudes and expectation values.

$$A_H(t) = U(t)^\dagger A_S U(t)$$
$$\langle q_f | U(t_f - t_i) | q_i \rangle = \int \mathcal{D}q e^{iS[q]/\hbar}$$

Equivalence requires compatible domains, measures, boundary conditions, and regularization; the formulations coincide only where they yield the same amplitudes or correlators.

4.3 Carrier factorization defines locality

This relation connects Quantum Entanglement, Quantum Information, Quantum Field Theory. Changing the tensor-product or algebraic decomposition changes which observables are local and which states are entangled. The retained physical relation is the global state and algebraic predictions.

$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$$

$$\rho_A = \text{Tr}_B \rho_{AB}$$

Equivalence requires a fixed subsystem algebra or tensor factorization because entanglement and locality are not invariant under arbitrary refactorization.

4.4 Constraints define physical observables

This relation connects Gauge Theory, Quantum Gravity, Measurement In Quantum Mechanics. Gauge or diffeomorphism constraints select the physical state space before outcome operators acquire observable meaning. The retained physical relation is predictions on the physical state space.

$$\hat{\mathcal{C}}_a |\psi_{\text{phys}}\rangle = 0$$

$$[\hat{\mathcal{O}}_{\text{phys}}, \hat{\mathcal{C}}_a] |\psi_{\text{phys}}\rangle = 0$$

Physical effects must descend to the constrained or quotient state space; gauge-dependent quantities do not define physical records.

4.5 Boundary conditions are spectral control variables

This relation connects Particle In A Box, Quantum Tunnelling, Scattering, Spectral Theory. The operator domain, boundary conditions, and asymptotic channels determine the spectrum and transmission amplitudes. The retained physical relation is the differential operator family and probability conservation.

$$H_D = -\frac{\hbar^2}{2m} \Delta_D + V$$

$$S : \mathcal{H}_{\text{in}} \rightarrow \mathcal{H}_{\text{out}}$$

Equivalence requires the same operator domain and conserved flux; identical differential expressions on different domains can have different spectra.

4.6 Measurement, channels, and error correction share an instrument calculus

This relation connects Quantum Channel, Measurement In Quantum Mechanics, Quantum Error Correction. A quantum instrument produces both an outcome probability and a conditional state; error correction uses the recorded outcome to select a recovery channel. The retained physical relation is complete positivity and total probability.

$$\mathcal{I}_i(\rho) = \sum_{\alpha} K_{i\alpha} \rho K_{i\alpha}^{\dagger}$$

$$\mathcal{E} = \sum_i \mathcal{I}_i$$

$$\mathcal{R}_i \circ \mathcal{I}_i$$

Equivalence requires complete positivity, normalization, and recovery fidelity with a reference system; a state-update rule alone is insufficient.

4.7 Duality and simulation require an intertwining map

This relation connects Quantum Simulator, Ads Cft Correspondence, Quantum Error Correction. An encoding represents one state space in another only when it intertwines the relevant dynamics and observables. The retained physical relation is a selected operator algebra and its correlators.

$$\begin{aligned} V H_{\text{target}} &\simeq H_{\text{carrier}} V \\ V O_{\text{target}} &\simeq O_{\text{carrier}} V \end{aligned}$$

Equivalence requires agreement of a generating set of observables or correlators within a stated approximation regime and error bound; state overlap alone is insufficient.

Chapter 5

Theory Extension By Transfer And Obstruction

A physical relation can survive in another state space, or its failure can identify the term missing from the target theory. Exact compatibility and structured incompatibility are therefore two outcomes of the same transformation.

5.1 Incomplete Mechanisms

A law can be stated before the physical degrees of freedom on which it acts are known. Conversely, a state space can be specified before its governing operation is fixed. These one-sided descriptions occur at the boundary between an abstract theory and a physical realization.

$$\begin{aligned}(\Omega, 0_\Xi) + (0_\Omega, \Xi') &\longrightarrow (\Omega, \Xi') \\ &\longrightarrow (\Omega, \Xi'; \hat{C}, \hat{R}, \hat{P}) \\ &\longrightarrow I'_{\text{real}}.\end{aligned}$$

The combined theory exists only if the operation has a domain on the target state space. Its closure must admit physical states, and its observable must produce normalized probabilities. These requirements determine the additional terms introduced by the new realization.

5.2 Exact Transfer

$$\Delta_{\alpha,\beta} = \Omega_B \alpha - \beta \Omega_A$$

The maps alpha and beta carry source states and outputs into the target description. A vanishing residual means that evolution followed by translation gives the same state as translation followed by target evolution. Agreement of the corresponding observables then establishes another physical realization of the retained mechanism. Schrodinger, Heisenberg, and path-integral descriptions provide standard examples. Their variables and intermediate objects differ, while amplitudes and expectation values agree on the common domain. Quantum simulation uses the same structure when an encoded target algebra reproduces selected correlators in another physical device.

5.3 Structured Failure

A nonzero residual separates two physical possibilities. An irregular residual reflects an incompatible comparison. A residual that recurs across states or scales and obeys a closure relation

has acquired the form of a physical operator.

$$\Delta \longmapsto \Omega', \Xi', C', R', P', \text{ or } A'.$$

Promotion into Ω introduces a correction law or interaction. Promotion into Ξ adds a hidden degree of freedom or state sector. Promotion into C changes the admissible domain. Promotion into P makes operation order part of the dynamics. The enlarged theory is fixed by the smallest promotion that restores predictive closure.

5.4 Observable Consequences

The promoted term becomes physical when it changes an observable that was not used to define it. Removing the obstructing coupling must remove that consequence. Curvature, memory, edge flux, and effective interactions are familiar outcomes of this logic: each retains information that the smaller state or transformation law had discarded.

Part II

State Spaces And Dynamical Laws

Chapter 6

State space, domain, and representation

The Hilbert or Fock space, operator domain, basis, gauge sector, and subsystem decomposition determine which states and operations exist. Changing this structure can change the theory before any Hamiltonian parameter is varied.

A Hilbert space and an operator domain determine which states and operations exist. Gauge sectors, bases, and subsystem decompositions refine that physical domain.

6.1 Topics

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Quantum differential calculus
Quantum mechanics
Old quantum theory
Fourier transform

6.2 Transformation theory (quantum mechanics)

Mechanism

Transformation theory (quantum mechanics) belongs at the first step of the constructor: it fixes the Hilbert space, operator domain, basis, representation, or preparation context before any probability statement is meaningful.

Transformation theory (quantum mechanics) is needed because a quantum equation has no fixed meaning until its state space, inner product, representation, and operator domains have been specified. These choices decide which states are admissible and which apparent changes are only changes of coordinates.

The state space, representation, basis, and operator domain determine which vectors are physical and which operators are defined.

Physical Construction

The state carrier is a Hilbert, Fock, or function space together with the operator domains and representation used in the calculation. The governing operation is a unitary or isometric change of basis, Fourier transform, coordinate map, or representation equivalence. Inner products, domains, normalization, and completeness relations must be preserved by a purely representational change. The calculated observables are Transition amplitudes, expectation values, spectra, and probabilities that remain invariant under an admissible representation change.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

A unitary or isometric change of representation carries the state and operator together. Amplitudes, expectation values, and spectra agree. If they do not, the physical model has changed rather than merely its notation.

A quantum state belongs to this state space, and every Hamiltonian and observable must act on its stated domain. These domain relations determine whether the resulting amplitudes and probabilities are defined.

Invariance And Realization

Transformation theory (quantum mechanics) supplies the admissible arena in which quantum states and operators are defined. Changing basis or representation should not change physical probabilities when the transformation is unitary. Normalization, domain conditions, and inner products remain part of the same formal container.

The local title, representation, and physical realization may change while the constructor role is preserved. The chosen basis, coordinate representation, or preparation convention can

change. The same calculation may be written with vectors, wave functions, density operators, or operator algebras. Physical realization enters later through boundary conditions, detectors, or fields.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Unitary changes of basis preserve Born probabilities; if probabilities change, the page has changed the physical context rather than only the representation. The operator domain and normalization conditions determine which questions are legal on the selected Hilbert space.

6.3 Mathematical formulation of quantum mechanics

Mechanism

Mathematical formulation of quantum mechanics belongs at the first step of the constructor: it fixes the Hilbert space, operator domain, basis, representation, or preparation context before any probability statement is meaningful.

This page is the entrance to the construction because quantum theory separates three objects that classical prose often mixes: a state encoding preparation, an operator encoding a physical question, and a probability rule connecting the two. The formalism is useful precisely because each object can change representation without changing the prediction.

The state space, representation, basis, and operator domain determine which vectors are physical and which operators are defined.

Physical Construction

The state carrier is a Hilbert, Fock, or function space together with the operator domains and representation used in the calculation. The governing operation is a unitary or isometric change of basis, Fourier transform, coordinate map, or representation equivalence. Inner products, domains, normalization, and completeness relations must be preserved by a purely representational change. The calculated observables are Transition amplitudes, expectation values, spectra, and probabilities that remain invariant under an admissible representation change.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, Hamiltonian, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (spectrum, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (uncertainty relation, complementarity, or commutator) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

A state vector or density operator carries preparation information; an observable supplies possible values; their pairing gives probabilities and expectation values. When a unitary map changes basis, the state and observable transform together. Their matrices change, but every probability remains the same.

A spin-one-half preparation can be written in the vertical basis or the horizontal basis. The two column vectors look different, and the spin operator has different matrix entries, yet a consistently transformed calculation predicts the same detector counts.

The Hilbert-space page now specifies the arena in which states, operators, inner products, and unitary transformations are defined.

Invariance And Realization

Mathematical formulation of quantum mechanics supplies the admissible arena in which quantum states and operators are defined. Changing basis or representation should not change physical probabilities when the transformation is unitary. Normalization, domain conditions, and inner products remain part of the same formal container.

The local title, representation, and physical realization may change while the constructor role is preserved. The chosen basis, coordinate representation, or preparation convention can change. The same calculation may be written with vectors, wave functions, density operators, or operator algebras. Physical realization enters later through boundary conditions, detectors, or fields.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Unitary changes of basis preserve Born probabilities; if probabilities change, the page has changed the physical context rather than only the representation. The operator domain and normalization conditions determine which questions are legal on the selected Hilbert space.

6.4 Hilbert space

Mechanism

Hilbert space is the admissible state carrier of quantum theory: it supplies the space in which states, operators, bases, spectra, and probabilities become legally defined.

Hilbert space is not merely a container for wave functions. Its inner product defines amplitudes and orthogonality, while the domains of unbounded operators decide whether expressions for position, momentum, and energy are mathematically and physically admissible.

Hilbert space is not physical space and not a geometric background in this book. It is the legal carrier of quantum identity. A state is a vector or density operator on it; the inner product gives amplitudes and norms; observables are self-adjoint operators on it; spectral projectors define possible answers; and unitary evolution preserves norm and probability. Hilbert space is therefore central because it binds state, probability, operator spectrum, and identity preservation into one formal carrier.

Physical Construction

The state carrier is a complex Hilbert space, or a density-operator state space built on it. The governing operation is Self-adjoint observables, unitary maps, spectral projectors, and domain-restricted generators defined on the carrier. Inner-product structure, normalization, positivity for density states, and operator-domain conditions make states and observables legal. The calculated observables are Born probabilities, spectral projectors, expectation values, and preserved norms.

Topic Equations

Standard constructor skeleton: normalized states, density states, spectral resolution, Born observable, and unitary identity preservation.

$$\begin{aligned} |\psi\rangle &\in \mathcal{H}, & \langle\psi|\psi\rangle &= 1 \\ \rho &\in \mathcal{S}(\mathcal{H}), & \rho &\geq 0, \quad \text{Tr } \rho = 1 \\ A &= A^\dagger, & A &= \int_{\sigma(A)} \lambda dE_A(\lambda) \\ \text{Pr}(\Delta \mid \rho, A) &= \text{Tr}(\rho E_A(\Delta)) \\ \rho_t &= U(t)\rho U(t)^\dagger, & U^\dagger U &= I \end{aligned}$$

Physical Meaning

The linear structure permits superposition, the inner product converts pairs of states into amplitudes, and completeness guarantees that convergent sequences of approximations remain inside the space. Operator domains must be carried with the operators; the same differential formula on another domain can describe a different physical system.

A qubit lives in a two-dimensional complex space, whereas a particle on a line is described in a space of square-integrable functions. Both obey the same Hilbert-space logic, but their operators, spectra, and boundary conditions are different.

After the arena is fixed, a quantum state selects one preparation or statistical ensemble within it.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

6.5 Quantum differential calculus

Mechanism

Quantum differential calculus belongs at the first step of the constructor: it fixes the Hilbert space, operator domain, basis, representation, or preparation context before any probability statement is meaningful.

Quantum differential calculus is needed because a quantum equation has no fixed meaning until its state space, inner product, representation, and operator domains have been specified. These choices decide which states are admissible and which apparent changes are only changes of coordinates.

The state space, representation, basis, and operator domain determine which vectors are physical and which operators are defined.

Physical Construction

The state carrier is a Hilbert, Fock, or function space together with the operator domains and representation used in the calculation. The governing operation is a unitary or isometric change of basis, Fourier transform, coordinate map, or representation equivalence. Inner products, domains, normalization, and completeness relations must be preserved by a purely representational change. The calculated observables are Transition amplitudes, expectation values, spectra, and probabilities that remain invariant under an admissible representation change.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

A unitary or isometric change of representation carries the state and operator together. Amplitudes, expectation values, and spectra agree. If they do not, the physical model has changed rather than merely its notation.

A quantum state belongs to this state space, and every Hamiltonian and observable must act on its stated domain. These domain relations determine whether the resulting amplitudes and probabilities are defined.

Invariance And Realization

Quantum differential calculus supplies the admissible arena in which quantum states and operators are defined. Changing basis or representation should not change physical probabilities when the transformation is unitary. Normalization, domain conditions, and inner products remain part of the same formal container.

The local title, representation, and physical realization may change while the constructor role is preserved. The chosen basis, coordinate representation, or preparation convention can

change. The same calculation may be written with vectors, wave functions, density operators, or operator algebras. Physical realization enters later through boundary conditions, detectors, or fields.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Unitary changes of basis preserve Born probabilities; if probabilities change, the page has changed the physical context rather than only the representation. The operator domain and normalization conditions determine which questions are legal on the selected Hilbert space.

6.6 Quantum mechanics

Mechanism

Quantum mechanics is the baseline constructor: states live in Hilbert space, physical questions are represented by operators, and probabilities are assigned to spectral projectors.

Quantum mechanics is needed because a quantum equation has no fixed meaning until its state space, inner product, representation, and operator domains have been specified. These choices decide which states are admissible and which apparent changes are only changes of coordinates.

The page supplies the general quantum assembly. A preparation gives a state vector or density operator. A self-adjoint observable or measurement operator family gives the possible outcome channels. The Born or trace rule assigns probabilities, while Hamiltonian evolution transports the state between preparation and observable.

Physical Construction

The state carrier is a Hilbert, Fock, or function space together with the operator domains and representation used in the calculation. The governing operation is a unitary or isometric change of basis, Fourier transform, coordinate map, or representation equivalence. Inner products, domains, normalization, and completeness relations must be preserved by a purely representational change. The calculated observables are Transition amplitudes, expectation values, spectra, and probabilities that remain invariant under an admissible representation change.

Topic Equations

Topic-specific constructor: the equations express state admissibility, spectral prediction, unitary evolution, and incompatibility.

$$\begin{aligned}\rho &\geq 0, & \text{Tr } \rho &= 1 \\ A &= \sum_a a P_a, & p(a) &= \text{Tr}(\rho P_a) \\ \rho(t) &= U(t)\rho(0)U(t)^\dagger, & U(t) &= e^{-iHt/\hbar} \\ [A, B] &\neq 0 & \Rightarrow & \text{no generic common sharp eigenbasis}\end{aligned}$$

Physical Meaning

A unitary or isometric change of representation carries the state and operator together. Amplitudes, expectation values, and spectra agree. If they do not, the physical model has changed rather than merely its notation.

A quantum state belongs to this state space, and every Hamiltonian and observable must act on its stated domain. These domain relations determine whether the resulting amplitudes and probabilities are defined.

Invariance And Realization

The relation between prepared states, observables, and spectral probability measures. The use of eigenvalues, projectors, modes, or outcome channels to represent admissible observations. The dependence of the observable on basis, domain, potential, preparation, or measurement context. The commutator structure that limits which observables can be jointly diagonalized.

The physical carrier: particle, wave, field mode, spin, qubit, detector, or excitation. The representation: wave mechanics, matrix mechanics, density matrices, path integrals, circuits, or

fields. Where time dependence is placed: on the state, on the operator, in a propagator, or in a path weight. The implementation of preparation, boundary condition, detector, or outcome channel.

Discriminating Consequences

A transfer target provides a state space, a transformation law, and a spectral or categorical observable, with one compatibility relation experimentally unresolved. A useful validation varies the basis, domain, or measurement context and measures whether the allowed observable changes while the underlying transformation law remains identifiable. A stronger validation contains two candidate observables whose predicted commutator controls joint resolvability.

6.7 Old quantum theory

Mechanism

Old quantum theory is the semiclassical quantization precursor to modern quantum mechanics: classical periodic motion is retained, while only selected actions and transition frequencies are admitted.

Old quantum theory is needed because a quantum equation has no fixed meaning until its state space, inner product, representation, and operator domains have been specified. These choices decide which states are admissible and which apparent changes are only changes of coordinates.

Old quantum theory predates the Hilbert-space formalism. Its carrier is classical phase space, its motion follows Hamiltonian trajectories, and its quantum restriction is imposed on periodic action integrals. The resulting discrete energies and transition frequencies anticipated quantum spectra but did not provide a general theory of states, observables, or noncommuting transformations.

Physical Construction

The state carrier is Classical phase space, especially periodic or multiply periodic Hamiltonian trajectories described by action variables. The governing operation is Hamiltonian flow together with Bohr–Sommerfeld action quantization; no general Hilbert-space operator calculus is assumed. Only trajectories satisfying quantized action conditions are retained, with correspondence to classical frequencies required at large quantum number. The calculated observables are Discrete energies, transition frequencies, and the spectral regularities inferred from them.

Topic Equations

Semiclassical constructor: Hamiltonian orbits are restricted by action quantization, and energy differences determine spectral frequencies.

$$\begin{aligned} J_k &= \oint p_k dq_k = n_k h \\ E_m - E_n &= h\nu_{mn} \\ \frac{\partial E}{\partial J_k} &= \nu_k^{\text{cl}} \quad \text{in the correspondence regime} \end{aligned}$$

Physical Meaning

A unitary or isometric change of representation carries the state and operator together. Amplitudes, expectation values, and spectra agree. If they do not, the physical model has changed rather than merely its notation.

A quantum state belongs to this state space, and every Hamiltonian and observable must act on its stated domain. These domain relations determine whether the resulting amplitudes and probabilities are defined.

Invariance And Realization

Hamiltonian trajectories and their action variables provide the classical carrier of the construction. Action quantization selects a discrete subset of otherwise continuous classical motions. Energy differences determine transition frequencies, with correspondence to classical orbital frequencies at large quantum number.

The orbit family, potential, number of action variables, and applicable quantization condition depend on the physical system. Modern semiclassical theory adds phase and turning-point corrections absent from the earliest quantization rules. The construction ceases to be adequate for generic nonintegrable motion, many-electron spectra, and intrinsically noncommuting questions.

Discriminating Consequences

The theory recovers the hydrogenic level and line spectrum in the regime where action quantization applies. A comparison of the semiclassical prediction with the corresponding Schrödinger spectrum and identify its controlled limit. Treat failures outside that limit as evidence for the modern state-and-operator formalism, rather than as adjustable orbit parameters.

6.8 Fourier transform

Mechanism

Fourier transform belongs at the first step of the constructor: it fixes the Hilbert space, operator domain, basis, representation, or preparation context before any probability statement is meaningful.

Fourier transform is needed because a quantum equation has no fixed meaning until its state space, inner product, representation, and operator domains have been specified. These choices decide which states are admissible and which apparent changes are only changes of coordinates.

The state space, representation, basis, and operator domain determine which vectors are physical and which operators are defined.

Physical Construction

The state carrier is a Hilbert, Fock, or function space together with the operator domains and representation used in the calculation. The governing operation is a unitary or isometric change of basis, Fourier transform, coordinate map, or representation equivalence. Inner products, domains, normalization, and completeness relations must be preserved by a purely representational change. The calculated observables are Transition amplitudes, expectation values, spectra, and probabilities that remain invariant under an admissible representation change.

Defining Relations

Preparation or domain terms (domain, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (unitary operator, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (uncertainty relation, commutator, or incompatible observable) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

A unitary or isometric change of representation carries the state and operator together. Amplitudes, expectation values, and spectra agree. If they do not, the physical model has changed rather than merely its notation.

A quantum state belongs to this state space, and every Hamiltonian and observable must act on its stated domain. These domain relations determine whether the resulting amplitudes and probabilities are defined.

Invariance And Realization

Fourier transform supplies the admissible arena in which quantum states and operators are defined. Changing basis or representation should not change physical probabilities when the transformation is unitary. Normalization, domain conditions, and inner products remain part of the same formal container.

The local title, representation, and physical realization may change while the constructor role is preserved. The chosen basis, coordinate representation, or preparation convention can change. The same calculation may be written with vectors, wave functions, density operators,

or operator algebras. Physical realization enters later through boundary conditions, detectors, or fields.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Unitary changes of basis preserve Born probabilities; if probabilities change, the page has changed the physical context rather than only the representation. The operator domain and normalization conditions determine which questions are legal on the selected Hilbert space.

Chapter 7

Quantum states and subsystem structure

A state is the probability-bearing element of the selected Hilbert space or its density-operator state space. Wave functions, density matrices, superpositions, and coherent states are different representations of this predictive carrier.

The state must retain every coordinate needed to determine later probabilities. Entanglement enters here because it is a property of a joint state relative to a stated subsystem decomposition.

7.1 Topics

Page

Density matrix

Quantum superposition

Quantum decoherence

Coherence (physics)

Superposition principle (*see Quantum Superposition*)

Wave function

Quantum state

Two-state quantum system

Qubit

Quantum number

Spin (physics)

Schrödinger's cat

Quantum entanglement

Quantum biology

Macroscopic quantum phenomena

Quantum fluctuation

7.2 Density matrix

Mechanism

Density matrix is the mixed-state constructor: it keeps probabilistic preparation, entanglement with unobserved degrees of freedom, and partial information in the same state formalism.

The density matrix is required whenever the preparation is statistical, part of an entangled system is ignored, or environmental coupling makes a state-vector description of the subsystem incomplete.

Density matrices generalize pure states without changing the state-to-spectrum probability assignment rule. They are the correct carrier when the preparation is statistical, when a subsystem is traced out, or when decoherence is being described.

Physical Construction

The state carrier is the mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register. The governing operation is Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states. The calculated observables are Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.

Topic Equations

Standard constructor skeleton: mixed state, trace rule, and subsystem reduction.

$$\begin{aligned}\rho &= \sum_a p_a |\psi_a\rangle\langle\psi_a|, & p_a &\geq 0, & \sum_a p_a &= 1 \\ \rho &\geq 0, & \text{Tr } \rho &= 1 \\ p(i) &= \text{Tr}(\rho P_i) \\ \rho_A &= \text{Tr}_B(\rho_{AB})\end{aligned}$$

Physical Meaning

Diagonal elements in a selected basis give populations and off-diagonal elements carry coherence in that basis. Positivity and unit trace make the Born probabilities legal. Partial tracing produces the state of a subsystem without assigning a pure vector to it.

Either randomly preparing two spin directions or discarding one member of an entangled pair can produce a mixed state. The density matrix records the observable statistics, even though the physical origins of the mixture differ.

With the state representation established, unitary and non-unitary maps specify how it changes.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system,

basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

7.3 Quantum superposition

Mechanism

Quantum superposition supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Quantum superposition specifies the object from which quantum probabilities are calculated. A Hamiltonian or an observable does not make a prediction by itself; it must act on a normalized state vector, density operator, or statistical sector that records the preparation.

The topic contributes the mathematical carrier of prediction: vector, wavefunction, density operator, register, coherent state, or field state.

Physical Construction

The state carrier is the mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register. The governing operation is Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states. The calculated observables are Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or superposition) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (observable operator, Hamiltonian, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Normalization guarantees that the probabilities sum to one, while positivity prevents negative probabilities. Pure vectors and density operators are not competing theories: the density-operator form also represents mixtures and reduced states obtained when unobserved degrees of freedom are traced out.

The Hamiltonian or channel evolves the prepared state. An observable and measurement map then convert that evolved state into outcome probabilities.

Invariance And Realization

The topic contributes the mathematical carrier of prediction: vector, wavefunction, density operator, register, coherent state, or field state. Quantum superposition carries the predictive information before a measurement question is asked. The same physical preparation may be represented as a vector, wave function, density matrix, or reduced state. Normalization and positivity are the admissibility checks that make the state usable for probability assignment.

The local title, representation, and physical realization may change while the constructor role is preserved. The state representation can change between position, momentum, spin, occupation, or density-operator forms. Pure-state and mixed-state descriptions may differ while

describing the same formal preparation. Subsystem descriptions change when degrees of freedom are traced out or ignored.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. A usable state gives normalized probabilities for every complete observable attached to the selected Hilbert space. Vector, wave-function, density-matrix, and reduced-state forms can describe the same preparation when connected by the appropriate representation map.

7.4 Quantum decoherence

Mechanism

Quantum decoherence is the loss of observable phase coherence in a subsystem when information about alternative amplitudes becomes encoded in environmental correlations.

Interference is lost when alternative system amplitudes become correlated with environmental states that can, even in principle, distinguish them. The missing coherence has then moved into correlations of the larger state rather than being erased by the subsystem equation alone.

The joint system and environment may evolve unitarily while the reduced system loses interference. Interaction correlates different system alternatives with distinguishable environmental states; tracing over the environment then suppresses off-diagonal terms in the selected pointer basis. Decoherence does not select one outcome, and it is not identical to memory. Markovian decoherence is possible when the reduced state still determines its future. Memory begins when discarded correlations return and two preparations with the same reduced state acquire different later statistics.

Physical Construction

The state carrier is a joint system-environment state together with the reduced density operator accessible to the observer. The governing operation is System-environment unitary evolution followed by partial trace, or the corresponding reduced dynamical map. The joint state remains normalized and positive; a Markovian reduction additionally requires future maps to be fixed by the present reduced state. The calculated observables are Off-diagonal coherence, interference visibility, purity, and history-dependent response.

Topic Equations

Joint unitary evolution, environmental distinguishability, reduced decoherence, and the semi-group test distinguish loss of interference from dynamical memory.

$$\begin{aligned}\rho_S(t) &= \text{Tr}_E[U_{SE}(t)\rho_{SE}(0)U_{SE}^\dagger(t)] \\ &= \frac{|0\rangle|E_0\rangle + |1\rangle|E_1\rangle}{\sqrt{2}}, \quad (\rho_S)_{01} \propto \langle E_1|E_0\rangle \\ \dot{\rho}_S &= -\frac{i}{\hbar}[H_S, \rho_S] + \sum_k \gamma_k \mathcal{D}[L_k]\rho_S \\ V_{t+s} &= V_t V_s \quad \text{only in the time-homogeneous Markov limit}\end{aligned}$$

Physical Meaning

Unitary system-environment evolution can suppress the off-diagonal elements of the reduced density matrix after the environment is traced out. The overlap of the corresponding environmental states fixes the remaining visibility. Decoherence can be Markovian; memory requires the stronger condition that hidden correlations later alter evolution from the same reduced state.

In a two-path interferometer, scattering one distinguishable environmental state from each path lowers the fringe visibility in proportion to their overlap. Erasing the distinguishing record can restore interference in suitable conditional measurements.

The distinction between lost visibility and returning correlations leads directly to quantum channels, non-Markovian dynamics, and the composition test for memory.

Consequences Forced By The Relation

Interference visibility falls with the overlap of the environmental states correlated with the interfering alternatives. The phase information can remain in the joint state even when it is absent from all reduced-system observables. When hidden correlations later alter the reduced dynamics, an auxiliary state coordinate or a memory kernel is required for predictive closure.

Domain Of The Construction

The preferred basis and decoherence rate depend on the interaction, environmental spectrum, and initial system-environment state. Suppression of off-diagonal terms does not by itself derive a unique recorded outcome.

Invariance And Realization

The joint state retains phase information transferred from the subsystem into system-environment correlations. Reduced interference visibility is fixed by the overlap of environmental states correlated with the alternatives. Equivalent system-environment dilations give the same reduced channel and system observables.

The preferred basis, decay rate, and recoherence depend on the interaction, environmental spectrum, and initial correlations. Changing the system-environment partition changes which correlations are hidden. A Markov approximation removes returning correlations; retaining them introduces auxiliary coordinates or a memory kernel.

Discriminating Consequences

Interference visibility is compared with a control in which the environment cannot distinguish the alternatives. Two preparations with the same reduced state determine whether hidden correlations alter later observables. Reversing or decoupling the interaction determines whether coherence remains recoverable in the joint state.

7.5 Coherence (physics)

Mechanism

Coherence (physics) supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Coherence (physics) specifies the object from which quantum probabilities are calculated. A Hamiltonian or an observable does not make a prediction by itself; it must act on a normalized state vector, density operator, or statistical sector that records the preparation.

The topic concerns quantum state transport under environmental coupling, coherence loss, biological or macroscopic boundary conditions, or effective dynamics outside an ideal closed system.

Physical Construction

The state carrier is a density operator, reduced state, coherence variable, bath-coupled state, or effective mesoscopic carrier. The governing operation is Hamiltonian plus environmental coupling, Lindbladian, memory kernel, stochastic map, or effective transport operator. Positivity, trace preservation, timescale separation, bath assumptions, and control over classical noise determine whether the model is legal. The calculated observables are Coherence, population transfer, relaxation rate, transport efficiency, noise spectrum, or macroscopic response.

Defining Relations

Preparation or domain terms (potential or domain condition, detector, or basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (unitary operator, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (spectrum, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Normalization guarantees that the probabilities sum to one, while positivity prevents negative probabilities. Pure vectors and density operators are not competing theories: the density-operator form also represents mixtures and reduced states obtained when unobserved degrees of freedom are traced out.

The Hamiltonian or channel evolves the prepared state. An observable and measurement map then convert that evolved state into outcome probabilities.

Invariance And Realization

The topic concerns quantum state transport under environmental coupling, coherence loss, biological or macroscopic boundary conditions, or effective dynamics outside an ideal closed system. Coherence (physics) carries the predictive information before a measurement question is asked. The same physical preparation may be represented as a vector, wave function, density matrix, or reduced state. Normalization and positivity are the admissibility checks that make the state usable for probability assignment.

The local title, representation, and physical realization may change while the constructor role is preserved. The state representation can change between position, momentum, spin, occupation, or density-operator forms. Pure-state and mixed-state descriptions may differ while describing the same formal preparation. Subsystem descriptions change when degrees of freedom are traced out or ignored.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. A usable state gives normalized probabilities for every complete observable attached to the selected Hilbert space. Vector, wave-function, density-matrix, and reduced-state forms can describe the same preparation when connected by the appropriate representation map.

7.6 Superposition principle

Superposition principle is an alternative name for Quantum Superposition. The physical derivation is given in [Quantum Superposition](#).

7.7 Wave function

Mechanism

Wave function is a basis-dependent representative of a pure-state ray; it is not identical to the abstract state or to physical configuration space.

The wave function is a representation of a state in a chosen continuous basis. Treating it as the state itself can obscure that a Fourier transform changes its shape while preserving all predictions.

For a configuration space Q with measure μ , the position wave function is the generalized-basis representative $\psi(x) = \langle x | \psi \rangle$ of a ray $[\psi]$ in $L^2(Q, \mu)$. Vectors that differ by a nonzero global phase represent the same pure state. Its modulus squared is a probability density only relative to the stated position measure; spin and particle statistics enlarge or constrain the carrier.

Physical Construction

The state carrier is the mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register. The governing operation is Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states. The calculated observables are Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.

Topic Equations

Topic-specific constructor: abstract state ray, position representation, measure-dependent Born probability, and internal spin carrier.

$$\begin{aligned}\mathcal{H} &= L^2(Q, d\mu), & \psi(x) &= \langle x | \psi \rangle \\ \int_Q |\psi(x)|^2 d\mu(x) &= 1, & |\psi\rangle &\sim e^{i\alpha} |\psi\rangle \\ \Pr(X \in \Delta \mid \psi) &= \langle \psi | E_X(\Delta) | \psi \rangle = \int_\Delta |\psi(x)|^2 d\mu(x) \\ \mathcal{H}_{\text{spin } s} &= L^2(Q, d\mu) \otimes \mathbb{C}^{2s+1}\end{aligned}$$

Physical Meaning

Its complex value is a probability amplitude. The squared magnitude gives a position density only in the position representation; phase differences remain essential because they control interference and momentum content. Normalization fixes the total probability.

A localized wave packet and its momentum-space Fourier transform emphasize different features of one preparation. Their forms differ, but either representation reproduces the same expectation values when the observables are transformed consistently.

Density operators extend this description to mixtures, subsystems, and open-system dynamics.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels.

The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

7.8 Quantum state

Mechanism

Quantum state supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

A quantum state is the minimal object that assigns probabilities to every admissible measurement in a fixed context. It is therefore defined by its predictive role, not by a preferred wave-function notation.

The topic contributes the mathematical carrier of prediction: vector, wavefunction, density operator, register, coherent state, or field state.

Physical Construction

The state carrier is the mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register. The governing operation is Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states. The calculated observables are Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.

Defining Relations

Preparation or domain terms (detector, preparation, or preparation context) determine which states are admissible. State terms (Quantum state, wave function, or superposition) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (observable operator, matrix, or Hamiltonian) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenstate, spectrum, or eigenvalue) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (uncertainty relation, incompatible observables, or commutator) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

A pure state is represented by a ray rather than by a unique vector, because an overall phase changes no probability. More general preparations are density operators. In either form, the state must be normalized and must yield non-negative probabilities.

Two laboratories may prepare the same polarization statistics by different procedures. If every allowed measurement has the same probability distribution, quantum theory assigns the same density operator to both preparations.

The wave-function and density-matrix pages develop the two principal representations of this predictive object.

Invariance And Realization

The topic contributes the mathematical carrier of prediction: vector, wavefunction, density operator, register, coherent state, or field state. Quantum state carries the predictive information before a measurement question is asked. The same physical preparation may be represented as a vector, wave function, density matrix, or reduced state. Normalization and positivity are the admissibility checks that make the state usable for probability assignment.

The local title, representation, and physical realization may change while the constructor role is preserved. The state representation can change between position, momentum, spin, occupation, or density-operator forms. Pure-state and mixed-state descriptions may differ while describing the same formal preparation. Subsystem descriptions change when degrees of freedom are traced out or ignored.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. A usable state gives normalized probabilities for every complete observable attached to the selected Hilbert space. Vector, wave-function, density-matrix, and reduced-state forms can describe the same preparation when connected by the appropriate representation map.

7.9 Two-state quantum system

Mechanism

Two-state quantum system supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Two-state quantum system specifies the object from which quantum probabilities are calculated. A Hamiltonian or an observable does not make a prediction by itself; it must act on a normalized state vector, density operator, or statistical sector that records the preparation.

The topic contributes the mathematical carrier of prediction: vector, wavefunction, density operator, register, coherent state, or field state.

Physical Construction

The state carrier is the mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register. The governing operation is Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states. The calculated observables are Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (wavefunction, state vector, or superposition) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (observable operator, Hamiltonian, or matrix) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Normalization guarantees that the probabilities sum to one, while positivity prevents negative probabilities. Pure vectors and density operators are not competing theories: the density-operator form also represents mixtures and reduced states obtained when unobserved degrees of freedom are traced out.

The Hamiltonian or channel evolves the prepared state. An observable and measurement map then convert that evolved state into outcome probabilities.

Invariance And Realization

The topic contributes the mathematical carrier of prediction: vector, wavefunction, density operator, register, coherent state, or field state. Two-state quantum system carries the predictive information before a measurement question is asked. The same physical preparation may be represented as a vector, wave function, density matrix, or reduced state. Normalization and positivity are the admissibility checks that make the state usable for probability assignment.

The local title, representation, and physical realization may change while the constructor role is preserved. The state representation can change between position, momentum, spin, occupation, or density-operator forms. Pure-state and mixed-state descriptions may differ while

describing the same formal preparation. Subsystem descriptions change when degrees of freedom are traced out or ignored.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. A usable state gives normalized probabilities for every complete observable attached to the selected Hilbert space. Vector, wave-function, density-matrix, and reduced-state forms can describe the same preparation when connected by the appropriate representation map.

7.10 Qubit

Mechanism

Qubit is the two-dimensional state-carrier constructor used when the admissible state space is $\mathbb{C}^2$.

Qubit specifies the object from which quantum probabilities are calculated. A Hamiltonian or an observable does not make a prediction by itself; it must act on a normalized state vector, density operator, or statistical sector that records the preparation.

A qubit is the minimal quantum state space with a basis, amplitudes, unitary control, and measurement observable. Bloch-vector language is a representation of the same two-dimensional carrier.

Physical Construction

The state carrier is the mathematical state object: vector, wavefunction, density operator, coherent state, field state, or register. The governing operation is Operators, maps, and observables become meaningful only after this carrier and its domain have been fixed. Normalization, positivity, inner product, representation, tensor factorization, or superselection conditions define legal states. The calculated observables are Probability distributions obtained by applying the appropriate observables or measurement maps to the carrier.

Topic Equations

Standard constructor skeleton: two-state carrier, Bloch representation, and basis observable.

$$\begin{aligned} |\psi\rangle &= \alpha|0\rangle + \beta|1\rangle, & |\alpha|^2 + |\beta|^2 &= 1 \\ \rho &= \frac{1}{2}(I + \mathbf{r} \cdot \boldsymbol{\sigma}), & |\mathbf{r}| &\leq 1 \\ p(0) &= |\langle 0|\psi\rangle|^2, & p(1) &= |\langle 1|\psi\rangle|^2 \end{aligned}$$

Physical Meaning

Normalization guarantees that the probabilities sum to one, while positivity prevents negative probabilities. Pure vectors and density operators are not competing theories: the density-operator form also represents mixtures and reduced states obtained when unobserved degrees of freedom are traced out.

The Hamiltonian or channel evolves the prepared state. An observable and measurement map then convert that evolved state into outcome probabilities.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

7.11 Quantum number

Mechanism

Quantum number supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Quantum number specifies the object from which quantum probabilities are calculated. A Hamiltonian or an observable does not make a prediction by itself; it must act on a normalized state vector, density operator, or statistical sector that records the preparation.

The state vector or density operator carries the preparation information from which probabilities are calculated.

Physical Construction

The state carrier is a state vector, wave function, density operator, field state, register state, or superselection sector. The governing operation is the transformations and observables defined on that state space. Normalization, positivity, inner-product, tensor-factorization, and superselection conditions define the allowed states. The calculated observables are Born probabilities, expectation values, reduced states, or correlation functions determined by the state.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (observable operator, Hamiltonian, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (spectrum, energy level, or eigenvalue) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (uncertainty relation, commutator, or incompatible observable) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Normalization guarantees that the probabilities sum to one, while positivity prevents negative probabilities. Pure vectors and density operators are not competing theories: the density-operator form also represents mixtures and reduced states obtained when unobserved degrees of freedom are traced out.

The Hamiltonian or channel evolves the prepared state. An observable and measurement map then convert that evolved state into outcome probabilities.

Invariance And Realization

Quantum number carries the predictive information before a measurement question is asked. The same physical preparation may be represented as a vector, wave function, density matrix, or reduced state. Normalization and positivity are the admissibility checks that make the state usable for probability assignment.

The local title, representation, and physical realization may change while the constructor role is preserved. The state representation can change between position, momentum, spin, occupation, or density-operator forms. Pure-state and mixed-state descriptions may differ while describing the same formal preparation. Subsystem descriptions change when degrees of freedom are traced out or ignored.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. A usable state gives normalized probabilities for every complete observable attached to the selected Hilbert space. Vector, wave-function, density-matrix, and reduced-state forms can describe the same preparation when connected by the appropriate representation map.

7.12 Spin (physics)

Mechanism

Spin (physics) supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Spin (physics) specifies the object from which quantum probabilities are calculated. A Hamiltonian or an observable does not make a prediction by itself; it must act on a normalized state vector, density operator, or statistical sector that records the preparation.

The state vector or density operator carries the preparation information from which probabilities are calculated.

Physical Construction

The state carrier is a state vector, wave function, density operator, field state, register state, or superselection sector. The governing operation is the transformations and observables defined on that state space. Normalization, positivity, inner-product, tensor-factorization, and superselection conditions define the allowed states. The calculated observables are Born probabilities, expectation values, reduced states, or correlation functions determined by the state.

Defining Relations

Preparation or domain terms (context, preparation context, or measurement basis) determine which states are admissible. State terms (wave function, quantum state, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Normalization guarantees that the probabilities sum to one, while positivity prevents negative probabilities. Pure vectors and density operators are not competing theories: the density-operator form also represents mixtures and reduced states obtained when unobserved degrees of freedom are traced out.

The Hamiltonian or channel evolves the prepared state. An observable and measurement map then convert that evolved state into outcome probabilities.

Invariance And Realization

Spin (physics) carries the predictive information before a measurement question is asked. The same physical preparation may be represented as a vector, wave function, density matrix, or reduced state. Normalization and positivity are the admissibility checks that make the state usable for probability assignment.

The local title, representation, and physical realization may change while the constructor role is preserved. The state representation can change between position, momentum, spin, occupation, or density-operator forms. Pure-state and mixed-state descriptions may differ while describing the same formal preparation. Subsystem descriptions change when degrees of freedom are traced out or ignored.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. A usable state gives normalized probabilities for every complete observable attached to the selected Hilbert space. Vector, wave-function, density-matrix, and reduced-state forms can describe the same preparation when connected by the appropriate representation map.

7.13 Schrödinger's cat

Mechanism

Schrödinger's cat supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Schrödinger's cat specifies the object from which quantum probabilities are calculated. A Hamiltonian or an observable does not make a prediction by itself; it must act on a normalized state vector, density operator, or statistical sector that records the preparation.

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement.

Physical Construction

The state carrier is a state vector, density operator, wavefunction, field state, or register on a specified domain. The governing operation is Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal. The calculated observables are Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.

Defining Relations

Preparation or domain terms (domain, preparation context, or measurement basis) determine which states are admissible. State terms (wave function, quantum state, or wavefunction) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (observable operator, Hamiltonian, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Normalization guarantees that the probabilities sum to one, while positivity prevents negative probabilities. Pure vectors and density operators are not competing theories: the density-operator form also represents mixtures and reduced states obtained when unobserved degrees of freedom are traced out.

The Hamiltonian or channel evolves the prepared state. An observable and measurement map then convert that evolved state into outcome probabilities.

Invariance And Realization

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement. Schrödinger's cat carries the predictive information before a measurement question is asked. The same physical preparation may be represented as a vector, wave function, density matrix, or reduced state. Normalization and positivity are the admissibility checks that make the state usable for probability assignment.

The local title, representation, and physical realization may change while the constructor role is preserved. The state representation can change between position, momentum, spin, occupation, or density-operator forms. Pure-state and mixed-state descriptions may differ while

describing the same formal preparation. Subsystem descriptions change when degrees of freedom are traced out or ignored.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. A usable state gives normalized probabilities for every complete observable attached to the selected Hilbert space. Vector, wave-function, density-matrix, and reduced-state forms can describe the same preparation when connected by the appropriate representation map.

7.14 Quantum entanglement

Mechanism

Quantum entanglement is a property of a composite state that cannot be written as a classical mixture of product states across a chosen subsystem decomposition; for a pure state, this reduces to failure to factor.

Entanglement is required when the state of a composite system cannot be assembled from independent states of its parts. It changes which correlations are possible and makes subsystem states mixed even when the total state is pure.

The tensor-product decomposition specifies what counts as subsystem A and subsystem B. A pure state is entangled when its Schmidt rank exceeds one. Each subsystem can then be mixed even though the joint state is pure, because partial tracing discards the correlations that purify it. Local basis changes preserve the Schmidt coefficients, while a different physical factorization can change whether the same vector is called entangled. Bell measurements test whether the resulting correlations admit a local hidden-variable model.

Physical Construction

The state carrier is a composite Hilbert space with a physically specified subsystem algebra or tensor-product factorization. The governing operation is Schmidt decomposition, partial trace, local observables, and joint correlation operators. The joint density operator is positive and normalized; separability is defined relative to the chosen subsystem structure. The calculated observables are Reduced-state spectra, entanglement entropy, correlation witnesses, and Bell parameters.

Topic Equations

Subsystem factorization, Schmidt spectrum, reduced-state entropy, and Bell correlations separate the state relation from its local observable.

$$\begin{aligned}\mathcal{H}_{AB} &= \mathcal{H}_A \otimes \mathcal{H}_B \\ |\Psi\rangle_{AB} &= \sum_k \sqrt{\lambda_k} |k_A\rangle |k_B\rangle, \quad \sum_k \lambda_k = 1 \\ \rho_A &= \text{Tr}_B |\Psi\rangle\langle\Psi|, \quad S_A = -\text{Tr}(\rho_A \log \rho_A) \\ |S_{\text{CHSH}}| &\leq 2, \quad |S_{\text{CHSH}}|_{\text{QM}} \leq 2\sqrt{2}\end{aligned}$$

Physical Meaning

The tensor-product structure defines the proposed subsystems. A state is entangled when it is not separable across that factorization. Partial traces describe each part, while joint observables expose correlations that no product state can reproduce.

In a Bell pair, each spin alone is maximally mixed, yet measurements on the pair are strongly correlated. The information resides in the relation between subsystems rather than in either local state.

Bell's theorem and contextuality determine which entangled correlations are incompatible with classical joint assignments.

Consequences Forced By The Relation

A pure bipartite state with more than one nonzero Schmidt coefficient gives mixed reduced states and correlations unavailable to product preparations. Local unitary changes of basis preserve

the Schmidt coefficients and the pure-state entanglement entropy. Suitable local measurements can violate a Bell inequality even though each subsystem alone carries no corresponding pure state.

Domain Of The Construction

Entanglement is defined relative to a physical subsystem algebra or tensor-product factorization. For mixed states, nonfactorization of one decomposition is insufficient; separability requires testing all convex product decompositions.

Invariance And Realization

Schmidt coefficients and pure-state entanglement entropy are invariant under local unitary changes of basis. The reduced-state spectra preserve the amount of bipartite pure-state entanglement. Correlations remain joint properties even when neither subsystem has a pure local state.

The subsystem factorization and accessible observable algebra determine which correlations count as entanglement. Noise, loss, and coarse graining can convert pure-state entanglement into mixed-state correlations. Photons, spins, atoms, modes, and encoded qubits can realize the same Schmidt structure.

Discriminating Consequences

Local measurements reconstruct a correlation witness or Bell parameter that product preparations cannot reproduce. Independent local-unitary rotations leave the inferred Schmidt spectrum unchanged. A separable-state control fixes the correlation background of the apparatus.

7.15 Quantum biology

Mechanism

Quantum biology supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Quantum biology specifies the object from which quantum probabilities are calculated. A Hamiltonian or an observable does not make a prediction by itself; it must act on a normalized state vector, density operator, or statistical sector that records the preparation.

The topic concerns quantum state transport under environmental coupling, coherence loss, biological or macroscopic boundary conditions, or effective dynamics outside an ideal closed system.

Physical Construction

The state carrier is a density operator, reduced state, coherence variable, bath-coupled state, or effective mesoscopic carrier. The governing operation is Hamiltonian plus environmental coupling, Lindbladian, memory kernel, stochastic map, or effective transport operator. Positivity, trace preservation, timescale separation, bath assumptions, and control over classical noise determine whether the model is legal. The calculated observables are Coherence, population transfer, relaxation rate, transport efficiency, noise spectrum, or macroscopic response.

Defining Relations

Preparation or domain terms (potential or domain condition, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (complementarity, uncertainty relation, or commutator) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Normalization guarantees that the probabilities sum to one, while positivity prevents negative probabilities. Pure vectors and density operators are not competing theories: the density-operator form also represents mixtures and reduced states obtained when unobserved degrees of freedom are traced out.

The Hamiltonian or channel evolves the prepared state. An observable and measurement map then convert that evolved state into outcome probabilities.

Invariance And Realization

The topic concerns quantum state transport under environmental coupling, coherence loss, biological or macroscopic boundary conditions, or effective dynamics outside an ideal closed system. Quantum biology carries the predictive information before a measurement question is asked. The same physical preparation may be represented as a vector, wave function, density matrix, or reduced state. Normalization and positivity are the admissibility checks that make the state usable for probability assignment.

The local title, representation, and physical realization may change while the constructor role is preserved. The state representation can change between position, momentum, spin, occupation, or density-operator forms. Pure-state and mixed-state descriptions may differ while describing the same formal preparation. Subsystem descriptions change when degrees of freedom are traced out or ignored.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. A usable state gives normalized probabilities for every complete observable attached to the selected Hilbert space. Vector, wave-function, density-matrix, and reduced-state forms can describe the same preparation when connected by the appropriate representation map.

7.16 Macroscopic quantum phenomena

Mechanism

Macroscopic quantum phenomena supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Macroscopic quantum phenomena specifies the object from which quantum probabilities are calculated. A Hamiltonian or an observable does not make a prediction by itself; it must act on a normalized state vector, density operator, or statistical sector that records the preparation.

The topic concerns quantum state transport under environmental coupling, coherence loss, biological or macroscopic boundary conditions, or effective dynamics outside an ideal closed system.

Physical Construction

The state carrier is a density operator, reduced state, coherence variable, bath-coupled state, or effective mesoscopic carrier. The governing operation is Hamiltonian plus environmental coupling, Lindbladian, memory kernel, stochastic map, or effective transport operator. Positivity, trace preservation, timescale separation, bath assumptions, and control over classical noise determine whether the model is legal. The calculated observables are Coherence, population transfer, relaxation rate, transport efficiency, noise spectrum, or macroscopic response.

Defining Relations

Preparation or domain terms (potential or domain condition, preparation context, or measurement basis) determine which states are admissible. State terms (wave function, quantum state, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Normalization guarantees that the probabilities sum to one, while positivity prevents negative probabilities. Pure vectors and density operators are not competing theories: the density-operator form also represents mixtures and reduced states obtained when unobserved degrees of freedom are traced out.

The Hamiltonian or channel evolves the prepared state. An observable and measurement map then convert that evolved state into outcome probabilities.

Invariance And Realization

The topic concerns quantum state transport under environmental coupling, coherence loss, biological or macroscopic boundary conditions, or effective dynamics outside an ideal closed system. Macroscopic quantum phenomena carries the predictive information before a measurement question is asked. The same physical preparation may be represented as a vector, wave function, density matrix, or reduced state. Normalization and positivity are the admissibility checks that make the state usable for probability assignment.

The local title, representation, and physical realization may change while the constructor role is preserved. The state representation can change between position, momentum, spin, occupation, or density-operator forms. Pure-state and mixed-state descriptions may differ while describing the same formal preparation. Subsystem descriptions change when degrees of freedom are traced out or ignored.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. A usable state gives normalized probabilities for every complete observable attached to the selected Hilbert space. Vector, wave-function, density-matrix, and reduced-state forms can describe the same preparation when connected by the appropriate representation map.

7.17 Quantum fluctuation

Mechanism

Quantum fluctuation supplies the carrier of prediction: the object that is propagated, transformed, restricted, or read out.

Quantum fluctuation specifies the object from which quantum probabilities are calculated. A Hamiltonian or an observable does not make a prediction by itself; it must act on a normalized state vector, density operator, or statistical sector that records the preparation.

The topic concerns quantum state transport under environmental coupling, coherence loss, biological or macroscopic boundary conditions, or effective dynamics outside an ideal closed system.

Physical Construction

The state carrier is a density operator, reduced state, coherence variable, bath-coupled state, or effective mesoscopic carrier. The governing operation is Hamiltonian plus environmental coupling, Lindbladian, memory kernel, stochastic map, or effective transport operator. Positivity, trace preservation, timescale separation, bath assumptions, and control over classical noise determine whether the model is legal. The calculated observables are Coherence, population transfer, relaxation rate, transport efficiency, noise spectrum, or macroscopic response.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (uncertainty relation, commutator, or incompatible observable) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Normalization guarantees that the probabilities sum to one, while positivity prevents negative probabilities. Pure vectors and density operators are not competing theories: the density-operator form also represents mixtures and reduced states obtained when unobserved degrees of freedom are traced out.

The Hamiltonian or channel evolves the prepared state. An observable and measurement map then convert that evolved state into outcome probabilities.

Invariance And Realization

The topic concerns quantum state transport under environmental coupling, coherence loss, biological or macroscopic boundary conditions, or effective dynamics outside an ideal closed system. Quantum fluctuation carries the predictive information before a measurement question is asked. The same physical preparation may be represented as a vector, wave function, density matrix, or reduced state. Normalization and positivity are the admissibility checks that make the state usable for probability assignment.

The local title, representation, and physical realization may change while the constructor role is preserved. The state representation can change between position, momentum, spin, occupation, or density-operator forms. Pure-state and mixed-state descriptions may differ while describing the same formal preparation. Subsystem descriptions change when degrees of freedom are traced out or ignored.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. A usable state gives normalized probabilities for every complete observable attached to the selected Hilbert space. Vector, wave-function, density-matrix, and reduced-state forms can describe the same preparation when connected by the appropriate representation map.

Chapter 8

Dynamics and transformations

Hamiltonians, unitary maps, equations of motion, and path weights describe the lawful transport of the state before a question is resolved.

Hamiltonians, channels, and path amplitudes determine how a prepared state changes. Equivalent formulations belong together when they preserve amplitudes or expectation values on a common domain.

8.1 Topics

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Unitary operator

Perturbation theory

Quantum dynamics

Path integral formulation (*see Path Integral*)

Hamiltonian mechanics

Path integral

Hamiltonian (quantum mechanics) (*see Hamiltonian Mechanics*)

Perturbation theory (quantum mechanics) (*see Perturbation Theory*)

Quantum chaos

Schrödinger equation

Symmetry in quantum mechanics

Heisenberg picture

Quantum stochastic calculus

Relativistic quantum mechanics

Schrödinger picture

Creation and annihilation operators

8.2 Unitary operator

Mechanism

Unitary operator is the reversible-map constructor: it changes the state while preserving inner products and probabilities.

Unitary operators express reversible quantum change and changes of representation with one mathematical condition: preservation of the inner product. That condition preserves normalization, orthogonality, and transition probabilities.

Unitary maps are the admissible reversible transformations of closed-system quantum theory. They preserve normalization and carry projective geometry of the state space through time or controlled operations.

Physical Construction

The state carrier is a state vector, density operator, wavefunction, field state, or register on a specified domain. The governing operation is Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal. The calculated observables are Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.

Topic Equations

Standard constructor skeleton: reversible state transformation and inner-product preservation.

$$\begin{aligned}U^\dagger U &= U U^\dagger = I \\ |\psi'\rangle &= U|\psi\rangle \\ \langle\psi'|\phi'\rangle &= \langle\psi|\phi\rangle\end{aligned}$$

Physical Meaning

Applying a unitary map to a state rotates it within Hilbert space without losing information. Conjugating an observable by the same map gives the equivalent description in the transformed representation. Continuous unitary evolution is generated by a self-adjoint operator.

A spin rotation changes the amplitudes assigned to vertical and horizontal outcomes, but a second inverse rotation restores the original state exactly. Decoherence cannot be represented by such a unitary map on the subsystem alone.

The Hamiltonian page identifies the generator that produces unitary time evolution.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

8.3 Perturbation theory

Mechanism

Perturbation theory belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Perturbation theory separates quantum kinematics from dynamics. The state space lists what can exist, whereas a Hamiltonian, action, Liouvillian, or channel generator specifies which changes are allowed and on what timescale.

The Hamiltonian, action, Liouvillian, or channel determines how an admissible state changes.

Physical Construction

The state carrier is a state vector, density operator, wave function, field state, or register on a specified domain. The governing operation is a Hamiltonian, action, Liouvillian, channel generator, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary conditions determine whether the evolution is legal. The calculated observables are Time-dependent probabilities, transition amplitudes, response functions, conserved quantities, or spectra implied by the dynamics.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The evolution law transports a state without redefining it. Closed-system evolution is unitary; effective open-system evolution must preserve trace and positivity. Equivalent Hamiltonian, propagator, and path-integral descriptions agree on transition amplitudes.

The evolved state becomes experimentally meaningful through an observable whose spectrum and expectation values expose the consequences of the dynamics.

Invariance And Realization

Perturbation theory specifies lawful change before measurement. The generator determines the propagator or path weight that carries the state between preparation and measurement. Conserved quantities and symmetries are read from the generator and its commutation relations.

The local title, representation, and physical realization may change while the constructor role is preserved. Time dependence can be assigned to states, operators, propagators, or path amplitudes. Perturbative, Hamiltonian, Lagrangian, and path-integral presentations can represent the same evolution. Approximation schemes change the calculational route without changing the target transition amplitude.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Lawful closed-system evolution preserves norm or trace; open-system evolution must preserve positivity and trace under the stated approximation. The short-time and classical limits identify whether the generator has the correct physical regime.

8.4 Quantum dynamics

Mechanism

Quantum dynamics belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Quantum dynamics separates quantum kinematics from dynamics. The state space lists what can exist, whereas a Hamiltonian, action, Liouvillian, or channel generator specifies which changes are allowed and on what timescale.

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement.

Physical Construction

The state carrier is a state vector, density operator, wavefunction, field state, or register on a specified domain. The governing operation is Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal. The calculated observables are Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (state vector, density matrix, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or matrix) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or incompatible observable) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The evolution law transports a state without redefining it. Closed-system evolution is unitary; effective open-system evolution must preserve trace and positivity. Equivalent Hamiltonian, propagator, and path-integral descriptions agree on transition amplitudes.

The evolved state becomes experimentally meaningful through an observable whose spectrum and expectation values expose the consequences of the dynamics.

Invariance And Realization

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement. Quantum dynamics specifies lawful change before measurement. The generator determines the propagator or path weight that carries the state between preparation and measurement. Conserved quantities and symmetries are read from the generator and its commutation relations.

The local title, representation, and physical realization may change while the constructor role is preserved. Time dependence can be assigned to states, operators, propagators, or path amplitudes. Perturbative, Hamiltonian, Lagrangian, and path-integral presentations can

represent the same evolution. Approximation schemes change the calculational route without changing the target transition amplitude.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Lawful closed-system evolution preserves norm or trace; open-system evolution must preserve positivity and trace under the stated approximation. The short-time and classical limits identify whether the generator has the correct physical regime.

8.5 Path integral formulation

Path integral formulation is an alternative name for Path Integral. The physical derivation is given in [Path Integral](#).

8.6 Hamiltonian mechanics

Mechanism

Hamiltonian mechanics belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Hamiltonian mechanics separates quantum kinematics from dynamics. The state space lists what can exist, whereas a Hamiltonian, action, Liouvillian, or channel generator specifies which changes are allowed and on what timescale.

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement.

Physical Construction

The state carrier is a state vector, density operator, wavefunction, field state, or register on a specified domain. The governing operation is Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal. The calculated observables are Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The evolution law transports a state without redefining it. Closed-system evolution is unitary; effective open-system evolution must preserve trace and positivity. Equivalent Hamiltonian, propagator, and path-integral descriptions agree on transition amplitudes.

The evolved state becomes experimentally meaningful through an observable whose spectrum and expectation values expose the consequences of the dynamics.

Invariance And Realization

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement. Hamiltonian mechanics specifies lawful change before measurement. The generator determines the propagator or path weight that carries the state between preparation and measurement. Conserved quantities and symmetries are read from the generator and its commutation relations.

The local title, representation, and physical realization may change while the constructor role is preserved. Time dependence can be assigned to states, operators, propagators, or path amplitudes. Perturbative, Hamiltonian, Lagrangian, and path-integral presentations can

represent the same evolution. Approximation schemes change the calculational route without changing the target transition amplitude.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Lawful closed-system evolution preserves norm or trace; open-system evolution must preserve positivity and trace under the stated approximation. The short-time and classical limits identify whether the generator has the correct physical regime.

8.7 Path integral

Mechanism

Path integral is an alternate generator constructor: transition amplitudes are obtained by summing phase weights over histories.

Path integral separates quantum kinematics from dynamics. The state space lists what can exist, whereas a Hamiltonian, action, Liouvillian, or channel generator specifies which changes are allowed and on what timescale.

The path integral does not replace the operator constructor. It repackages the generator step as a weighted sum over histories between boundary conditions. It is especially useful when action, symmetry, and field degrees of freedom are more natural than state-vector evolution.

Physical Construction

The state carrier is a state vector, density operator, wavefunction, field state, or register on a specified domain. The governing operation is Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal. The calculated observables are Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.

Topic Equations

Standard constructor skeleton: boundary-to-boundary transition amplitude through action weights.

$$K(x_f, t_f; x_i, t_i) = \int_{x_i}^{x_f} \mathcal{D}x(t) \exp\left(\frac{i}{\hbar} S[x]\right)$$
$$\psi(x_f, t_f) = \int K(x_f, t_f; x_i, t_i) \psi(x_i, t_i) dx_i$$

Physical Meaning

The evolution law transports a state without redefining it. Closed-system evolution is unitary; effective open-system evolution must preserve trace and positivity. Equivalent Hamiltonian, propagator, and path-integral descriptions agree on transition amplitudes.

The evolved state becomes experimentally meaningful through an observable whose spectrum and expectation values expose the consequences of the dynamics.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

8.8 Hamiltonian (quantum mechanics)

Hamiltonian (quantum mechanics) is an alternative name for Hamiltonian Mechanics. The physical derivation is given in [Hamiltonian Mechanics](#).

8.9 Perturbation theory (quantum mechanics)

Perturbation theory (quantum mechanics) is an alternative name for Perturbation Theory. The physical derivation is given in [Perturbation Theory](#).

8.10 Quantum chaos

Mechanism

Quantum chaos belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Quantum chaos separates quantum kinematics from dynamics. The state space lists what can exist, whereas a Hamiltonian, action, Liouvillian, or channel generator specifies which changes are allowed and on what timescale.

The Hamiltonian, action, Liouvillian, or channel determines how an admissible state changes.

Physical Construction

The state carrier is a state vector, density operator, wave function, field state, or register on a specified domain. The governing operation is a Hamiltonian, action, Liouvillian, channel generator, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary conditions determine whether the evolution is legal. The calculated observables are Time-dependent probabilities, transition amplitudes, response functions, conserved quantities, or spectra implied by the dynamics.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or matrix) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The evolution law transports a state without redefining it. Closed-system evolution is unitary; effective open-system evolution must preserve trace and positivity. Equivalent Hamiltonian, propagator, and path-integral descriptions agree on transition amplitudes.

The evolved state becomes experimentally meaningful through an observable whose spectrum and expectation values expose the consequences of the dynamics.

Invariance And Realization

Quantum chaos specifies lawful change before measurement. The generator determines the propagator or path weight that carries the state between preparation and measurement. Conserved quantities and symmetries are read from the generator and its commutation relations.

The local title, representation, and physical realization may change while the constructor role is preserved. Time dependence can be assigned to states, operators, propagators, or path amplitudes. Perturbative, Hamiltonian, Lagrangian, and path-integral presentations can represent the same evolution. Approximation schemes change the calculational route without changing the target transition amplitude.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Lawful closed-system evolution preserves norm or trace; open-system evolution must preserve positivity and trace under the stated approximation. The short-time and classical limits identify whether the generator has the correct physical regime.

8.11 Schrödinger equation

Mechanism

Schrödinger equation is the state-transport constructor: the Hamiltonian generates lawful change of the state before measurement.

The Schrodinger equation turns a Hamiltonian into a local rule for the time dependence of a state. It is the point at which the chosen state space, boundary conditions, and interaction model become a calculable prediction.

The Schrödinger equation is not a measurement rule. It is the generator step of the quantum constructor. It evolves the predictive carrier while preserving normalization when the Hamiltonian is self-adjoint.

Physical Construction

The state carrier is a state vector, density operator, wavefunction, field state, or register on a specified domain. The governing operation is Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal. The calculated observables are Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.

Topic Equations

Standard constructor skeleton: Hamiltonian transport and norm preservation.

$$\begin{aligned}i\hbar \partial_t |\psi(t)\rangle &= H |\psi(t)\rangle \\ |\psi(t)\rangle &= U(t) |\psi(0)\rangle, \quad U(t) = e^{-iHt/\hbar} \\ \frac{d}{dt} \langle \psi(t) | \psi(t) \rangle &= 0 \quad (H = H^\dagger)\end{aligned}$$

Physical Meaning

The time derivative of the state is fixed by the Hamiltonian acting on that state. For a self-adjoint Hamiltonian, the equation preserves norm. Stationary solutions separate time dependence from the spatial eigenvalue problem, but a general preparation is their superposition.

A wave packet in a potential well spreads and interferes according to the same equation whose stationary solutions define the well's energy levels.

Observables determine which consequences of the evolved state are compared with experiment.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

8.12 Symmetry in quantum mechanics

Mechanism

Symmetry in quantum mechanics belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Symmetry in quantum mechanics separates quantum kinematics from dynamics. The state space lists what can exist, whereas a Hamiltonian, action, Liouvillian, or channel generator specifies which changes are allowed and on what timescale.

The Hamiltonian, action, Liouvillian, or channel determines how an admissible state changes.

Physical Construction

The state carrier is a state vector, density operator, wave function, field state, or register on a specified domain. The governing operation is a Hamiltonian, action, Liouvillian, channel generator, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary conditions determine whether the evolution is legal. The calculated observables are Time-dependent probabilities, transition amplitudes, response functions, conserved quantities, or spectra implied by the dynamics.

Defining Relations

Preparation or domain terms (context, preparation context, or measurement basis) determine which states are admissible. State terms (wavefunction, quantum state, or wave function) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, generator, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (non-commuting observables, commutator, or uncertainty relation) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The evolution law transports a state without redefining it. Closed-system evolution is unitary; effective open-system evolution must preserve trace and positivity. Equivalent Hamiltonian, propagator, and path-integral descriptions agree on transition amplitudes.

The evolved state becomes experimentally meaningful through an observable whose spectrum and expectation values expose the consequences of the dynamics.

Invariance And Realization

Symmetry in quantum mechanics specifies lawful change before measurement. The generator determines the propagator or path weight that carries the state between preparation and measurement. Conserved quantities and symmetries are read from the generator and its commutation relations.

The local title, representation, and physical realization may change while the constructor role is preserved. Time dependence can be assigned to states, operators, propagators, or path amplitudes. Perturbative, Hamiltonian, Lagrangian, and path-integral presentations can represent the same evolution. Approximation schemes change the calculational route without changing the target transition amplitude.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Lawful closed-system evolution preserves norm or trace; open-system evolution must preserve positivity and trace under the stated approximation. The short-time and classical limits identify whether the generator has the correct physical regime.

8.13 Heisenberg picture

Mechanism

Heisenberg picture belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Heisenberg picture separates quantum kinematics from dynamics. The state space lists what can exist, whereas a Hamiltonian, action, Liouvillian, or channel generator specifies which changes are allowed and on what timescale.

The Hamiltonian, action, Liouvillian, or channel determines how an admissible state changes.

Physical Construction

The state carrier is a state vector, density operator, wave function, field state, or register on a specified domain. The governing operation is a Hamiltonian, action, Liouvillian, channel generator, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary conditions determine whether the evolution is legal. The calculated observables are Time-dependent probabilities, transition amplitudes, response functions, conserved quantities, or spectra implied by the dynamics.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or commutator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or incompatible observable) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The evolution law transports a state without redefining it. Closed-system evolution is unitary; effective open-system evolution must preserve trace and positivity. Equivalent Hamiltonian, propagator, and path-integral descriptions agree on transition amplitudes.

The evolved state becomes experimentally meaningful through an observable whose spectrum and expectation values expose the consequences of the dynamics.

Invariance And Realization

Heisenberg picture specifies lawful change before measurement. The generator determines the propagator or path weight that carries the state between preparation and measurement. Conserved quantities and symmetries are read from the generator and its commutation relations.

The local title, representation, and physical realization may change while the constructor role is preserved. Time dependence can be assigned to states, operators, propagators, or path amplitudes. Perturbative, Hamiltonian, Lagrangian, and path-integral presentations can represent the same evolution. Approximation schemes change the calculational route without changing the target transition amplitude.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Lawful closed-system evolution preserves norm or trace; open-system evolution must preserve positivity and trace under the stated approximation. The short-time and classical limits identify whether the generator has the correct physical regime.

8.14 Quantum stochastic calculus

Mechanism

Quantum stochastic calculus belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Quantum stochastic calculus separates quantum kinematics from dynamics. The state space lists what can exist, whereas a Hamiltonian, action, Liouvillian, or channel generator specifies which changes are allowed and on what timescale.

The Hamiltonian, action, Liouvillian, or channel determines how an admissible state changes.

Physical Construction

The state carrier is a state vector, density operator, wave function, field state, or register on a specified domain. The governing operation is a Hamiltonian, action, Liouvillian, channel generator, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary conditions determine whether the evolution is legal. The calculated observables are Time-dependent probabilities, transition amplitudes, response functions, conserved quantities, or spectra implied by the dynamics.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, commutator, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (mode, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or incompatible observable) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The evolution law transports a state without redefining it. Closed-system evolution is unitary; effective open-system evolution must preserve trace and positivity. Equivalent Hamiltonian, propagator, and path-integral descriptions agree on transition amplitudes.

The evolved state becomes experimentally meaningful through an observable whose spectrum and expectation values expose the consequences of the dynamics.

Invariance And Realization

Quantum stochastic calculus specifies lawful change before measurement. The generator determines the propagator or path weight that carries the state between preparation and measurement. Conserved quantities and symmetries are read from the generator and its commutation relations.

The local title, representation, and physical realization may change while the constructor role is preserved. Time dependence can be assigned to states, operators, propagators, or path amplitudes. Perturbative, Hamiltonian, Lagrangian, and path-integral presentations can represent the same evolution. Approximation schemes change the calculational route without changing the target transition amplitude.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Lawful closed-system evolution preserves norm or trace; open-system evolution must preserve positivity and trace under the stated approximation. The short-time and classical limits identify whether the generator has the correct physical regime.

8.15 Relativistic quantum mechanics

Mechanism

Relativistic quantum mechanics belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Relativistic quantum mechanics separates quantum kinematics from dynamics. The state space lists what can exist, whereas a Hamiltonian, action, Liouvillian, or channel generator specifies which changes are allowed and on what timescale.

The Hamiltonian, action, Liouvillian, or channel determines how an admissible state changes.

Physical Construction

The state carrier is a state vector, density operator, wave function, field state, or register on a specified domain. The governing operation is a Hamiltonian, action, Liouvillian, channel generator, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary conditions determine whether the evolution is legal. The calculated observables are Time-dependent probabilities, transition amplitudes, response functions, conserved quantities, or spectra implied by the dynamics.

Defining Relations

Preparation or domain terms (potential or domain condition, context, or preparation context) determine which states are admissible. State terms (wavefunction, wave function, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, matrix, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The evolution law transports a state without redefining it. Closed-system evolution is unitary; effective open-system evolution must preserve trace and positivity. Equivalent Hamiltonian, propagator, and path-integral descriptions agree on transition amplitudes.

The evolved state becomes experimentally meaningful through an observable whose spectrum and expectation values expose the consequences of the dynamics.

Invariance And Realization

Relativistic quantum mechanics specifies lawful change before measurement. The generator determines the propagator or path weight that carries the state between preparation and measurement. Conserved quantities and symmetries are read from the generator and its commutation relations.

The local title, representation, and physical realization may change while the constructor role is preserved. Time dependence can be assigned to states, operators, propagators, or path amplitudes. Perturbative, Hamiltonian, Lagrangian, and path-integral presentations can represent the same evolution. Approximation schemes change the calculational route without changing the target transition amplitude.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Lawful closed-system evolution preserves norm or trace; open-system evolution must preserve positivity and trace under the stated approximation. The short-time and classical limits identify whether the generator has the correct physical regime.

8.16 Schrödinger picture

Mechanism

Schrödinger picture belongs to the lawful-change step: it specifies how the state changes before a question is asked.

Schrödinger picture separates quantum kinematics from dynamics. The state space lists what can exist, whereas a Hamiltonian, action, Liouvillian, or channel generator specifies which changes are allowed and on what timescale.

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement.

Physical Construction

The state carrier is a state vector, density operator, wavefunction, field state, or register on a specified domain. The governing operation is Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal. The calculated observables are Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.

Defining Relations

Preparation or domain terms (potential or domain condition, preparation context, or measurement basis) determine which states are admissible. State terms (state vector, wavefunction, or wave function) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, unitary operator, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The evolution law transports a state without redefining it. Closed-system evolution is unitary; effective open-system evolution must preserve trace and positivity. Equivalent Hamiltonian, propagator, and path-integral descriptions agree on transition amplitudes.

The evolved state becomes experimentally meaningful through an observable whose spectrum and expectation values expose the consequences of the dynamics.

Invariance And Realization

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement. Schrödinger picture specifies lawful change before measurement. The generator determines the propagator or path weight that carries the state between preparation and measurement. Conserved quantities and symmetries are read from the generator and its commutation relations.

The local title, representation, and physical realization may change while the constructor role is preserved. Time dependence can be assigned to states, operators, propagators, or path amplitudes. Perturbative, Hamiltonian, Lagrangian, and path-integral presentations can

represent the same evolution. Approximation schemes change the calculational route without changing the target transition amplitude.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Lawful closed-system evolution preserves norm or trace; open-system evolution must preserve positivity and trace under the stated approximation. The short-time and classical limits identify whether the generator has the correct physical regime.

8.17 Creation and annihilation operators

Mechanism

Creation and annihilation operators are sector-changing operators: they add or remove one quantum from a mode and make many-body or field descriptions executable.

Creation and annihilation operators separates quantum kinematics from dynamics. The state space lists what can exist, whereas a Hamiltonian, action, Liouvillian, or channel generator specifies which changes are allowed and on what timescale.

The page is about the algebraic move that changes occupation number. Creation raises the population of a mode, annihilation lowers it, and the commutation or anticommutation rule determines the statistics. The number operator gives the spectral prediction.

Physical Construction

The state carrier is a state vector, density operator, wave function, field state, or register on a specified domain. The governing operation is a Hamiltonian, action, Liouvillian, channel generator, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary conditions determine whether the evolution is legal. The calculated observables are Time-dependent probabilities, transition amplitudes, response functions, conserved quantities, or spectra implied by the dynamics.

Topic Equations

Topic-specific constructor: the equations express raising, lowering, and occupation-number observable.

$$\begin{aligned}a_i^\dagger|\dots, n_i, \dots\rangle &= \sqrt{n_i + 1}|\dots, n_i + 1, \dots\rangle \\a_i|\dots, n_i, \dots\rangle &= \sqrt{n_i}|\dots, n_i - 1, \dots\rangle \\N_i &= a_i^\dagger a_i\end{aligned}$$

Physical Meaning

The evolution law transports a state without redefining it. Closed-system evolution is unitary; effective open-system evolution must preserve trace and positivity. Equivalent Hamiltonian, propagator, and path-integral descriptions agree on transition amplitudes.

The evolved state becomes experimentally meaningful through an observable whose spectrum and expectation values expose the consequences of the dynamics.

Invariance And Realization

The relation between prepared states, observables, and spectral probability measures. The use of eigenvalues, projectors, modes, or outcome channels to represent admissible observations. The dependence of the observable on basis, domain, potential, preparation, or measurement context. The commutator structure that limits which observables can be jointly diagonalized.

The physical carrier: particle, wave, field mode, spin, qubit, detector, or excitation. The representation: wave mechanics, matrix mechanics, density matrices, path integrals, circuits, or fields. Where time dependence is placed: on the state, on the operator, in a propagator, or in a path weight. The implementation of preparation, boundary condition, detector, or outcome channel.

Discriminating Consequences

A transfer target provides a state space, a transformation law, and a spectral or categorical observable, with one compatibility relation experimentally unresolved. A useful validation varies the basis, domain, or measurement context and measures whether the allowed observable changes while the underlying transformation law remains identifiable. A stronger validation contains two candidate observables whose predicted commutator controls joint resolvability.

Part III

Observables, Compatibility, And Probability

Chapter 9

Observables and spectra

An observable is a permitted question whose operator form determines the possible numerical answers.

An observable determines possible outcomes through its spectral measure. The state then fixes their probabilities.

9.1 Topics

Page
Spectral theory
Pauli matrices
Angular momentum operator
Observable
Self-adjoint operator
Operator theory
Operator (physics) (<i>see Operator Theory</i>)
Eigenvalues and eigenvectors
Heisenberg group
Von Neumann algebra
Quantum logic

9.2 Spectral theory

Mechanism

Spectral theory belongs to the question step: it turns a physical question into an operator with admissible answers.

Spectral theory states which physical question is being asked. The same state supports many incompatible questions, so a prediction requires an operator, spectral measure, or effect family in addition to the state itself.

An observable represents a physical quantity; its spectrum gives the possible sharp values and the state fixes their probabilities.

Physical Construction

The state carrier is an admissible quantum state space on which the physical quantity is represented. The governing operation is a self-adjoint operator, operator-valued measure, or algebra element representing the physical question. Domain, self-adjointness, gauge invariance, and spectral conditions determine whether the quantity is a physical observable. The calculated observables are Eigenvalues, spectral measures, expectation values, moments, and response functions associated with the observable.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, spectrum, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The operator's spectrum lists possible sharp values, while the state determines their weights. Matrix entries depend on basis, but the spectrum, expectation values, and probability distribution are unchanged by an equivalent representation. Domain and self-adjointness conditions are part of the physical definition.

An observable defines possible outcomes. The measurement chapter adds the probability rule and, when needed, the physical interaction that records one of those outcomes.

Invariance And Realization

Spectral theory defines the legal question being asked of the state. The measurable answers are encoded by the operator spectrum, projectors, or spectral measure. The operator role is preserved across equivalent bases even when matrix entries change.

The local title, representation, and physical realization may change while the constructor role is preserved. The same observable may be represented by matrices, differential operators, projectors, or algebraic elements. Degeneracy, basis choice, and domain conditions can change how the spectrum is displayed. Detector implementation changes the physical realization, not the operator role itself.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Self-adjointness, or the appropriate POVM positivity condition, is what makes the question a legal observable. A complete spectral resolution supplies all outcome channels for the question being asked.

9.3 Pauli matrices

Mechanism

Pauli matrices belongs to the question step: it turns a physical question into an operator with admissible answers.

Pauli matrices states which physical question is being asked. The same state supports many incompatible questions, so a prediction requires an operator, spectral measure, or effect family in addition to the state itself.

An observable represents a physical quantity; its spectrum gives the possible sharp values and the state fixes their probabilities.

Physical Construction

The state carrier is an admissible quantum state space on which the physical quantity is represented. The governing operation is a self-adjoint operator, operator-valued measure, or algebra element representing the physical question. Domain, self-adjointness, gauge invariance, and spectral conditions determine whether the quantity is a physical observable. The calculated observables are Eigenvalues, spectral measures, expectation values, moments, and response functions associated with the observable.

Defining Relations

Preparation or domain terms (basis, context, or preparation context) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, observable operator, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The operator's spectrum lists possible sharp values, while the state determines their weights. Matrix entries depend on basis, but the spectrum, expectation values, and probability distribution are unchanged by an equivalent representation. Domain and self-adjointness conditions are part of the physical definition.

An observable defines possible outcomes. The measurement chapter adds the probability rule and, when needed, the physical interaction that records one of those outcomes.

Invariance And Realization

Pauli matrices defines the legal question being asked of the state. The measurable answers are encoded by the operator spectrum, projectors, or spectral measure. The operator role is preserved across equivalent bases even when matrix entries change.

The local title, representation, and physical realization may change while the constructor role is preserved. The same observable may be represented by matrices, differential operators, projectors, or algebraic elements. Degeneracy, basis choice, and domain conditions can change how the spectrum is displayed. Detector implementation changes the physical realization, not the operator role itself.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Self-adjointness, or the appropriate POVM positivity condition, is what makes the question a legal observable. A complete spectral resolution supplies all outcome channels for the question being asked.

9.4 Angular momentum operator

Mechanism

Angular momentum operator belongs to the question step: it turns a physical question into an operator with admissible answers.

Angular momentum operator states which physical question is being asked. The same state supports many incompatible questions, so a prediction requires an operator, spectral measure, or effect family in addition to the state itself.

An observable represents a physical quantity; its spectrum gives the possible sharp values and the state fixes their probabilities.

Physical Construction

The state carrier is an admissible quantum state space on which the physical quantity is represented. The governing operation is a self-adjoint operator, operator-valued measure, or algebra element representing the physical question. Domain, self-adjointness, gauge invariance, and spectral conditions determine whether the quantity is a physical observable. The calculated observables are Eigenvalues, spectral measures, expectation values, moments, and response functions associated with the observable.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (observable operator, commutator, or Hamiltonian) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenstate, eigenvalue, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or incompatible observable) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The operator's spectrum lists possible sharp values, while the state determines their weights. Matrix entries depend on basis, but the spectrum, expectation values, and probability distribution are unchanged by an equivalent representation. Domain and self-adjointness conditions are part of the physical definition.

An observable defines possible outcomes. The measurement chapter adds the probability rule and, when needed, the physical interaction that records one of those outcomes.

Invariance And Realization

Angular momentum operator defines the legal question being asked of the state. The measurable answers are encoded by the operator spectrum, projectors, or spectral measure. The operator role is preserved across equivalent bases even when matrix entries change.

The local title, representation, and physical realization may change while the constructor role is preserved. The same observable may be represented by matrices, differential operators, projectors, or algebraic elements. Degeneracy, basis choice, and domain conditions can change how the spectrum is displayed. Detector implementation changes the physical realization, not the operator role itself.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Self-adjointness, or the appropriate POVM positivity condition, is what makes the question a legal observable. A complete spectral resolution supplies all outcome channels for the question being asked.

9.5 Observable

Mechanism

Observable is the legal-question constructor: it turns a physical question into an operator with spectral outcome channels.

An observable converts a broad physical question such as position, energy, or spin into an operator with a defined domain and spectrum. Without that operator, the state alone does not specify which distribution is being predicted.

An observable is the mathematical form of a question that can be asked of a state. Its spectral decomposition defines the possible answers.

Physical Construction

The state carrier is an admissible quantum state space on which the physical quantity is represented. The governing operation is a self-adjoint operator, operator-valued measure, or algebra element representing the physical question. Domain, self-adjointness, gauge invariance, and spectral conditions determine whether the quantity is a physical observable. The calculated observables are Eigenvalues, spectral measures, expectation values, moments, and response functions associated with the observable.

Topic Equations

Standard constructor skeleton: self-adjoint question, spectral projectors, and Born probabilities.

$$\begin{aligned} A &= A^\dagger \\ A &= \sum_i a_i P_i \\ p(a_i) &= \text{Tr}(\rho P_i) \end{aligned}$$

Physical Meaning

The spectral measure decomposes the observable into possible outcome sectors. Pairing those sectors with the state gives probabilities, and weighting them by their spectral values gives expectation values. Degenerate eigenspaces correspond to outcomes that do not resolve every state component.

A spin state has different probability distributions for measurements along different axes. The preparation is unchanged; the observable selects the question.

The Born rule supplies the probability assigned to each spectral sector, and measurement theory describes its physical registration.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

9.6 Self-adjoint operator

Mechanism

Self-adjoint operator is the admissible-observable condition: it gives real spectra and well-defined spectral measures.

Self-adjoint operator states which physical question is being asked. The same state supports many incompatible questions, so a prediction requires an operator, spectral measure, or effect family in addition to the state itself.

Self-adjointness is not a technical decoration. It is the condition that makes an operator a legitimate spectral question in ordinary quantum mechanics.

Physical Construction

The state carrier is an admissible quantum state space on which the physical quantity is represented. The governing operation is a self-adjoint operator, operator-valued measure, or algebra element representing the physical question. Domain, self-adjointness, gauge invariance, and spectral conditions determine whether the quantity is a physical observable. The calculated observables are Eigenvalues, spectral measures, expectation values, moments, and response functions associated with the observable.

Topic Equations

Standard constructor skeleton: spectral theorem form of a legitimate observable.

$$\begin{aligned} A &= A^\dagger \\ A &= \int_{\sigma(A)} \lambda dE_A(\lambda) \\ \Pr(\Delta) &= \text{Tr}(\rho E_A(\Delta)) \end{aligned}$$

Physical Meaning

The operator's spectrum lists possible sharp values, while the state determines their weights. Matrix entries depend on basis, but the spectrum, expectation values, and probability distribution are unchanged by an equivalent representation. Domain and self-adjointness conditions are part of the physical definition.

An observable defines possible outcomes. The measurement chapter adds the probability rule and, when needed, the physical interaction that records one of those outcomes.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

9.7 Operator theory

Mechanism

Operator theory belongs to the question step: it turns a physical question into an operator with admissible answers.

Operator theory states which physical question is being asked. The same state supports many incompatible questions, so a prediction requires an operator, spectral measure, or effect family in addition to the state itself.

An observable represents a physical quantity; its spectrum gives the possible sharp values and the state fixes their probabilities.

Physical Construction

The state carrier is an admissible quantum state space on which the physical quantity is represented. The governing operation is a self-adjoint operator, operator-valued measure, or algebra element representing the physical question. Domain, self-adjointness, gauge invariance, and spectral conditions determine whether the quantity is a physical observable. The calculated observables are Eigenvalues, spectral measures, expectation values, moments, and response functions associated with the observable.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, unitary operator, or Hamiltonian) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (Spectrum, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The operator's spectrum lists possible sharp values, while the state determines their weights. Matrix entries depend on basis, but the spectrum, expectation values, and probability distribution are unchanged by an equivalent representation. Domain and self-adjointness conditions are part of the physical definition.

An observable defines possible outcomes. The measurement chapter adds the probability rule and, when needed, the physical interaction that records one of those outcomes.

Invariance And Realization

Operator theory defines the legal question being asked of the state. The measurable answers are encoded by the operator spectrum, projectors, or spectral measure. The operator role is preserved across equivalent bases even when matrix entries change.

The local title, representation, and physical realization may change while the constructor role is preserved. The same observable may be represented by matrices, differential operators, projectors, or algebraic elements. Degeneracy, basis choice, and domain conditions can change how the spectrum is displayed. Detector implementation changes the physical realization, not the operator role itself.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Self-adjointness, or the appropriate POVM positivity condition, is what makes the question a legal observable. A complete spectral resolution supplies all outcome channels for the question being asked.

9.8 Operator (physics)

Operator (physics) is an alternative name for Operator Theory. The physical derivation is given in [Operator Theory](#).

9.9 Eigenvalues and eigenvectors

Mechanism

Eigenvalues and eigenvectors belongs to the question step: it turns a physical question into an operator with admissible answers.

Eigenvalues and eigenvectors states which physical question is being asked. The same state supports many incompatible questions, so a prediction requires an operator, spectral measure, or effect family in addition to the state itself.

An observable represents a physical quantity; its spectrum gives the possible sharp values and the state fixes their probabilities.

Physical Construction

The state carrier is an admissible quantum state space on which the physical quantity is represented. The governing operation is a self-adjoint operator, operator-valued measure, or algebra element representing the physical question. Domain, self-adjointness, gauge invariance, and spectral conditions determine whether the quantity is a physical observable. The calculated observables are Eigenvalues, spectral measures, expectation values, moments, and response functions associated with the observable.

Defining Relations

Preparation or domain terms (basis, domain, or context) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, eigenstate, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The operator's spectrum lists possible sharp values, while the state determines their weights. Matrix entries depend on basis, but the spectrum, expectation values, and probability distribution are unchanged by an equivalent representation. Domain and self-adjointness conditions are part of the physical definition.

An observable defines possible outcomes. The measurement chapter adds the probability rule and, when needed, the physical interaction that records one of those outcomes.

Invariance And Realization

Eigenvalues and eigenvectors defines the legal question being asked of the state. The measurable answers are encoded by the operator spectrum, projectors, or spectral measure. The operator role is preserved across equivalent bases even when matrix entries change.

The local title, representation, and physical realization may change while the constructor role is preserved. The same observable may be represented by matrices, differential operators, projectors, or algebraic elements. Degeneracy, basis choice, and domain conditions can change how the spectrum is displayed. Detector implementation changes the physical realization, not the operator role itself.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Self-adjointness, or the appropriate POVM positivity condition, is what makes the question a legal observable. A complete spectral resolution supplies all outcome channels for the question being asked.

9.10 Heisenberg group

Mechanism

Heisenberg group belongs to the question step: it turns a physical question into an operator with admissible answers.

Heisenberg group states which physical question is being asked. The same state supports many incompatible questions, so a prediction requires an operator, spectral measure, or effect family in addition to the state itself.

An observable represents a physical quantity; its spectrum gives the possible sharp values and the state fixes their probabilities.

Physical Construction

The state carrier is an admissible quantum state space on which the physical quantity is represented. The governing operation is a self-adjoint operator, operator-valued measure, or algebra element representing the physical question. Domain, self-adjointness, gauge invariance, and spectral conditions determine whether the quantity is a physical observable. The calculated observables are Eigenvalues, spectral measures, expectation values, moments, and response functions associated with the observable.

Defining Relations

Preparation or domain terms (context, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, unitary operator, or Hamiltonian) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The operator's spectrum lists possible sharp values, while the state determines their weights. Matrix entries depend on basis, but the spectrum, expectation values, and probability distribution are unchanged by an equivalent representation. Domain and self-adjointness conditions are part of the physical definition.

An observable defines possible outcomes. The measurement chapter adds the probability rule and, when needed, the physical interaction that records one of those outcomes.

Invariance And Realization

Heisenberg group defines the legal question being asked of the state. The measurable answers are encoded by the operator spectrum, projectors, or spectral measure. The operator role is preserved across equivalent bases even when matrix entries change.

The local title, representation, and physical realization may change while the constructor role is preserved. The same observable may be represented by matrices, differential operators, projectors, or algebraic elements. Degeneracy, basis choice, and domain conditions can change how the spectrum is displayed. Detector implementation changes the physical realization, not the operator role itself.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Self-adjointness, or the appropriate POVM positivity condition, is what makes the question a legal observable. A complete spectral resolution supplies all outcome channels for the question being asked.

9.11 Von Neumann algebra

Mechanism

Von Neumann algebra belongs to the question step: it turns a physical question into an operator with admissible answers.

Von Neumann algebra states which physical question is being asked. The same state supports many incompatible questions, so a prediction requires an operator, spectral measure, or effect family in addition to the state itself.

An observable represents a physical quantity; its spectrum gives the possible sharp values and the state fixes their probabilities.

Physical Construction

The state carrier is an admissible quantum state space on which the physical quantity is represented. The governing operation is a self-adjoint operator, operator-valued measure, or algebra element representing the physical question. Domain, self-adjointness, gauge invariance, and spectral conditions determine whether the quantity is a physical observable. The calculated observables are Eigenvalues, spectral measures, expectation values, moments, and response functions associated with the observable.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (unitary operator, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The operator's spectrum lists possible sharp values, while the state determines their weights. Matrix entries depend on basis, but the spectrum, expectation values, and probability distribution are unchanged by an equivalent representation. Domain and self-adjointness conditions are part of the physical definition.

An observable defines possible outcomes. The measurement chapter adds the probability rule and, when needed, the physical interaction that records one of those outcomes.

Invariance And Realization

Von Neumann algebra defines the legal question being asked of the state. The measurable answers are encoded by the operator spectrum, projectors, or spectral measure. The operator role is preserved across equivalent bases even when matrix entries change.

The local title, representation, and physical realization may change while the constructor role is preserved. The same observable may be represented by matrices, differential operators, projectors, or algebraic elements. Degeneracy, basis choice, and domain conditions can change how the spectrum is displayed. Detector implementation changes the physical realization, not the operator role itself.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Self-adjointness, or the appropriate POVM positivity condition, is what makes the question a legal observable. A complete spectral resolution supplies all outcome channels for the question being asked.

9.12 Quantum logic

Mechanism

Quantum logic belongs to the question step: it turns a physical question into an operator with admissible answers.

Quantum logic states which physical question is being asked. The same state supports many incompatible questions, so a prediction requires an operator, spectral measure, or effect family in addition to the state itself.

An observable represents a physical quantity; its spectrum gives the possible sharp values and the state fixes their probabilities.

Physical Construction

The state carrier is an admissible quantum state space on which the physical quantity is represented. The governing operation is a self-adjoint operator, operator-valued measure, or algebra element representing the physical question. Domain, self-adjointness, gauge invariance, and spectral conditions determine whether the quantity is a physical observable. The calculated observables are Eigenvalues, spectral measures, expectation values, moments, and response functions associated with the observable.

Defining Relations

Preparation or domain terms (potential or domain condition, basis, or preparation context) determine which states are admissible. State terms (wavefunction, superposition, or wave function) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (observable operator, Hamiltonian, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, eigenstate, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (uncertainty relation, commutator, or incompatible observable) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The operator's spectrum lists possible sharp values, while the state determines their weights. Matrix entries depend on basis, but the spectrum, expectation values, and probability distribution are unchanged by an equivalent representation. Domain and self-adjointness conditions are part of the physical definition.

An observable defines possible outcomes. The measurement chapter adds the probability rule and, when needed, the physical interaction that records one of those outcomes.

Invariance And Realization

Quantum logic defines the legal question being asked of the state. The measurable answers are encoded by the operator spectrum, projectors, or spectral measure. The operator role is preserved across equivalent bases even when matrix entries change.

The local title, representation, and physical realization may change while the constructor role is preserved. The same observable may be represented by matrices, differential operators, projectors, or algebraic elements. Degeneracy, basis choice, and domain conditions can change how the spectrum is displayed. Detector implementation changes the physical realization, not the operator role itself.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Self-adjointness, or the appropriate POVM positivity condition, is what makes the question a legal observable. A complete spectral resolution supplies all outcome channels for the question being asked.

Chapter 10

Noncommuting observables

Two observables need not possess a common sharp spectral resolution. Their commutator and uncertainty relations quantify this algebraic incompatibility. Entanglement belongs instead to the structure of composite states, while Bell experiments join that state structure to local measurements.

Noncommuting observables lack a common sharp spectral resolution. Commutators and uncertainty relations quantify this algebraic fact.

10.1 Topics

Page

Commutator

Uncertainty principle

Canonical commutation relation

10.2 Commutator

Mechanism

Commutator measures the physical consequence of exchanging two operations: it governs joint measurability, generated motion, and the leading difference between reversed protocols.

The commutator measures whether the order of two transformations matters. For observables it also tests whether a common eigenbasis, and hence a joint sharp description, can exist.

For two operators on a common domain, $[A,B]=AB-BA$ compares the two possible orders. For observables, a nonzero commutator obstructs a common sharp spectral resolution and enters uncertainty bounds. For a Hamiltonian H , $[H,A]$ gives the dynamical change of A . For short control pulses, $[A,B]$ is the first term that distinguishes the two reversed sequences. One algebraic object therefore connects compatibility, dynamics, and order-sensitive response.

Physical Construction

The state carrier is a common state space on which two transformations, observables, or questions are both defined. The governing operation is the ordered products AB and BA , compared through the obstruction $[A,B]=AB-BA$. A nonzero commutator marks an order-dependence or compatibility limit; a zero commutator permits a common sharp refinement only when the remaining spectral conditions hold. The calculated observables are Compatibility tests, uncertainty bounds, common eigenspaces, or canonical commutation relations.

Topic Equations

Order exchange, Heisenberg evolution, uncertainty, and the leading closed-sequence response are four consequences of the same commutator.

$$\begin{aligned} &= AB - BA \\ \frac{dA}{dt} &= \frac{i}{\hbar}[H, A] + \left(\frac{\partial A}{\partial t}\right) \\ \Delta A \Delta B &\geq \frac{1}{2} |\langle [A, B] \rangle| \\ e^{\epsilon A} e^{\epsilon B} e^{-\epsilon A} e^{-\epsilon B} &= I + \epsilon^2 [A, B] + O(\epsilon^3) \end{aligned}$$

Physical Meaning

A zero commutator is a compatibility condition under the stated domains; a nonzero commutator identifies an algebraic obstruction. The value can also generate symmetry transformations, so the same operation links compatibility, dynamics, and conservation.

Position followed by momentum is not the same operation as momentum followed by position. Their canonical commutator fixes the lower bound in the corresponding uncertainty relation.

The uncertainty-principle page converts this algebraic obstruction into a state-dependent bound on simultaneous dispersion.

Consequences Forced By The Relation

A nonzero commutator prevents a generic common sharp eigenbasis and imposes state-dependent uncertainty bounds. The commutator with the Hamiltonian determines whether an observable is conserved and how it changes in time. Reversing two short operations produces a difference proportional to their commutator at leading nontrivial order.

Domain Of The Construction

For unbounded operators, the common domain belongs to the physical statement; a formal commutator without a domain need not define an observable relation. A nonzero commutator is not by itself a new interaction. It becomes physical through a state and an observable consequence.

Invariance And Realization

The commutator transforms covariantly under a simultaneous unitary change of representation. Joint spectral compatibility is unchanged by a consistent representation change. The leading order-sensitive response survives when different physical controls realize the same operator algebra.

The matrix entries, basis, carrier, and physical implementation of the two operations may change. Domains can change the meaning of formal commutation relations for unbounded operators. The observable consequence depends on the prepared state and on how the operator difference is read out.

Discriminating Consequences

Reversing two calibrated operations isolates the response generated by their commutator. A commuting control pair removes the order-sensitive contribution without changing the individual operations. Agreement requires the same operator domain and measured observable, not merely similar notation.

10.3 Uncertainty principle

Mechanism

Uncertainty principle is a compatibility-limit theorem: non-commuting observables impose lower bounds on joint sharpness.

The uncertainty principle states a quantitative consequence of incompatibility. It is not a claim about poor apparatus: it constrains the spreads of outcome distributions for every state in the relevant operator domains.

Uncertainty is not detector imperfection. It is a structural consequence of state variance and non-commuting observables.

Physical Construction

The state carrier is One state space or a multipartite state space on which several questions can be asked. The governing operation is Two or more observables, contexts, correlation operators, or hidden-variable assignments being compared. Commutators, uncertainty bounds, contextuality constraints, or Bell-type inequalities decide which joint assignments are possible. The calculated observables are Joint spectra, correlations, inequality violations, uncertainty products, or incompatible outcome statistics.

Topic Equations

Standard constructor skeleton: variance bound from commutator structure.

$$\Delta_{\rho}A \Delta_{\rho}B \geq \frac{1}{2} |\text{Tr}(\rho[A, B])|$$
$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

Physical Meaning

The lower bound depends on the expectation value of the commutator and can be strengthened by covariance terms. A small spread in one observable therefore requires a compensating spread in its incompatible partner for the same prepared state.

Narrowing a particle's position wave packet broadens its momentum distribution because the two amplitudes are Fourier transforms, not because the momentum detector disturbs a pre-existing sharp value.

Multipartite correlations generalize incompatibility beyond pairs of observables and lead to entanglement and Bell tests.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

10.4 Canonical commutation relation

Mechanism

Canonical commutation relation belongs to the compatibility step: it marks when two valid questions cannot be jointly sharpened in one basis.

Canonical commutation relation is needed because individually valid observables need not admit a common set of definite values. Quantum theory therefore requires a separate compatibility analysis rather than treating every collection of questions as classical.

Commutators and correlation constraints determine which observables can share a sharp assignment.

Physical Construction

The state carrier is One state space or a multipartite state space on which several observables or contexts are defined. The governing operation is Commutators, correlation operators, joint measurements, or hidden-variable assignments under comparison. Uncertainty relations, contextuality constraints, and Bell-type inequalities restrict possible joint assignments. The calculated observables are Joint spectra, uncertainty products, correlation functions, and inequality violations.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, commutator, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or incompatible observable) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

A nonzero commutator obstructs a common eigenbasis for the corresponding sharp observables. Uncertainty, contextuality, and Bell inequalities express related obstructions under different assumptions. The assumptions must be stated because the mathematical conclusion changes when the measurement context or factorization changes.

Compatibility limits are then carried into concrete realizations. Boundaries, interfaces, and tensor factorizations decide which observables and correlations can actually be prepared and compared.

Invariance And Realization

Canonical commutation relation identifies when otherwise legal quantum questions cannot be made jointly sharp. The stable object is the obstruction: non-commutation, non-factorization, contextuality, or failure of a joint assignment. The page belongs to the compatibility layer because it limits which spectra can be read together.

The local title, representation, and physical realization may change while the constructor role is preserved. The obstruction may be written as a commutator, inequality, correlation bound, uncertainty relation, or contextuality test. Different experiments realize the same compatibility

limit with different observables and detectors. The language of paradox can change while the formal obstruction remains.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. A common eigenbasis or joint probability model exists only when the relevant compatibility conditions are satisfied. Commutators, uncertainty bounds, Bell inequalities, and contextuality tests are different forms of the same joint-observable obstruction.

Chapter 11

Measurement, instruments, and probabilities

Measurement connects a state and an observable to recorded frequencies. Projection, POVMs, Born weights, and collapse language are alternative ways of presenting this state-to-spectrum probability assignment.

A quantum instrument joins an outcome probability to the corresponding state change. Bell experiments enter here because local measurements interrogate correlations in a composite state.

11.1 Topics

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Quantum nonlocality
Delayed-choice quantum eraser
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11.2 POVM

Mechanism

POVM is the generalized-observable constructor: outcome effects need not be orthogonal projectors.

POVM connects the formal state and observable to experimental frequencies. It distinguishes the probability assigned to an outcome from the conditional state change that may follow a recorded event.

POVMs separate the probability assignment from the idealized projection assumption. They are the natural mechanism for noisy, coarse-grained, indirect, or open-system measurements.

Physical Construction

The state carrier is a state vector or density operator together with the measurement context in which outcome channels are defined. The governing operation is a projection-valued measure, POVM, update map, or instrument map connecting state to record. Outcome probabilities must be positive, normalized, and tied to a specified measurement map rather than to informal observer language. The calculated observables are Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.

Topic Equations

Standard constructor skeleton: positive effects and generalized Born rule.

$$\begin{aligned} E_i &\geq 0, & \sum_i E_i &= I \\ p(i) &= \text{Tr}(\rho E_i) \\ E_i &= \sum_{\alpha} K_{i\alpha}^{\dagger} K_{i\alpha} \end{aligned}$$

Physical Meaning

Each positive effect represents an outcome channel and the effects sum to the identity, which enforces normalized probabilities. A projective measurement is a special case. A complete detector model may further specify a quantum instrument, whose maps describe both the outcome probability and the corresponding post-measurement state.

Once the probability rule is explicit, incompatibility can be tested rather than asserted. Commutators, uncertainty relations, and Bell-type constraints identify when several measurement questions cannot share one sharp assignment.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

11.3 Measurement in quantum mechanics

Mechanism

Measurement in quantum mechanics is the complete measurement junction: it combines a state, a measurement model, probabilities, and sometimes an update rule.

Measurement theory must describe both an outcome distribution and the physical operation that produces a record. Keeping these roles separate prevents an interpretive account of state change from being mistaken for the probability law itself.

Measurement is not the root of quantum theory in this book. It is the junction where a prepared state and an observable or POVM are converted into probabilities and recorded outcomes.

Physical Construction

The state carrier is a state vector or density operator together with the measurement context in which outcome channels are defined. The governing operation is a projection-valued measure, POVM, update map, or instrument map connecting state to record. Outcome probabilities must be positive, normalized, and tied to a specified measurement map rather than to informal observer language. The calculated observables are Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.

Topic Equations

Standard constructor skeleton: generalized measurement probability and conditional update.

$$\begin{aligned} p(i) &= \text{Tr}(\rho E_i) \\ \rho \mapsto \rho_i &= \frac{K_i \rho K_i^\dagger}{\text{Tr}(K_i \rho K_i^\dagger)} \\ E_i &= K_i^\dagger K_i \end{aligned}$$

Physical Meaning

A POVM gives outcome probabilities, whereas a quantum instrument gives the corresponding conditional transformations. Projective measurement is an ideal sharp limit. Real detectors are calibrated by showing that their effects are positive, complete, and consistent with observed frequencies.

A photon counter may report click or no click with non-unit efficiency. A two-effect POVM models those probabilities; the associated instrument is needed only when the state after the event matters.

The incompatibility chapter asks which families of such measurements can be jointly realized or assigned simultaneous sharp values.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

11.4 Quantum jump

Mechanism

Quantum jump belongs to the measurement step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Quantum jump connects the formal state and observable to experimental frequencies. It distinguishes the probability assigned to an outcome from the conditional state change that may follow a recorded event.

A measurement couples a prepared state to recorded outcomes and, when conditioning is retained, to the resulting state change.

Physical Construction

The state carrier is a prepared state together with the measurement context and any apparatus degrees of freedom retained in the model. The governing operation is a projection-valued measure, POVM, quantum instrument, or detector interaction. Outcome probabilities are positive and normalized; conditional state changes must define completely positive maps. The calculated observables are Outcome probabilities, detector records, ensemble frequencies, and conditional post-measurement states.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (unitary operator, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (energy level, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Each positive effect represents an outcome channel and the effects sum to the identity, which enforces normalized probabilities. A projective measurement is a special case. A complete detector model may further specify a quantum instrument, whose maps describe both the outcome probability and the corresponding post-measurement state.

Once the probability rule is explicit, incompatibility can be tested rather than asserted. Commutators, uncertainty relations, and Bell-type constraints identify when several measurement questions cannot share one sharp assignment.

Invariance And Realization

Quantum jump connects the state and the spectral question to observed probabilities. The invariant step is the map from state plus measurement operators to a normalized probability distribution. Projection-valued and POVM observables preserve the same role: outcome channels weighted by the state.

The local title, representation, and physical realization may change while the constructor role is preserved. The detector model, basis, and update convention can change. State-vector,

density-matrix, projective, and generalized-measurement forms may present the observable differently. Interpretive language about collapse or information update can vary without changing the probability rule.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Outcome probabilities are non-negative and normalized because the observable acts on a valid state with a complete effect family. Projective measurement is the sharp limit of the same probability rule when effects become orthogonal projectors.

11.5 Wave function collapse

Mechanism

Wave function collapse belongs to the measurement step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Wave function collapse connects the formal state and observable to experimental frequencies. It distinguishes the probability assigned to an outcome from the conditional state change that may follow a recorded event.

The topic modifies how a state is connected to recorded outcomes. The stable machinery is the spectral measure or POVM together with the probability rule.

Physical Construction

The state carrier is a state vector or density operator together with the measurement context in which outcome channels are defined. The governing operation is a projection-valued measure, POVM, update map, or instrument map connecting state to record. Outcome probabilities must be positive, normalized, and tied to a specified measurement map rather than to informal observer language. The calculated observables are Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.

Defining Relations

Preparation or domain terms (detector, measurement apparatus, or basis) determine which states are admissible. State terms (Wave function, state vector, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (observable operator, Hamiltonian, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenstate, eigenvalue, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Each positive effect represents an outcome channel and the effects sum to the identity, which enforces normalized probabilities. A projective measurement is a special case. A complete detector model may further specify a quantum instrument, whose maps describe both the outcome probability and the corresponding post-measurement state.

Once the probability rule is explicit, incompatibility can be tested rather than asserted. Commutators, uncertainty relations, and Bell-type constraints identify when several measurement questions cannot share one sharp assignment.

Invariance And Realization

The topic modifies how a state is connected to recorded outcomes. The stable machinery is the spectral measure or POVM together with the probability rule. Wave function collapse connects the state and the spectral question to observed probabilities. The invariant step is the map from state plus measurement operators to a normalized probability distribution. Projection-valued and POVM observables preserve the same role: outcome channels weighted by the state.

The local title, representation, and physical realization may change while the constructor role is preserved. The detector model, basis, and update convention can change. State-vector,

density-matrix, projective, and generalized-measurement forms may present the observable differently. Interpretive language about collapse or information update can vary without changing the probability rule.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Outcome probabilities are non-negative and normalized because the observable acts on a valid state with a complete effect family. Projective measurement is the sharp limit of the same probability rule when effects become orthogonal projectors.

11.6 Born rule

Mechanism

Born rule is the probability-observable constructor: it maps a state and a spectral channel to an observed probability.

The Born rule is the bridge from complex amplitudes to experimentally testable probabilities. It is an additional postulate: linear state evolution alone does not say how often a detector outcome should occur.

The Born rule is the point where the constructor becomes predictive. It does not name an object; it connects state preparation and a legal question to frequencies over outcome channels.

Physical Construction

The state carrier is a state vector or density operator together with the measurement context in which outcome channels are defined. The governing operation is a projection-valued measure, POVM, update map, or instrument map connecting state to record. Outcome probabilities must be positive, normalized, and tied to a specified measurement map rather than to informal observer language. The calculated observables are Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.

Topic Equations

Standard constructor skeleton: probability assignment for projective and position observables.

$$\begin{aligned} p(i|\rho, \{P_i\}) &= \text{Tr}(\rho P_i) \\ \Pr(X \in \Delta|\psi) &= \int_{\Delta} |\psi(x)|^2 d\mu(x) \\ \sum_i p(i) &= 1 \end{aligned}$$

Physical Meaning

For a sharp measurement, the probability of an outcome is the state weight in the corresponding eigenspace. The density-operator form extends the same rule to mixtures and generalized measurements. Completeness of the outcome operators makes the probabilities sum to one.

A spin prepared equally between two vertical outcomes gives one-half for each vertical detector channel, even though each individual run records only one result.

Measurement theory adds detector coupling and conditional state change without altering this probability assignment.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

11.7 Measurement problem

Mechanism

Measurement problem is the junction between unitary system–apparatus coupling, probability assignment, and conditional state update; these are distinct maps and need not be identified.

Measurement problem connects the formal state and observable to experimental frequencies. It distinguishes the probability assigned to an outcome from the conditional state change that may follow a recorded event.

A measurement model first couples the system to an apparatus or environment. A POVM or instrument then assigns outcome probabilities, and a conditional map specifies the post-record state. The foundational problem concerns the relation between these operations and a definite record, not the absence of a probability formula.

Physical Construction

The state carrier is a joint system–apparatus state, possibly enlarged by environmental degrees of freedom. The governing operation is a premeasurement interaction followed by a measurement instrument whose components label possible records. The instrument maps are completely positive and their sum is trace preserving; the outcome effects sum to the identity. The calculated observables are Outcome probabilities and conditional post-record states must be stated separately.

Topic Equations

Topic-specific constructor: premeasurement coupling, outcome probability, conditional update, and unconditioned evolution are separated.

$$\begin{aligned}\rho'_{SA} &= U_{SA}(\rho_S \otimes \rho_A)U_{SA}^\dagger \\ p(i) &= \text{Tr}[\mathcal{I}_i(\rho_S)] = \text{Tr}(\rho_S E_i) \\ \rho_{S|i} &= \frac{\mathcal{I}_i(\rho_S)}{p(i)}, \quad \rho'_S = \sum_i \mathcal{I}_i(\rho_S)\end{aligned}$$

Physical Meaning

Each positive effect represents an outcome channel and the effects sum to the identity, which enforces normalized probabilities. A projective measurement is a special case. A complete detector model may further specify a quantum instrument, whose maps describe both the outcome probability and the corresponding post-measurement state.

Once the probability rule is explicit, incompatibility can be tested rather than asserted. Commutators, uncertainty relations, and Bell-type constraints identify when several measurement questions cannot share one sharp assignment.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system,

basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

11.8 Quantum eraser experiment

Mechanism

Quantum eraser experiment belongs to the measurement step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Quantum eraser experiment connects the formal state and observable to experimental frequencies. It distinguishes the probability assigned to an outcome from the conditional state change that may follow a recorded event.

The topic tests whether separately legal questions can be resolved together. The mechanism is a restriction on joint spectra, correlations, or admissible hidden-variable assignments.

Physical Construction

The state carrier is One state space or a multipartite state space on which several questions can be asked. The governing operation is Two or more observables, contexts, correlation operators, or hidden-variable assignments being compared. Commutators, uncertainty bounds, contextuality constraints, or Bell-type inequalities decide which joint assignments are possible. The calculated observables are Joint spectra, correlations, inequality violations, uncertainty products, or incompatible outcome statistics.

Defining Relations

Preparation or domain terms (detector, experimental setup, or preparation context) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (mode, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (complementarity, commutator, or uncertainty relation) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Each positive effect represents an outcome channel and the effects sum to the identity, which enforces normalized probabilities. A projective measurement is a special case. A complete detector model may further specify a quantum instrument, whose maps describe both the outcome probability and the corresponding post-measurement state.

Once the probability rule is explicit, incompatibility can be tested rather than asserted. Commutators, uncertainty relations, and Bell-type constraints identify when several measurement questions cannot share one sharp assignment.

Invariance And Realization

The topic tests whether separately legal questions can be resolved together. The mechanism is a restriction on joint spectra, correlations, or admissible hidden-variable assignments. Quantum eraser experiment connects the state and the spectral question to observed probabilities. The invariant step is the map from state plus measurement operators to a normalized probability distribution. Projection-valued and POVM observables preserve the same role: outcome channels weighted by the state.

The local title, representation, and physical realization may change while the constructor role is preserved. The detector model, basis, and update convention can change. State-vector, density-matrix, projective, and generalized-measurement forms may present the observable differently. Interpretive language about collapse or information update can vary without changing the probability rule.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Outcome probabilities are non-negative and normalized because the observable acts on a valid state with a complete effect family. Projective measurement is the sharp limit of the same probability rule when effects become orthogonal projectors.

11.9 Projection-valued measure

Mechanism

Projection-valued measure is the sharp-observable constructor: mutually exclusive outcome projectors partition the identity.

Projection-valued measure connects the formal state and observable to experimental frequencies. It distinguishes the probability assigned to an outcome from the conditional state change that may follow a recorded event.

A projection-valued measure encodes an ideal sharp measurement. It defines outcome channels that are orthogonal and exhaustive.

Physical Construction

The state carrier is a state vector or density operator together with the measurement context in which outcome channels are defined. The governing operation is a projection-valued measure, POVM, update map, or instrument map connecting state to record. Outcome probabilities must be positive, normalized, and tied to a specified measurement map rather than to informal observer language. The calculated observables are Born probabilities, detector records, post-measurement states, ensemble frequencies, or decision probabilities.

Topic Equations

Standard constructor skeleton: sharp outcome channels, probability, and projective update.

$$\begin{aligned} P_i P_j &= \delta_{ij} P_i, & \sum_i P_i &= I \\ p(i) &= \text{Tr}(\rho P_i) \\ \rho &\mapsto \frac{P_i \rho P_i}{\text{Tr}(\rho P_i)} \end{aligned}$$

Physical Meaning

Each positive effect represents an outcome channel and the effects sum to the identity, which enforces normalized probabilities. A projective measurement is a special case. A complete detector model may further specify a quantum instrument, whose maps describe both the outcome probability and the corresponding post-measurement state.

Once the probability rule is explicit, incompatibility can be tested rather than asserted. Commutators, uncertainty relations, and Bell-type constraints identify when several measurement questions cannot share one sharp assignment.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

11.10 Quantum amplifier

Mechanism

Quantum amplifier belongs to the measurement step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Quantum amplifier connects the formal state and observable to experimental frequencies. It distinguishes the probability assigned to an outcome from the conditional state change that may follow a recorded event.

The mechanism is an apparatus-coupled observable: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate.

Physical Construction

The state carrier is a probe state, sample state, field mode, detector state, or estimation register. The governing operation is an interaction Hamiltonian, transfer map, measurement channel, reconstruction map, or estimator. The instrument must separate sample signal from preparation, detector response, calibration, noise, and reconstruction artifacts. The calculated observables are Counts, images, spectra, phase shifts, trajectories, intensity maps, correlation data, or parameter estimates.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, state vector, or superposition) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (unitary operator, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (mode, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (uncertainty relation, commutator, or incompatible observable) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Each positive effect represents an outcome channel and the effects sum to the identity, which enforces normalized probabilities. A projective measurement is a special case. A complete detector model may further specify a quantum instrument, whose maps describe both the outcome probability and the corresponding post-measurement state.

Once the probability rule is explicit, incompatibility can be tested rather than asserted. Commutators, uncertainty relations, and Bell-type constraints identify when several measurement questions cannot share one sharp assignment.

Invariance And Realization

The mechanism is an apparatus-coupled observable: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate. Quantum amplifier connects the state and the spectral question to observed probabilities. The invariant step is the map from state plus measurement operators to a normalized probability distribution.

Projection-valued and POVM observables preserve the same role: outcome channels weighted by the state.

The local title, representation, and physical realization may change while the constructor role is preserved. The detector model, basis, and update convention can change. State-vector, density-matrix, projective, and generalized-measurement forms may present the observable differently. Interpretive language about collapse or information update can vary without changing the probability rule.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Outcome probabilities are non-negative and normalized because the observable acts on a valid state with a complete effect family. Projective measurement is the sharp limit of the same probability rule when effects become orthogonal projectors.

11.11 Bell's theorem

Mechanism

Bell's theorem is a compatibility/locality stress test: quantum correlations violate bounds satisfied by local hidden-variable assignments.

Bell's theorem connects the formal state and observable to experimental frequencies. It distinguishes the probability assigned to an outcome from the conditional state change that may follow a recorded event.

Bell's theorem is not a page about a mysterious object. It is a falsifier for a classical joint-assignment model of measurement outcomes.

Physical Construction

The state carrier is One state space or a multipartite state space on which several questions can be asked. The governing operation is Two or more observables, contexts, correlation operators, or hidden-variable assignments being compared. Commutators, uncertainty bounds, contextuality constraints, or Bell-type inequalities decide which joint assignments are possible. The calculated observables are Joint spectra, correlations, inequality violations, uncertainty products, or incompatible outcome statistics.

Topic Equations

Standard constructor skeleton: CHSH inequality and quantum violation bound.

$$|E(a, b) + E(a, b') + E(a', b) - E(a', b')| \leq 2$$

quantum prediction can reach $2\sqrt{2}$

Physical Meaning

Each positive effect represents an outcome channel and the effects sum to the identity, which enforces normalized probabilities. A projective measurement is a special case. A complete detector model may further specify a quantum instrument, whose maps describe both the outcome probability and the corresponding post-measurement state.

Once the probability rule is explicit, incompatibility can be tested rather than asserted. Commutators, uncertainty relations, and Bell-type constraints identify when several measurement questions cannot share one sharp assignment.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

11.12 Quantum nonlocality

Mechanism

Quantum nonlocality belongs to the measurement step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Quantum nonlocality connects the formal state and observable to experimental frequencies. It distinguishes the probability assigned to an outcome from the conditional state change that may follow a recorded event.

The topic tests whether separately legal questions can be resolved together. The mechanism is a restriction on joint spectra, correlations, or admissible hidden-variable assignments.

Physical Construction

The state carrier is One state space or a multipartite state space on which several questions can be asked. The governing operation is Two or more observables, contexts, correlation operators, or hidden-variable assignments being compared. Commutators, uncertainty bounds, contextuality constraints, or Bell-type inequalities decide which joint assignments are possible. The calculated observables are Joint spectra, correlations, inequality violations, uncertainty products, or incompatible outcome statistics.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (wavefunction, quantum state, or wave function) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Each positive effect represents an outcome channel and the effects sum to the identity, which enforces normalized probabilities. A projective measurement is a special case. A complete detector model may further specify a quantum instrument, whose maps describe both the outcome probability and the corresponding post-measurement state.

Once the probability rule is explicit, incompatibility can be tested rather than asserted. Commutators, uncertainty relations, and Bell-type constraints identify when several measurement questions cannot share one sharp assignment.

Invariance And Realization

The topic tests whether separately legal questions can be resolved together. The mechanism is a restriction on joint spectra, correlations, or admissible hidden-variable assignments. Quantum nonlocality connects the state and the spectral question to observed probabilities. The invariant step is the map from state plus measurement operators to a normalized probability distribution. Projection-valued and POVM observables preserve the same role: outcome channels weighted by the state.

The local title, representation, and physical realization may change while the constructor role is preserved. The detector model, basis, and update convention can change. State-vector, density-matrix, projective, and generalized-measurement forms may present the observable differently. Interpretive language about collapse or information update can vary without changing the probability rule.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Outcome probabilities are non-negative and normalized because the observable acts on a valid state with a complete effect family. Projective measurement is the sharp limit of the same probability rule when effects become orthogonal projectors.

11.13 Delayed-choice quantum eraser

Mechanism

Delayed-choice quantum eraser belongs to the measurement step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Delayed-choice quantum eraser connects the formal state and observable to experimental frequencies. It distinguishes the probability assigned to an outcome from the conditional state change that may follow a recorded event.

The topic tests whether separately legal questions can be resolved together. The mechanism is a restriction on joint spectra, correlations, or admissible hidden-variable assignments.

Physical Construction

The state carrier is One state space or a multipartite state space on which several questions can be asked. The governing operation is Two or more observables, contexts, correlation operators, or hidden-variable assignments being compared. Commutators, uncertainty bounds, contextuality constraints, or Bell-type inequalities decide which joint assignments are possible. The calculated observables are Joint spectra, correlations, inequality violations, uncertainty products, or incompatible outcome statistics.

Defining Relations

Preparation or domain terms (detector, experimental setup, or basis) determine which states are admissible. State terms (superposition, wave function, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (mode, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (complementarity, uncertainty relation, or commutator) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Each positive effect represents an outcome channel and the effects sum to the identity, which enforces normalized probabilities. A projective measurement is a special case. A complete detector model may further specify a quantum instrument, whose maps describe both the outcome probability and the corresponding post-measurement state.

Once the probability rule is explicit, incompatibility can be tested rather than asserted. Commutators, uncertainty relations, and Bell-type constraints identify when several measurement questions cannot share one sharp assignment.

Invariance And Realization

The topic tests whether separately legal questions can be resolved together. The mechanism is a restriction on joint spectra, correlations, or admissible hidden-variable assignments. Delayed-choice quantum eraser connects the state and the spectral question to observed probabilities. The invariant step is the map from state plus measurement operators to a normalized probability distribution. Projection-valued and POVM observables preserve the same role: outcome channels weighted by the state.

The local title, representation, and physical realization may change while the constructor role is preserved. The detector model, basis, and update convention can change. State-vector,

density-matrix, projective, and generalized-measurement forms may present the observable differently. Interpretive language about collapse or information update can vary without changing the probability rule.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Outcome probabilities are non-negative and normalized because the observable acts on a valid state with a complete effect family. Projective measurement is the sharp limit of the same probability rule when effects become orthogonal projectors.

11.14 Quantum imaging

Mechanism

Quantum imaging belongs to the measurement step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Quantum imaging connects the formal state and observable to experimental frequencies. It distinguishes the probability assigned to an outcome from the conditional state change that may follow a recorded event.

The mechanism is an apparatus-coupled observable: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate.

Physical Construction

The state carrier is a probe state, sample state, field mode, detector state, or estimation register. The governing operation is an interaction Hamiltonian, transfer map, measurement channel, reconstruction map, or estimator. The instrument must separate sample signal from preparation, detector response, calibration, noise, and reconstruction artifacts. The calculated observables are Counts, images, spectra, phase shifts, trajectories, intensity maps, correlation data, or parameter estimates.

Defining Relations

Preparation or domain terms (potential or domain condition, detector, or preparation context) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (mode, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Each positive effect represents an outcome channel and the effects sum to the identity, which enforces normalized probabilities. A projective measurement is a special case. A complete detector model may further specify a quantum instrument, whose maps describe both the outcome probability and the corresponding post-measurement state.

Once the probability rule is explicit, incompatibility can be tested rather than asserted. Commutators, uncertainty relations, and Bell-type constraints identify when several measurement questions cannot share one sharp assignment.

Invariance And Realization

The mechanism is an apparatus-coupled observable: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate. Quantum imaging connects the state and the spectral question to observed probabilities. The invariant step is the map from state plus measurement operators to a normalized probability distribution.

Projection-valued and POVM observables preserve the same role: outcome channels weighted by the state.

The local title, representation, and physical realization may change while the constructor role is preserved. The detector model, basis, and update convention can change. State-vector, density-matrix, projective, and generalized-measurement forms may present the observable differently. Interpretive language about collapse or information update can vary without changing the probability rule.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Outcome probabilities are non-negative and normalized because the observable acts on a valid state with a complete effect family. Projective measurement is the sharp limit of the same probability rule when effects become orthogonal projectors.

11.15 Wave–particle duality

Mechanism

Wave–particle duality belongs to the measurement step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Wave–particle duality connects the formal state and observable to experimental frequencies. It distinguishes the probability assigned to an outcome from the conditional state change that may follow a recorded event.

The mechanism is the field/mode version of the constructor: a state space is decomposed into modes or sectors, operators create, annihilate, or constrain those modes, and observables are occupation, charge, spin, momentum, energy, or scattering response.

Physical Construction

The state carrier is Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data. The governing operation is Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal. The calculated observables are Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.

Defining Relations

Preparation or domain terms (potential or domain condition, detector, or preparation context) determine which states are admissible. State terms (probability amplitude, wave function, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (spectrum, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Each positive effect represents an outcome channel and the effects sum to the identity, which enforces normalized probabilities. A projective measurement is a special case. A complete detector model may further specify a quantum instrument, whose maps describe both the outcome probability and the corresponding post-measurement state.

Once the probability rule is explicit, incompatibility can be tested rather than asserted. Commutators, uncertainty relations, and Bell-type constraints identify when several measurement questions cannot share one sharp assignment.

Invariance And Realization

The mechanism is the field/mode version of the constructor: a state space is decomposed into modes or sectors, operators create, annihilate, or constrain those modes, and observables are occupation, charge, spin, momentum, energy, or scattering response. Wave–particle duality connects the state and the spectral question to observed probabilities. The invariant step is the map from state plus measurement operators to a normalized probability distribution.

Projection-valued and POVM observables preserve the same role: outcome channels weighted by the state.

The local title, representation, and physical realization may change while the constructor role is preserved. The detector model, basis, and update convention can change. State-vector, density-matrix, projective, and generalized-measurement forms may present the observable differently. Interpretive language about collapse or information update can vary without changing the probability rule.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Outcome probabilities are non-negative and normalized because the observable acts on a valid state with a complete effect family. Projective measurement is the sharp limit of the same probability rule when effects become orthogonal projectors.

11.16 Einstein–Podolsky–Rosen paradox

Mechanism

Einstein–Podolsky–Rosen paradox belongs to the measurement step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Einstein–Podolsky–Rosen paradox connects the formal state and observable to experimental frequencies. It distinguishes the probability assigned to an outcome from the conditional state change that may follow a recorded event.

A measurement couples a prepared state to recorded outcomes and, when conditioning is retained, to the resulting state change.

Physical Construction

The state carrier is a prepared state together with the measurement context and any apparatus degrees of freedom retained in the model. The governing operation is a projection-valued measure, POVM, quantum instrument, or detector interaction. Outcome probabilities are positive and normalized; conditional state changes must define completely positive maps. The calculated observables are Outcome probabilities, detector records, ensemble frequencies, and conditional post-measurement states.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (wave function, quantum state, or superposition) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (measurement outcome, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (incompatible observables, uncertainty relation, or commutator) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Each positive effect represents an outcome channel and the effects sum to the identity, which enforces normalized probabilities. A projective measurement is a special case. A complete detector model may further specify a quantum instrument, whose maps describe both the outcome probability and the corresponding post-measurement state.

Once the probability rule is explicit, incompatibility can be tested rather than asserted. Commutators, uncertainty relations, and Bell-type constraints identify when several measurement questions cannot share one sharp assignment.

Invariance And Realization

Einstein–Podolsky–Rosen paradox connects the state and the spectral question to observed probabilities. The invariant step is the map from state plus measurement operators to a normalized probability distribution. Projection-valued and POVM observables preserve the same role: outcome channels weighted by the state.

The local title, representation, and physical realization may change while the constructor role is preserved. The detector model, basis, and update convention can change. State-vector,

density-matrix, projective, and generalized-measurement forms may present the observable differently. Interpretive language about collapse or information update can vary without changing the probability rule.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Outcome probabilities are non-negative and normalized because the observable acts on a valid state with a complete effect family. Projective measurement is the sharp limit of the same probability rule when effects become orthogonal projectors.

11.17 Electron microscope

Mechanism

Electron microscope belongs to the measurement step: it connects a prepared state and an operator spectrum to probabilities or state updates.

Electron microscope connects the formal state and observable to experimental frequencies. It distinguishes the probability assigned to an outcome from the conditional state change that may follow a recorded event.

The mechanism is an apparatus-coupled observable: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate.

Physical Construction

The state carrier is a probe state, sample state, field mode, detector state, or estimation register. The governing operation is an interaction Hamiltonian, transfer map, measurement channel, reconstruction map, or estimator. The instrument must separate sample signal from preparation, detector response, calibration, noise, and reconstruction artifacts. The calculated observables are Counts, images, spectra, phase shifts, trajectories, intensity maps, correlation data, or parameter estimates.

Representative Relation

The equation identifies the state, operation, and observable relation associated with this physical role.

$$\rho_{\text{probe}} \mapsto \mathcal{E}_{\text{sample}}(\rho_{\text{probe}}), \quad p(y) = \text{Tr}(M_y \mathcal{E}_{\text{sample}}(\rho_{\text{probe}})), \quad \hat{s} = R(\{y_i\})$$

Physical Meaning

Each positive effect represents an outcome channel and the effects sum to the identity, which enforces normalized probabilities. A projective measurement is a special case. A complete detector model may further specify a quantum instrument, whose maps describe both the outcome probability and the corresponding post-measurement state.

Once the probability rule is explicit, incompatibility can be tested rather than asserted. Commutators, uncertainty relations, and Bell-type constraints identify when several measurement questions cannot share one sharp assignment.

Invariance And Realization

The mechanism is an apparatus-coupled observable: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate. Electron microscope connects the state and the spectral question to observed probabilities. The invariant step is the map from state plus measurement operators to a normalized probability distribution. Projection-valued and POVM observables preserve the same role: outcome channels weighted by the state.

The local title, representation, and physical realization may change while the constructor role is preserved. The detector model, basis, and update convention can change. State-vector, density-matrix, projective, and generalized-measurement forms may present the observable differently. Interpretive language about collapse or information update can vary without changing the probability rule.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Outcome probabilities are non-negative and normalized because the observable acts on a valid state with a complete effect family. Projective measurement is the sharp limit of the same probability rule when effects become orthogonal projectors.

Part IV

Controlled And Spatial Realizations

Chapter 12

Control sequences and quantum channels

Quantum computing, channels, circuits, algorithms, networks, sensors, and error correction turn the same formal machinery into controlled sequences of operations.

An ordered sequence becomes dynamics when its composition defines the implemented channel. Quantum control and information theory exploit this promotion directly.

12.1 Topics

Page

Quantum computing
Quantum information science (*see Quantum Information*)
Quantum network
Quantum algorithm
Quantum error correction
Quantum channel
Quantum logic gate
Quantum neural network
Quantum circuit
Quantum information
Quantum key distribution
Quantum sensor
Quantum machine
Quantum teleportation
Quantum finite automaton
Quantum machine learning
Quantum programming
Quantum simulator
Quantum cryptography
Quantum metrology
Quantum bus
Quantum image processing
Quantum engineering
Quantum cellular automaton

12.2 Quantum computing

Mechanism

Quantum computing belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum computing specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state.

Physical Construction

The state carrier is an input state, register, channel state, error syndrome, key, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective. The calculated observables are Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, superposition, or wave function) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. Quantum computing turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.3 Quantum information science

Quantum information science is an alternative name for Quantum Information. The physical derivation is given in [Quantum Information](#).

12.4 Quantum network

Mechanism

Quantum network belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum network specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state.

Physical Construction

The state carrier is an input state, register, channel state, error syndrome, key, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective. The calculated observables are Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (potential or domain condition, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (mode, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. Quantum network turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.5 Quantum algorithm

Mechanism

Quantum algorithm belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum algorithm specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state.

Physical Construction

The state carrier is an input state, register, channel state, error syndrome, key, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective. The calculated observables are Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (domain, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, superposition, or wave function) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, matrix, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. Quantum algorithm turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.6 Quantum error correction

Mechanism

Quantum error correction belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum error correction protects a subspace against a family of noise operations without learning the encoded amplitudes. It is a mechanism design problem involving encoding, error syndromes, conditional correction, and a final logical observable.

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state.

Physical Construction

The state carrier is an input state, register, channel state, error syndrome, key, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective. The calculated observables are Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (preparation, preparation context, or measurement basis) determine which states are admissible. State terms (superposition, wave function, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, unitary operator, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The error-correction conditions require different errors either to act identically on the code space or to move it into distinguishable syndrome sectors. A recovery map then restores the logical state while preserving superpositions.

A repetition-style code can diagnose one class of flips by comparing parity checks. The syndrome identifies the error location without measuring the unknown logical amplitudes themselves.

Fault-tolerant protocols extend this logic by constraining how errors propagate through an entire sequence of gates and measurements.

Invariance And Realization

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. Quantum error correction turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.7 Quantum channel

Mechanism

Quantum channel is the open-system protocol constructor: it maps input states to output states while preserving complete positivity and trace.

A quantum channel is the general transformation available to a state when an environment, uncontrolled degree of freedom, or measurement outcome is not retained. It extends unitary dynamics without abandoning positivity or probability conservation.

A channel is the mechanism for noisy transformations, measurements with forgotten outcomes, and subsystem evolution.

Physical Construction

The state carrier is Input and output density operators, possibly on different Hilbert spaces or subsystem carriers. The governing operation is a completely positive trace-preserving map, often represented by Kraus operators or by a Stinespring dilation. Complete positivity and trace preservation are the legal conditions; non-trace-preserving maps require an explicitly conditioned outcome. The calculated observables are Output state, final POVM probabilities, fidelity, capacity, error rate, or recovered subsystem statistics.

Topic Equations

Standard constructor skeleton: completely positive trace-preserving map and observable.

$$\begin{aligned}\mathcal{E}(\rho) &= \sum_a K_a \rho K_a^\dagger \\ \sum_a K_a^\dagger K_a &= I \\ p(y) &= \text{Tr}(M_y \mathcal{E}(\rho))\end{aligned}$$

Physical Meaning

Complete positivity guarantees that the map remains physical when the input is entangled with an untouched system, and trace preservation guarantees total probability. A Kraus representation displays one realization, but different Kraus sets can describe the same channel.

Loss of phase coherence can be represented by a dephasing channel. Its action suppresses off-diagonal density-matrix elements while leaving the corresponding populations unchanged.

Quantum error correction asks whether information can be encoded so that a specified family of channels is detectable and reversible on the code space.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

12.8 Quantum logic gate

Mechanism

Quantum logic gate belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum logic gate specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state.

Physical Construction

The state carrier is an input state, register, channel state, error syndrome, key, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective. The calculated observables are Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, Hamiltonian, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. Quantum logic gate turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.9 Quantum neural network

Mechanism

Quantum neural network belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum neural network specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state.

Physical Construction

The state carrier is an input state, register, channel state, error syndrome, key, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective. The calculated observables are Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (superposition, wave function, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (unitary operator, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. Quantum neural network turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.10 Quantum circuit

Mechanism

Quantum circuit is the engineered-composition constructor: a finite sequence of admissible maps prepares, transforms, and measures a register.

Quantum circuit specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

A circuit is the protocol layer of the same state-operator-observable machinery. Gates are controlled unitary or channel maps; measurement converts final states into output probabilities.

Physical Construction

The state carrier is an input state, register, channel state, error syndrome, key, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective. The calculated observables are Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.

Topic Equations

Standard constructor skeleton: composed gates and final measurement.

$$\begin{aligned}\rho_{\text{out}} &= U_m \cdots U_2 U_1 \rho_{\text{in}} U_1^\dagger U_2^\dagger \cdots U_m^\dagger \\ p(y) &= \text{Tr}(M_y \rho_{\text{out}})\end{aligned}$$

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

12.11 Quantum information

Mechanism

Quantum information belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum information specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

Circuits, controls, and sensing sequences are ordered compositions of physical transformations and measurements.

Physical Construction

The state carrier is an input state, register, encoded subspace, channel state, key, syndrome, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, encodings, corrections, or feedback maps. Each step must belong to the claimed map class and the composition must preserve normalization and positivity. The calculated observables are Output state, fidelity, error rate, key rate, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (preparation, basis, or preparation context) determine which states are admissible. State terms (quantum state, density matrix, or wave function) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (observable operator, matrix, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenstate, eigenvalue, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (uncertainty relation, non-commuting observables, or commutator) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

Quantum information turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized

map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.12 Quantum key distribution

Mechanism

Quantum key distribution belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum key distribution specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

Circuits, controls, and sensing sequences are ordered compositions of physical transformations and measurements.

Physical Construction

The state carrier is an input state, register, encoded subspace, channel state, key, syndrome, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, encodings, corrections, or feedback maps. Each step must belong to the claimed map class and the composition must preserve normalization and positivity. The calculated observables are Output state, fidelity, error rate, key rate, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (Mode, eigenstate, or eigenvalue) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

Quantum key distribution turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized

map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.13 Quantum sensor

Mechanism

Quantum sensor belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum sensor specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

The mechanism is an apparatus-coupled observable: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate.

Physical Construction

The state carrier is a probe state, sample state, field mode, detector state, or estimation register. The governing operation is an interaction Hamiltonian, transfer map, measurement channel, reconstruction map, or estimator. The instrument must separate sample signal from preparation, detector response, calibration, noise, and reconstruction artifacts. The calculated observables are Counts, images, spectra, phase shifts, trajectories, intensity maps, correlation data, or parameter estimates.

Defining Relations

Preparation or domain terms (potential or domain condition, preparation context, or measurement basis) determine which states are admissible. State terms (superposition, wave function, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (mode, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (uncertainty relation, commutator, or incompatible observable) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

The mechanism is an apparatus-coupled observable: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate. Quantum sensor turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary

gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.14 Quantum machine

Mechanism

Quantum machine belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum machine specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

Circuits, controls, and sensing sequences are ordered compositions of physical transformations and measurements.

Physical Construction

The state carrier is an input state, register, encoded subspace, channel state, key, syndrome, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, encodings, corrections, or feedback maps. Each step must belong to the claimed map class and the composition must preserve normalization and positivity. The calculated observables are Output state, fidelity, error rate, key rate, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, superposition, or wave function) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (observable operator, Hamiltonian, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

Quantum machine turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized

map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.15 Quantum teleportation

Mechanism

Quantum teleportation belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum teleportation specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

Circuits, controls, and sensing sequences are ordered compositions of physical transformations and measurements.

Physical Construction

The state carrier is an input state, register, encoded subspace, channel state, key, syndrome, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, encodings, corrections, or feedback maps. Each step must belong to the claimed map class and the composition must preserve normalization and positivity. The calculated observables are Output state, fidelity, error rate, key rate, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, superposition, or wave function) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

Quantum teleportation turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized

map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.16 Quantum finite automaton

Mechanism

Quantum finite automaton belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum finite automaton specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

Circuits, controls, and sensing sequences are ordered compositions of physical transformations and measurements.

Physical Construction

The state carrier is an input state, register, encoded subspace, channel state, key, syndrome, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, encodings, corrections, or feedback maps. Each step must belong to the claimed map class and the composition must preserve normalization and positivity. The calculated observables are Output state, fidelity, error rate, key rate, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, state vector, or wave function) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, unitary operator, or Hamiltonian) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

Quantum finite automaton turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized

map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.17 Quantum machine learning

Mechanism

Quantum machine learning belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum machine learning specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

Circuits, controls, and sensing sequences are ordered compositions of physical transformations and measurements.

Physical Construction

The state carrier is an input state, register, encoded subspace, channel state, key, syndrome, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, encodings, corrections, or feedback maps. Each step must belong to the claimed map class and the composition must preserve normalization and positivity. The calculated observables are Output state, fidelity, error rate, key rate, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (preparation, basis, or preparation context) determine which states are admissible. State terms (quantum state, superposition, or wave function) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, Hamiltonian, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

Quantum machine learning turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized

map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.18 Quantum programming

Mechanism

Quantum programming belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum programming specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state.

Physical Construction

The state carrier is an input state, register, channel state, error syndrome, key, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective. The calculated observables are Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (unitary operator, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. Quantum programming turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.19 Quantum simulator

Mechanism

Quantum simulator is a target–carrier–validation construction: a controllable physical system encodes another model, and selected observables test whether the encoded dynamics is faithful.

Quantum simulator specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

The simulator Hamiltonian is not by itself the target theory. The claim also needs an encoding between target and device states, a correspondence between their generators or channels, and validation observables with an error budget over the stated time and parameter range.

Physical Construction

The state carrier is a controllable device state space together with an explicit encoding of the target state space. The governing operation is Device Hamiltonians, channels, or gate sequences intended to reproduce target dynamics under the encoding. Control errors, leakage, finite size, noise, and approximation order define the regime in which the correspondence is claimed. The calculated observables are Encoded target observables compared with independently predicted or calibrated device measurements.

Topic Equations

Topic-specific constructor: encoding, dynamical correspondence, and observable validation are separate obligations.

$$\begin{aligned} V : \mathcal{H}_{\text{target}} &\hookrightarrow \mathcal{H}_{\text{device}} \\ \|U_{\text{device}}(t)V - VU_{\text{target}}(t)\| &\leq \varepsilon(t) \\ \left| \langle O \rangle_{\text{target}} - \langle VOV^\dagger \rangle_{\text{device}} \right| &\leq \delta_O \end{aligned}$$

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

12.20 Quantum cryptography

Mechanism

Quantum cryptography belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum cryptography specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state.

Physical Construction

The state carrier is an input state, register, channel state, error syndrome, key, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective. The calculated observables are Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (potential or domain condition, basis, or preparation context) determine which states are admissible. State terms (quantum state, wave function, or superposition) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. Quantum cryptography turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.21 Quantum metrology

Mechanism

Quantum metrology belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum metrology specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

The mechanism is an apparatus-coupled observable: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate.

Physical Construction

The state carrier is a probe state, sample state, field mode, detector state, or estimation register. The governing operation is an interaction Hamiltonian, transfer map, measurement channel, reconstruction map, or estimator. The instrument must separate sample signal from preparation, detector response, calibration, noise, and reconstruction artifacts. The calculated observables are Counts, images, spectra, phase shifts, trajectories, intensity maps, correlation data, or parameter estimates.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, unitary operator, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (mode, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

The mechanism is an apparatus-coupled observable: a prepared probe state interacts with a sample or field, the interaction changes phase, momentum, intensity, or counting statistics, and the instrument reconstructs an image, spectrum, trajectory, or estimate. Quantum metrology turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement.

Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.22 Quantum bus

Mechanism

Quantum bus belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum bus specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

Circuits, controls, and sensing sequences are ordered compositions of physical transformations and measurements.

Physical Construction

The state carrier is an input state, register, encoded subspace, channel state, key, syndrome, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, encodings, corrections, or feedback maps. Each step must belong to the claimed map class and the composition must preserve normalization and positivity. The calculated observables are Output state, fidelity, error rate, key rate, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (superposition, wave function, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (hamiltonian, observable operator, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

Quantum bus turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized

map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.23 Quantum image processing

Mechanism

Quantum image processing belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum image processing specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

Circuits, controls, and sensing sequences are ordered compositions of physical transformations and measurements.

Physical Construction

The state carrier is an input state, register, encoded subspace, channel state, key, syndrome, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, encodings, corrections, or feedback maps. Each step must belong to the claimed map class and the composition must preserve normalization and positivity. The calculated observables are Output state, fidelity, error rate, key rate, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

Quantum image processing turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized

map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.24 Quantum engineering

Mechanism

Quantum engineering belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum engineering specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

Circuits, controls, and sensing sequences are ordered compositions of physical transformations and measurements.

Physical Construction

The state carrier is an input state, register, encoded subspace, channel state, key, syndrome, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, encodings, corrections, or feedback maps. Each step must belong to the claimed map class and the composition must preserve normalization and positivity. The calculated observables are Output state, fidelity, error rate, key rate, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, superposition, or wave function) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (generator, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

Quantum engineering turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized

map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.25 Quantum cellular automaton

Mechanism

Quantum cellular automaton belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum cellular automaton specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

Circuits, controls, and sensing sequences are ordered compositions of physical transformations and measurements.

Physical Construction

The state carrier is an input state, register, encoded subspace, channel state, key, syndrome, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, encodings, corrections, or feedback maps. Each step must belong to the claimed map class and the composition must preserve normalization and positivity. The calculated observables are Output state, fidelity, error rate, key rate, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (context, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (unitary operator, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

Quantum cellular automaton turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized

map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

12.26 Quantum complexity theory

Mechanism

Quantum complexity theory belongs to the protocol layer: it packages the quantum constructor into engineered sequences of admissible transformations and observables.

Quantum complexity theory specifies an ordered sequence of operations. Order is physical whenever the maps do not commute, so a list of available gates or channels is insufficient to define an algorithm, sensor, communication scheme, or correction cycle.

Circuits, controls, and sensing sequences are ordered compositions of physical transformations and measurements.

Physical Construction

The state carrier is an input state, register, encoded subspace, channel state, key, syndrome, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, encodings, corrections, or feedback maps. Each step must belong to the claimed map class and the composition must preserve normalization and positivity. The calculated observables are Output state, fidelity, error rate, key rate, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (context, preparation context, or measurement basis) determine which states are admissible. State terms (state vector, wave function, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The ordered composition carries a prepared input to a final state. Every intermediate map must preserve its stated physical conditions, and conditional operations are tied to explicit measurement outcomes. Performance is quantified through fidelity, error rate, capacity, precision, or success probability.

This is the executable end of the mechanism tree. It also closes the loop: failed predictions can be traced backward to the operation order, the generator, the state preparation, or the mathematical domain rather than attributed to the topic as a whole.

Invariance And Realization

Quantum complexity theory turns the quantum constructor into an ordered operation sequence. The stable role is compositional: admissible maps transform an input state into an output state before measurement. Unitary gates, channels, measurements, correction steps, and algorithms are protocolized versions of the same state-map-observable logic.

The local title, representation, and physical realization may change while the constructor role is preserved. The implementation can be a circuit, channel, network, sensor, automaton, or cryptographic protocol. Noise, measurement timing, and correction rules change the realized

map. Different hardware can implement the same abstract sequence of completely positive or unitary operations.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Each operation in the sequence is constrained by the map class it claims: unitary, completely positive, trace preserving, measurement, correction, or conditional update. The composed protocol is defined by its output state and outcome probabilities, not only by the names of the gates.

Chapter 13

Boundaries and operator domains

Walls, interfaces, asymptotic conditions, and media become part of the mechanism when they select the operator domain. The resulting self-adjoint extension fixes spectra, scattering channels, tunnelling amplitudes, and edge states.

A boundary becomes physical closure when it fixes an operator domain. Tunnelling, confinement, scattering, and edge spectra follow from that domain.

13.1 Topics

Page
Potential well
Scattering
Particle in a box
Quantum optics
Wave interference
Spectral line
Quantum tunnelling
S-matrix
Quantum harmonic oscillator
Quantum metamaterial

13.2 Potential well

Mechanism

Potential well belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Potential well makes the operator domain physical. The same differential expression can have different spectra, resonances, and scattering channels when its boundary conditions, potential, or asymptotic states are changed.

The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel.

Physical Construction

The state carrier is a Hilbert space with a selected domain, potential, interface, asymptotic channel, cavity, well, or boundary condition. The governing operation is a Hamiltonian, wave operator, transfer operator, or scattering map whose domain is changed by the boundary. Boundary conditions and matching conditions determine allowed states, resonances, transmission amplitudes, and spectra. The calculated observables are Eigenvalues, resonances, tunnelling probabilities, phase shifts, reflection/transmission amplitudes, or scattering data.

Defining Relations

Preparation or domain terms (potential or domain condition, preparation context, or measurement basis) determine which states are admissible. State terms (wave function, quantum state, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (spectrum, energy level, or mode) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The eigenvalue or scattering equation must be read together with its domain and matching conditions. Confinement selects discrete modes; open channels produce transmission, reflection, and resonances. Removing the boundary should recover the appropriate free or infinite-domain limit.

Boundary-shaped single-particle mechanisms extend naturally to many modes and fields. The field-theory chapter replaces a fixed particle number with occupation sectors, local fields, and symmetry constraints.

Invariance And Realization

The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel. Potential well shows how a context, domain, potential, or boundary changes the allowed quantum channels. The invariant role is boundary-shaped spectral selection: the operator is the same kind of object, but its domain changes. Transmission, confinement, scattering, and mode selection are read as consequences of admissible boundary conditions.

The local title, representation, and physical realization may change while the constructor role is preserved. The potential, geometry, asymptotic condition, or detector arrangement can change. The same boundary role may appear as a box, barrier, cavity, interface, or scattering region. Changing the boundary can change the spectrum without changing the general quantum constructor.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. The boundary changes the operator domain, and therefore the allowed modes, transmission amplitudes, or scattering channels. The same operator can have different spectra when the admissible domain changes.

13.3 Scattering

Mechanism

Scattering belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Scattering makes the operator domain physical. The same differential expression can have different spectra, resonances, and scattering channels when its boundary conditions, potential, or asymptotic states are changed.

The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel.

Physical Construction

The state carrier is a Hilbert space with a selected domain, potential, interface, asymptotic channel, cavity, well, or boundary condition. The governing operation is a Hamiltonian, wave operator, transfer operator, or scattering map whose domain is changed by the boundary. Boundary conditions and matching conditions determine allowed states, resonances, transmission amplitudes, and spectra. The calculated observables are Eigenvalues, resonances, tunnelling probabilities, phase shifts, reflection/transmission amplitudes, or scattering data.

Defining Relations

Preparation or domain terms (boundary condition, potential or domain condition, or preparation context) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Matrix, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The eigenvalue or scattering equation must be read together with its domain and matching conditions. Confinement selects discrete modes; open channels produce transmission, reflection, and resonances. Removing the boundary should recover the appropriate free or infinite-domain limit.

Boundary-shaped single-particle mechanisms extend naturally to many modes and fields. The field-theory chapter replaces a fixed particle number with occupation sectors, local fields, and symmetry constraints.

Invariance And Realization

The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel. Scattering shows how a context, domain, potential, or boundary changes the allowed quantum channels. The invariant role is boundary-shaped spectral selection: the operator is the same kind of object, but its domain changes. Transmission, confinement, scattering, and mode selection are read as consequences of admissible boundary conditions.

The local title, representation, and physical realization may change while the constructor role is preserved. The potential, geometry, asymptotic condition, or detector arrangement can change. The same boundary role may appear as a box, barrier, cavity, interface, or scattering region. Changing the boundary can change the spectrum without changing the general quantum constructor.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. The boundary changes the operator domain, and therefore the allowed modes, transmission amplitudes, or scattering channels. The same operator can have different spectra when the admissible domain changes.

13.4 Particle in a box

Mechanism

Particle in a box is a boundary-spectrum constructor: a spatial domain and boundary condition discretize the allowed energy spectrum.

The particle in a box shows in the simplest form that a boundary condition is part of the Hamiltonian's physical definition. Confinement turns a continuous free-particle spectrum into discrete allowed energies.

The page shows how a boundary condition changes the domain of the Hamiltonian and therefore the allowed spectra.

Physical Construction

The state carrier is Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data. The governing operation is Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal. The calculated observables are Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.

Topic Equations

Standard constructor skeleton: boundary condition and discrete spectrum.

$$\begin{aligned}\psi(0) &= \psi(L) = 0 \\ \psi_n(x) &= \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L} \\ E_n &= \frac{\hbar^2 \pi^2 n^2}{2mL^2}\end{aligned}$$

Physical Meaning

The wave function satisfies the same differential equation inside the box as a free particle, but vanishing boundary values select standing waves. Only wavelengths fitting the interval are admissible, and their curvature fixes the quantized energies.

Doubling the box length reduces every level spacing by a factor of four. This follows from the boundary-selected wavelength and provides a direct check of the construction.

Replacing an impenetrable wall by a finite barrier leads to tunnelling, resonances, and scattering channels.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

13.5 Quantum optics

Mechanism

Quantum optics belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Quantum optics makes the operator domain physical. The same differential expression can have different spectra, resonances, and scattering channels when its boundary conditions, potential, or asymptotic states are changed.

The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel.

Physical Construction

The state carrier is a Hilbert space with a selected domain, potential, interface, asymptotic channel, cavity, well, or boundary condition. The governing operation is a Hamiltonian, wave operator, transfer operator, or scattering map whose domain is changed by the boundary. Boundary conditions and matching conditions determine allowed states, resonances, transmission amplitudes, and spectra. The calculated observables are Eigenvalues, resonances, tunnelling probabilities, phase shifts, reflection/transmission amplitudes, or scattering data.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (wavefunction, wave function, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (spectrum, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The eigenvalue or scattering equation must be read together with its domain and matching conditions. Confinement selects discrete modes; open channels produce transmission, reflection, and resonances. Removing the boundary should recover the appropriate free or infinite-domain limit.

Boundary-shaped single-particle mechanisms extend naturally to many modes and fields. The field-theory chapter replaces a fixed particle number with occupation sectors, local fields, and symmetry constraints.

Invariance And Realization

The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel. Quantum optics shows how a context, domain, potential, or boundary changes the allowed quantum channels. The invariant role is boundary-shaped spectral selection: the operator is the same kind of object, but its domain changes. Transmission, confinement, scattering, and mode selection are read as consequences of admissible boundary conditions.

The local title, representation, and physical realization may change while the constructor role is preserved. The potential, geometry, asymptotic condition, or detector arrangement can change. The same boundary role may appear as a box, barrier, cavity, interface, or scattering region. Changing the boundary can change the spectrum without changing the general quantum constructor.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. The boundary changes the operator domain, and therefore the allowed modes, transmission amplitudes, or scattering channels. The same operator can have different spectra when the admissible domain changes.

13.6 Wave interference

Mechanism

Wave interference belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Wave interference makes the operator domain physical. The same differential expression can have different spectra, resonances, and scattering channels when its boundary conditions, potential, or asymptotic states are changed.

Boundary conditions, interfaces, and asymptotic channels enter the operator domain and alter the allowed solutions and spectra.

Physical Construction

The state carrier is a state space equipped with a domain, potential, interface, cavity, or asymptotic channel. The governing operation is a Hamiltonian, wave operator, transfer operator, or scattering map whose domain depends on the boundary data. Boundary and matching conditions determine the allowed modes, resonances, and conserved fluxes. The calculated observables are Energy levels, resonances, transmission and reflection amplitudes, phase shifts, or scattering cross-sections.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (superposition, wave function, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (observable operator, Hamiltonian, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The eigenvalue or scattering equation must be read together with its domain and matching conditions. Confinement selects discrete modes; open channels produce transmission, reflection, and resonances. Removing the boundary should recover the appropriate free or infinite-domain limit.

Boundary-shaped single-particle mechanisms extend naturally to many modes and fields. The field-theory chapter replaces a fixed particle number with occupation sectors, local fields, and symmetry constraints.

Invariance And Realization

Wave interference shows how a context, domain, potential, or boundary changes the allowed quantum channels. The invariant role is boundary-shaped spectral selection: the operator is the same kind of object, but its domain changes. Transmission, confinement, scattering, and mode selection are read as consequences of admissible boundary conditions.

The local title, representation, and physical realization may change while the constructor role is preserved. The potential, geometry, asymptotic condition, or detector arrangement can change. The same boundary role may appear as a box, barrier, cavity, interface, or scattering

region. Changing the boundary can change the spectrum without changing the general quantum constructor.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. The boundary changes the operator domain, and therefore the allowed modes, transmission amplitudes, or scattering channels. The same operator can have different spectra when the admissible domain changes.

13.7 Spectral line

Mechanism

Spectral line belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Spectral line makes the operator domain physical. The same differential expression can have different spectra, resonances, and scattering channels when its boundary conditions, potential, or asymptotic states are changed.

The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel.

Physical Construction

The state carrier is a Hilbert space with a selected domain, potential, interface, asymptotic channel, cavity, well, or boundary condition. The governing operation is a Hamiltonian, wave operator, transfer operator, or scattering map whose domain is changed by the boundary. Boundary conditions and matching conditions determine allowed states, resonances, transmission amplitudes, and spectra. The calculated observables are Eigenvalues, resonances, tunnelling probabilities, phase shifts, reflection/transmission amplitudes, or scattering data.

Defining Relations

Preparation or domain terms (potential or domain condition, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (spectrum, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (uncertainty relation, commutator, or incompatible observable) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The eigenvalue or scattering equation must be read together with its domain and matching conditions. Confinement selects discrete modes; open channels produce transmission, reflection, and resonances. Removing the boundary should recover the appropriate free or infinite-domain limit.

Boundary-shaped single-particle mechanisms extend naturally to many modes and fields. The field-theory chapter replaces a fixed particle number with occupation sectors, local fields, and symmetry constraints.

Invariance And Realization

The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel. Spectral line shows how a context, domain, potential, or boundary changes the allowed quantum channels. The invariant role is boundary-shaped spectral selection: the operator is the same kind of object, but its domain changes. Transmission, confinement, scattering, and mode selection are read as consequences of admissible boundary conditions.

The local title, representation, and physical realization may change while the constructor role is preserved. The potential, geometry, asymptotic condition, or detector arrangement can change. The same boundary role may appear as a box, barrier, cavity, interface, or scattering region. Changing the boundary can change the spectrum without changing the general quantum constructor.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. The boundary changes the operator domain, and therefore the allowed modes, transmission amplitudes, or scattering channels. The same operator can have different spectra when the admissible domain changes.

13.8 Quantum tunnelling

Mechanism

Quantum tunnelling is a boundary-realization constructor: a state has nonzero transmission through a classically forbidden region.

Quantum tunnelling tests the consequence of wave evolution across a classically forbidden region. The effect depends jointly on the generator, barrier geometry, matching conditions, and incident state.

Tunnelling shows that the realization layer matters. A potential barrier changes the admissible wave solutions and produces transmission even where classical kinetic energy would be negative.

Physical Construction

The state carrier is a Hilbert space with a selected domain, potential, interface, asymptotic channel, cavity, well, or boundary condition. The governing operation is a Hamiltonian, wave operator, transfer operator, or scattering map whose domain is changed by the boundary. Boundary conditions and matching conditions determine allowed states, resonances, transmission amplitudes, and spectra. The calculated observables are Eigenvalues, resonances, tunnelling probabilities, phase shifts, reflection/transmission amplitudes, or scattering data.

Topic Equations

Standard constructor skeleton: barrier-domain Schrödinger equation and WKB transmission.

$$T \sim \exp\left(-2 \int_{x_1}^{x_2} \sqrt{\frac{2m(V(x) - E)}{\hbar^2}} dx\right)$$
$$-\frac{\hbar^2}{2m}\psi''(x) + V(x)\psi(x) = E\psi(x)$$

Physical Meaning

Inside the barrier the stationary wave function is evanescent rather than oscillatory. Continuity and flux conditions connect it to incoming and outgoing waves, producing a nonzero transmission amplitude that falls approximately exponentially with barrier width in the semiclassical regime.

Changing the barrier width while holding its height and the incident energy fixed isolates the predicted exponential dependence and separates tunnelling from an over-barrier contribution.

Scattering theory generalizes the same matching construction to many incoming, outgoing, and resonant channels.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

13.9 S-matrix

Mechanism

S-matrix belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

S-matrix makes the operator domain physical. The same differential expression can have different spectra, resonances, and scattering channels when its boundary conditions, potential, or asymptotic states are changed.

The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel.

Physical Construction

The state carrier is a Hilbert space with a selected domain, potential, interface, asymptotic channel, cavity, well, or boundary condition. The governing operation is a Hamiltonian, wave operator, transfer operator, or scattering map whose domain is changed by the boundary. Boundary conditions and matching conditions determine allowed states, resonances, transmission amplitudes, and spectra. The calculated observables are Eigenvalues, resonances, tunnelling probabilities, phase shifts, reflection/transmission amplitudes, or scattering data.

Defining Relations

Preparation or domain terms (potential or domain condition, context, or basis) determine which states are admissible. State terms (wave function, probability amplitude, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, Hamiltonian, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The eigenvalue or scattering equation must be read together with its domain and matching conditions. Confinement selects discrete modes; open channels produce transmission, reflection, and resonances. Removing the boundary should recover the appropriate free or infinite-domain limit.

Boundary-shaped single-particle mechanisms extend naturally to many modes and fields. The field-theory chapter replaces a fixed particle number with occupation sectors, local fields, and symmetry constraints.

Invariance And Realization

The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel. S-matrix shows how a context, domain, potential, or boundary changes the allowed quantum channels. The invariant role is boundary-shaped spectral selection: the operator is the same kind of object, but its domain changes. Transmission, confinement, scattering, and mode selection are read as consequences of admissible boundary conditions.

The local title, representation, and physical realization may change while the constructor role is preserved. The potential, geometry, asymptotic condition, or detector arrangement can change. The same boundary role may appear as a box, barrier, cavity, interface, or scattering region. Changing the boundary can change the spectrum without changing the general quantum constructor.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. The boundary changes the operator domain, and therefore the allowed modes, transmission amplitudes, or scattering channels. The same operator can have different spectra when the admissible domain changes.

13.10 Quantum harmonic oscillator

Mechanism

Quantum harmonic oscillator belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Quantum harmonic oscillator makes the operator domain physical. The same differential expression can have different spectra, resonances, and scattering channels when its boundary conditions, potential, or asymptotic states are changed.

Boundary conditions, interfaces, and asymptotic channels enter the operator domain and alter the allowed solutions and spectra.

Physical Construction

The state carrier is a state space equipped with a domain, potential, interface, cavity, or asymptotic channel. The governing operation is a Hamiltonian, wave operator, transfer operator, or scattering map whose domain depends on the boundary data. Boundary and matching conditions determine the allowed modes, resonances, and conserved fluxes. The calculated observables are Energy levels, resonances, transmission and reflection amplitudes, phase shifts, or scattering cross-sections.

Defining Relations

Preparation or domain terms (potential or domain condition, basis, or preparation context) determine which states are admissible. State terms (wave function, wavefunction, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (energy level, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (uncertainty relation, commutator, or incompatible observable) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The eigenvalue or scattering equation must be read together with its domain and matching conditions. Confinement selects discrete modes; open channels produce transmission, reflection, and resonances. Removing the boundary should recover the appropriate free or infinite-domain limit.

Boundary-shaped single-particle mechanisms extend naturally to many modes and fields. The field-theory chapter replaces a fixed particle number with occupation sectors, local fields, and symmetry constraints.

Invariance And Realization

Quantum harmonic oscillator shows how a context, domain, potential, or boundary changes the allowed quantum channels. The invariant role is boundary-shaped spectral selection: the operator is the same kind of object, but its domain changes. Transmission, confinement, scattering, and mode selection are read as consequences of admissible boundary conditions.

The local title, representation, and physical realization may change while the constructor role is preserved. The potential, geometry, asymptotic condition, or detector arrangement can

change. The same boundary role may appear as a box, barrier, cavity, interface, or scattering region. Changing the boundary can change the spectrum without changing the general quantum constructor.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. The boundary changes the operator domain, and therefore the allowed modes, transmission amplitudes, or scattering channels. The same operator can have different spectra when the admissible domain changes.

13.11 Quantum metamaterial

Mechanism

Quantum metamaterial belongs to realization: it shows how the abstract state-operator construction becomes legal on a domain, interface, potential, detector geometry, or scattering boundary.

Quantum metamaterial makes the operator domain physical. The same differential expression can have different spectra, resonances, and scattering channels when its boundary conditions, potential, or asymptotic states are changed.

The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel.

Physical Construction

The state carrier is a Hilbert space with a selected domain, potential, interface, asymptotic channel, cavity, well, or boundary condition. The governing operation is a Hamiltonian, wave operator, transfer operator, or scattering map whose domain is changed by the boundary. Boundary conditions and matching conditions determine allowed states, resonances, transmission amplitudes, and spectra. The calculated observables are Eigenvalues, resonances, tunnelling probabilities, phase shifts, reflection/transmission amplitudes, or scattering data.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

The eigenvalue or scattering equation must be read together with its domain and matching conditions. Confinement selects discrete modes; open channels produce transmission, reflection, and resonances. Removing the boundary should recover the appropriate free or infinite-domain limit.

Boundary-shaped single-particle mechanisms extend naturally to many modes and fields. The field-theory chapter replaces a fixed particle number with occupation sectors, local fields, and symmetry constraints.

Invariance And Realization

The topic changes the admissible domain or boundary condition and thereby changes the allowed spectrum, transmission amplitude, resonance, or scattering channel. Quantum metamaterial shows how a context, domain, potential, or boundary changes the allowed quantum channels. The invariant role is boundary-shaped spectral selection: the operator is the same kind of object, but its domain changes. Transmission, confinement, scattering, and mode selection are read as consequences of admissible boundary conditions.

The local title, representation, and physical realization may change while the constructor role is preserved. The potential, geometry, asymptotic condition, or detector arrangement can change. The same boundary role may appear as a box, barrier, cavity, interface, or scattering region. Changing the boundary can change the spectrum without changing the general quantum constructor.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. The boundary changes the operator domain, and therefore the allowed modes, transmission amplitudes, or scattering channels. The same operator can have different spectra when the admissible domain changes.

Part V

Fields, Constraints, And Scale

Chapter 14

Fields, constraints, and scale

Quantum field theory promotes fields to operator-valued degrees of freedom. Gauge constraints select the physical Hilbert space, particle statistics select admissible sectors, and renormalization makes the effective law depend on scale.

Fields enlarge the state space to variable occupation and local operator algebras. Gauge constraints, statistics, and scale then select the physical sectors and effective law.

14.1 Topics

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Fock space

14.2 Gauge theory

Mechanism

Gauge theory defines how internal states are compared at different spacetime points: a connection relates neighboring frames, and curvature records the path dependence that no single gauge choice can remove.

Gauge theory separates physical states from multiple mathematical descriptions related by local transformations. The redundancy is useful because it makes locality and interaction structure explicit, but constraints are required to remove unphysical degrees of freedom.

A local gauge transformation changes the field coordinates used at each point without changing the physical state. Ordinary derivatives compare fields in different local frames and therefore cease to transform covariantly. The gauge connection repairs that comparison. Its commutator gives the field strength, while Wilson loops measure the accumulated transport around a closed path. Gauss constraints select physical states and charges. The connection is representation dependent; curvature, loop observables, and gauge-invariant amplitudes carry the physical content.

Physical Construction

The state carrier is Matter and gauge fields modulo local gauge equivalence, restricted to the physical constraint sector. The governing operation is a covariant derivative and connection whose commutator gives the field strength. Gauss constraints, gauge covariance, operator domains, and boundary conditions select the physical states and charges. The calculated observables are Field strengths, Wilson loops, conserved charges, scattering amplitudes, and other gauge-invariant quantities.

Topic Equations

The connection defines local comparison, curvature measures its infinitesimal path dependence, and the constraint selects physical states.

$$\begin{aligned}D_\mu &= \partial_\mu + igA_\mu \\[D_\mu, D_\nu] &= igF_{\mu\nu} \\W(\gamma) &= \text{Tr } \mathcal{P} \exp \left(ig \oint_\gamma A_\mu dx^\mu \right) \\G^a |\Psi_{\text{phys}}\rangle &= 0\end{aligned}$$

Physical Meaning

Gauge-related field configurations represent the same physical state. Observable quantities must therefore be gauge invariant, or transform covariantly within a construction whose final predictions are invariant. Gauge fixing chooses one representative without changing the physical equivalence class.

Changing the electromagnetic scalar and vector potentials by a gauge transformation changes their formulas but leaves electric and magnetic fields, phase-consistent amplitudes, and measured forces unchanged.

Renormalization adds a second kind of transformation: changing the scale at which the same theory is parametrized.

Consequences Forced By The Relation

Gauge-related potentials give identical gauge-invariant amplitudes, charges, field strengths, and loop observables. A nontrivial Wilson loop can retain global transport information that is absent from any one local gauge representative. The Gauss constraint removes redundant state vectors and fixes the physical charge sector.

Domain Of The Construction

The gauge group, representation, matter content, dimension, and boundary conditions distinguish different physical theories. A gauge-dependent potential becomes observable only through a gauge-invariant relation such as field strength, phase difference, charge, or loop holonomy.

Invariance And Realization

Gauge-equivalent potentials give the same gauge-invariant amplitudes, field strengths, charges, and loop observables. Curvature records the infinitesimal holonomy of the connection and cannot be removed by a local gauge choice. The physical state belongs to the constraint sector rather than to an arbitrary field-coordinate representation.

The gauge potential, local basis, gauge-fixing condition, and coordinate description may change. The gauge group, representation, matter content, dimension, and boundary conditions specify different physical theories. Topological sectors and boundary charges can survive even where the local field strength vanishes.

Discriminating Consequences

A closed-loop phase or Wilson observable distinguishes nontrivial holonomy from a removable local gauge choice. Gauge-related descriptions give identical probabilities for the same physical preparation and observable. An additional field or interaction changes a gauge-invariant observable rather than only the gauge potential.

14.3 Photon

Mechanism

Photon is a field-mode constructor: a one-quantum excitation of the electromagnetic field, constrained by massless dispersion and transverse polarization.

Photon places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

The native photon mechanism is field quantization. The electromagnetic field is decomposed into modes, creation and annihilation operators act on those modes, and a one-photon state is created from the vacuum. The relevant observables are occupation number, energy, momentum, polarization, and detector clicks; the constraints are dispersion and gauge-compatible transversality.

Physical Construction

The state carrier is Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data. The governing operation is Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal. The calculated observables are Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.

Topic Equations

Topic-specific constructor: the equations express massless dispersion, one-mode occupation, number observable, and transverse polarization.

$$\begin{aligned}E &= \hbar\omega, & \mathbf{p} &= \hbar\mathbf{k}, & \omega &= c|\mathbf{k}| \\|1_{\mathbf{k},\lambda}\rangle &= a_{\mathbf{k},\lambda}^\dagger|0\rangle \\ \hat{N}_{\mathbf{k},\lambda} &= a_{\mathbf{k},\lambda}^\dagger a_{\mathbf{k},\lambda} \\ \mathbf{k} \cdot \boldsymbol{\epsilon}_{\mathbf{k},\lambda} &= 0\end{aligned}$$

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

The relation between prepared states, observables, and spectral probability measures. The use of eigenvalues, projectors, modes, or outcome channels to represent admissible observations. The dependence of the observable on basis, domain, potential, preparation, or measurement context. The commutator structure that limits which observables can be jointly diagonalized.

The physical carrier: particle, wave, field mode, spin, qubit, detector, or excitation. The representation: wave mechanics, matrix mechanics, density matrices, path integrals, circuits, or fields. Where time dependence is placed: on the state, on the operator, in a propagator, or in a path weight. The implementation of preparation, boundary condition, detector, or outcome channel.

Discriminating Consequences

A transfer target provides a state space, a transformation law, and a spectral or categorical observable, with one compatibility relation experimentally unresolved. A useful validation varies the basis, domain, or measurement context and measures whether the allowed observable changes while the underlying transformation law remains identifiable. A stronger validation contains two candidate observables whose predicted commutator controls joint resolvability.

14.4 Renormalization

Mechanism

Renormalization relates physical descriptions at different resolutions by changing couplings and operator weights while preserving long-distance predictions.

Renormalization explains how a theory preserves predictions while its effective parameters and degrees of freedom change with scale. It is therefore a transformation between descriptions, not merely a device for removing infinities.

Coarse graining removes short-distance variables and generates every operator allowed by the remaining symmetries. Their coefficients flow with scale. Relevant directions grow, irrelevant directions decay, and fixed points organize scale-invariant behaviour. Microscopically different systems can therefore share critical exponents and scaling functions when their flows approach the same fixed point. Universality is the invariance class of this scale transformation, not a visual similarity between equations.

Physical Construction

The state carrier is Effective fields and degrees of freedom defined at a stated resolution or cutoff. The governing operation is a coarse-graining transformation and beta functions acting on the coefficients of an effective operator expansion. Symmetry, dimensionality, locality assumptions, relevant directions, and renormalization conditions determine the allowed flow. The calculated observables are Running couplings, correlation functions, scaling dimensions, critical exponents, and corrections to scaling.

Topic Equations

Beta functions, scale-independent observables, fixed points, and scaling directions define the transport of a theory across resolution.

$$\begin{aligned}\mu \frac{dg_i}{d\mu} &= \beta_i(\{g\}) \\ (\mu \partial_\mu + \beta_i \partial_{g_i} + n\gamma) G^{(n)} &= 0 \\ \beta_i(g_*) &= 0, \quad \delta g_i(b) = b^{y_i} \delta g_i \\ \mathcal{L}_{\text{eff}}(\mu) &= \sum_i c_i(\mu) \mathcal{O}_i\end{aligned}$$

Physical Meaning

A change of scale moves the couplings along a flow. Predictions remain fixed when explicit scale dependence and coupling dependence compensate. Fixed points and relevant directions then organize universality across microscopically different systems.

Different lattice models can approach the same critical exponents because coarse-graining removes microscopic details while preserving the long-distance transformation structure.

Effective field theory uses this scale organization to decide which operators must be retained for a specified accuracy.

Consequences Forced By The Relation

Observable predictions are independent of the arbitrary renormalization scale when explicit scale dependence and coupling flow compensate. Systems approaching the same fixed point with the same relevant directions share critical exponents and scaling functions. Relevant perturbations drive the flow away from a fixed point and select the accessible phases or crossover regimes.

Domain Of The Construction

Universality requires matching dimensionality, symmetries, conserved quantities, interaction range, and relevant directions; a shared dimensionless group alone is insufficient. An effective theory is tied to a scale and accuracy, because omitted operators return through controlled corrections.

Invariance And Realization

Observable predictions remain independent of the arbitrary renormalization scale when couplings and fields flow consistently. Critical exponents and scaling functions are shared by systems approaching the same fixed point with the same relevant directions. Symmetry and dimensionality constrain the effective operators generated by coarse graining.

Couplings, field normalizations, effective degrees of freedom, and operator coefficients depend on scale. Microscopic Hamiltonians can differ while their long-distance flows enter the same universality class. A relevant perturbation can drive the system away from one fixed point toward another phase or crossover regime.

Discriminating Consequences

Measurements at several scales determine whether running couplings follow one beta function. Microscopically different realizations determine whether critical exponents and scaling functions coincide. Corrections to scaling distinguish approach to a fixed point from an exact scale-invariant law.

14.5 Quantum field theory

Mechanism

Quantum field theory is the many-mode local-field extension of the quantum constructor.

Quantum field theory is needed when relativity, locality, and particle creation must be described together. Fields become the primary operator-valued objects, while particles appear as excitations of their modes.

Quantum field theory lifts the state-operator-spectrum construction to fields, local operators, creation and annihilation modes, and scattering amplitudes. Particles become stable excitation/observable roles of fields.

Physical Construction

The state carrier is Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data. The governing operation is Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal. The calculated observables are Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.

Topic Equations

Standard constructor skeleton: field expansion, mode algebra, and correlation observable.

$$\begin{aligned}\Phi(x) &= \sum_k \left(a_k u_k(x) + a_k^\dagger u_k^*(x) \right) \\ [a_k, a_l^\dagger]_{\mp} &= \delta_{kl} \\ \langle 0 | T \{ \Phi(x) \Phi(y) \} | 0 \rangle\end{aligned}$$

Physical Meaning

The state lives in a many-particle or field space, local operators create correlations, and the action or Hamiltonian generates their evolution. Perturbative amplitudes are expansions of these operator relations, not independent pictorial rules.

A photon is represented as an excitation of the electromagnetic field. Processes that change photon number are natural in Fock space but cannot be represented in a fixed one-particle Hilbert space.

Gauge theory identifies redundant field descriptions, and renormalization connects parameters measured at different scales.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

14.6 Fermi–Dirac statistics

Mechanism

Fermi–Dirac statistics is an antisymmetric-state constructor: exchange statistics restricts which many-fermion occupation patterns are admissible.

Fermi–Dirac statistics is needed because identical fermions do not occupy many-particle states independently. Antisymmetry under particle exchange, and the exclusion principle that follows from it, changes the allowed occupation patterns before any Hamiltonian dynamics is considered.

Fermi–Dirac statistics is an admissibility rule, not a generator of time evolution. Anticommutation and antisymmetry imply single occupation of each one-particle mode, while the thermal distribution states the mean occupation of those modes.

Physical Construction

The state carrier is a fermionic Fock space assembled from antisymmetric many-particle sectors or occupation-number modes. The governing operation is Creation, annihilation, and number operators obeying canonical anticommutation relations. Exchange antisymmetry restricts each one-particle mode to occupation zero or one for each internal state. The calculated observables are Mode occupations, Fermi energy, particle density, pressure, heat capacity, and other equilibrium response functions.

Topic Equations

Topic-specific construction: antisymmetric mode algebra, exclusion, and equilibrium occupation.

$$\{a_i, a_j^\dagger\} = \delta_{ij}, \quad n_i \in \{0, 1\}$$
$$\bar{n}_i = \frac{1}{e^{\beta(\varepsilon_i - \mu)} + 1}$$

Physical Meaning

Single-particle energy levels provide the available modes, while each mode can be occupied at most once per internal state. The Fermi–Dirac distribution gives the mean occupation at thermal equilibrium. At zero temperature it fills modes up to the Fermi energy.

The degeneracy pressure of an electron gas follows from filling successively higher momentum states even without a repulsive force between the electrons. The effect is a consequence of antisymmetric state construction.

Bose–Einstein statistics changes the exchange rule from antisymmetric to symmetric and therefore permits unlimited occupation of one mode.

Invariance And Realization

Exchange antisymmetry and canonical anticommutation define the fermionic many-particle sectors. Each one-particle mode has occupation zero or one for each internal state. The Fermi–Dirac function gives the equilibrium mean occupation once energy, temperature, and chemical potential are specified.

The dispersion relation, dimensionality, degeneracy, density of states, and interaction approximation depend on the physical system. Electrons, atoms, nucleons, and fermionic quasiparticles realize the same exchange rule with different Hamiltonians and observables. Finite temperature smooths the occupation edge that is sharp at the Fermi energy in the ideal zero-temperature limit.

Discriminating Consequences

The theory recovers occupations restricted to zero or one and the ideal zero-temperature Fermi sea. The theory recovers the Maxwell-Boltzmann distribution in the dilute low-fugacity limit. Integrate the mode occupations against the density of states and verify the specified particle number.

14.7 Dirac equation

Mechanism

Dirac equation belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Dirac equation places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement.

Physical Construction

The state carrier is a state vector, density operator, wavefunction, field state, or register on a specified domain. The governing operation is Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal. The calculated observables are Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.

Defining Relations

Preparation or domain terms (context, preparation context, or measurement basis) determine which states are admissible. State terms (wavefunction, wave function, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (spectrum, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement. Dirac equation extends the state-operator-spectrum constructor to many modes, fields, particles, gauge structure, or scale. Particle identity is treated as a stable excitation or representation role rather than as the starting object. Creation/annihilation, field operators, gauge constraints, and scale flow preserve operator structure across realizations.

The local title, representation, and physical realization may change while the constructor role is preserved. The carrier can be a field state, occupation-number state, gauge orbit, spin network, or effective theory. The same formal role may be displayed through particles, modes, amplitudes, correlation functions, or boundary dictionaries. Scale and geometry can change the realization while preserving operator or spectral content.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Commutation, anticommutation, gauge, and occupation rules define which many-mode states are admissible. The field or many-mode construction must reduce to the appropriate single-particle, quasiparticle, or low-energy limit when those limits exist.

14.8 Quantum electrodynamics

Mechanism

Quantum electrodynamics belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Quantum electrodynamics places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement.

Physical Construction

The state carrier is a state vector, density operator, wavefunction, field state, or register on a specified domain. The governing operation is Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal. The calculated observables are Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.

Defining Relations

Preparation or domain terms (experimental setup, preparation context, or measurement basis) determine which states are admissible. State terms (probability amplitude, wave function, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (observable operator, Hamiltonian, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (incompatible observables, commutator, or uncertainty relation) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement. Quantum electrodynamics extends the state-operator-spectrum constructor to many modes, fields, particles, gauge structure, or scale. Particle identity is treated as a stable excitation or representation role rather than as the starting object. Creation/annihilation, field operators, gauge constraints, and scale flow preserve operator structure across realizations.

The local title, representation, and physical realization may change while the constructor role is preserved. The carrier can be a field state, occupation-number state, gauge orbit, spin network, or effective theory. The same formal role may be displayed through particles, modes, amplitudes, correlation functions, or boundary dictionaries. Scale and geometry can change the realization while preserving operator or spectral content.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Commutation, anticommutation, gauge, and occupation rules define which many-mode states are admissible. The field or many-mode construction must reduce to the appropriate single-particle, quasiparticle, or low-energy limit when those limits exist.

14.9 Boson

Mechanism

Boson is an exchange-symmetry constructor: identical bosons live in symmetric sectors and can share the same mode.

Boson places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

The mechanism is symmetric exchange. Commuting creation and annihilation operators allow arbitrary nonnegative occupation of a mode, which supports field modes, coherent states, condensates, and photon-like observables.

Physical Construction

The state carrier is Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data. The governing operation is Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal. The calculated observables are Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.

Topic Equations

Topic-specific constructor: the equations express symmetric sectors, commutation, and unrestricted mode occupation.

$$\begin{aligned}\mathcal{F}_+(\mathcal{H}) &= \bigoplus_{n=0}^{\infty} \text{Sym}^n \mathcal{H} \\ [a_i, a_j^\dagger] &= \delta_{ij}, \quad [a_i, a_j] = 0 \\ n_i &= a_i^\dagger a_i \in \{0, 1, 2, \dots\}\end{aligned}$$

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

The relation between prepared states, observables, and spectral probability measures. The use of eigenvalues, projectors, modes, or outcome channels to represent admissible observations. The dependence of the observable on basis, domain, potential, preparation, or measurement context. The commutator structure that limits which observables can be jointly diagonalized.

The physical carrier: particle, wave, field mode, spin, qubit, detector, or excitation. The representation: wave mechanics, matrix mechanics, density matrices, path integrals, circuits, or fields. Where time dependence is placed: on the state, on the operator, in a propagator, or in

a path weight. The implementation of preparation, boundary condition, detector, or outcome channel.

Discriminating Consequences

A transfer target provides a state space, a transformation law, and a spectral or categorical observable, with one compatibility relation experimentally unresolved. A useful validation varies the basis, domain, or measurement context and measures whether the allowed observable changes while the underlying transformation law remains identifiable. A stronger validation contains two candidate observables whose predicted commutator controls joint resolvability.

14.10 Fermion

Mechanism

Fermion is an exchange-antisymmetry construction: exchanging two identical fermions reverses the many-body amplitude, so coincident one-particle states are removed from the admissible state space.

A many-particle Hilbert space does not specify how identical particles are exchanged. Choosing the antisymmetric representation removes coincidence states and changes the spectrum, pressure, correlations, and admissible collective phases before a particular interaction Hamiltonian is chosen.

The defining object is the antisymmetric many-body sector, not a particle name. Its nodal set contains every coincidence configuration, and the canonical anticommutation relations preserve that restriction while particles are added or removed. Filling distinct one-particle modes then produces an exchange hole, a Fermi surface and degeneracy pressure even in the absence of a repulsive potential.

Physical Construction

The state carrier is Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data. The governing operation is Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal. The calculated observables are Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.

Topic Equations

The exchange sign fixes the admissible state space; exterior Fock space and anticommutation extend it to variable particle number.

$$\begin{aligned}\Psi(\dots, x_i, \dots, x_j, \dots) &= -\Psi(\dots, x_j, \dots, x_i, \dots), & \Psi(\dots, x, \dots, x, \dots) &= 0 \\ \mathcal{F}_-(\mathcal{H}) &= \bigoplus_{n=0}^{\infty} \wedge^n \mathcal{H} \\ \{a_i, a_j^\dagger\} &= \delta_{ij}, & \{a_i, a_j\} &= 0 \\ n_i &= a_i^\dagger a_i \in \{0, 1\}\end{aligned}$$

Physical Meaning

The minus sign acquired under exchange makes every Slater determinant vanish when two columns coincide. In occupation language the same restriction is encoded by anticommutation and by eigenvalues zero or one of each mode-number operator. These are equivalent descriptions of one state-space constraint.

A zero-temperature ideal Fermi gas fills all momentum modes up to the Fermi momentum. Compressing the gas therefore forces particles into higher-momentum states and raises its pressure even when the interaction potential is set to zero.

Fermi–Dirac statistics adds thermal occupation to this exchange-constrained state space. Pairing, bosonization and Jordan–Wigner transformations then test which consequences of fermionic algebra survive on a different carrier.

Consequences Forced By The Relation

Antisymmetry forces the wave function to vanish when two identical fermions occupy the same one-particle state. The resulting exchange hole is present before a dynamical interaction is specified. At finite density, distinct momentum states fill up to the Fermi surface. The associated kinetic energy and degeneracy pressure arise from state counting rather than from pairwise repulsion. Pairing changes the exchange sector: a bound state of two fermions has even fermion parity and can acquire bosonic collective behaviour, as in superconductors and superfluid helium-3.

Transformations To Other Physical Realizations

In one dimension, hard-core bosons and free fermions can share the same density spectrum. The map changes exchange phases and off-diagonal correlations, so equality of energies does not imply identity of all observables. A Jordan–Wigner map carries local fermionic occupation into spins by adding a parity string. The exchange algebra survives, but locality is transferred into an ordered, generally nonlocal operator. In two dimensions, exchanges are classified by braids rather than only by permutations. Anyonic statistics therefore extends, rather than merely interpolates between, the boson and fermion constructions.

Domain Of The Construction

The relativistic spin–statistics theorem additionally requires locality, positive energy and the relativistic field framework. Antisymmetric lattice or effective quasiparticle models do not by themselves establish that theorem. Fermi surfaces and degeneracy pressure require a many-mode spectrum and a specified density or particle-number constraint; they do not follow from the word fermion alone.

Invariance And Realization

Exchange antisymmetry, exterior-product state space, canonical anticommutation and zero-or-one mode occupation are equivalent forms of the fermionic restriction. The coincidence node and exchange hole survive changes between first-quantized wave functions, Slater determinants and second-quantized fields. Fermion parity remains meaningful when particle number changes, including in paired and superconducting states.

Mass, charge, dispersion, dimensionality, interaction law and gauge representation belong to the physical realization and are not fixed by exchange statistics. A change of carrier can turn local fermion operators into nonlocal strings, as in the Jordan–Wigner transformation. Two-dimensional braid statistics and composite-particle structure alter the exchange construction beyond the elementary boson–fermion dichotomy.

Discriminating Consequences

The defining relation is antisymmetry or canonical anticommutation and the resulting zero-or-one occupation spectrum. At fixed one-particle spectrum, compare with distinguishable-particle and sign-erased controls to isolate exchange. For a claimed transfer, compare correlations and locality as well as energies; spectral agreement alone is insufficient.

14.11 Quantum chromodynamics

Mechanism

Quantum chromodynamics belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Quantum chromodynamics places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement.

Physical Construction

The state carrier is a state vector, density operator, wavefunction, field state, or register on a specified domain. The governing operation is Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal. The calculated observables are Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.

Defining Relations

Preparation or domain terms (context, basis, or preparation context) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement. Quantum chromodynamics extends the state-operator-spectrum constructor to many modes, fields, particles, gauge structure, or scale. Particle identity is treated as a stable excitation or representation role rather than as the starting object. Creation/annihilation, field operators, gauge constraints, and scale flow preserve operator structure across realizations.

The local title, representation, and physical realization may change while the constructor role is preserved. The carrier can be a field state, occupation-number state, gauge orbit, spin network, or effective theory. The same formal role may be displayed through particles, modes, amplitudes, correlation functions, or boundary dictionaries. Scale and geometry can change the realization while preserving operator or spectral content.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Commutation, anticommutation, gauge, and occupation rules define which many-mode states are admissible. The field or many-mode construction must reduce to the appropriate single-particle, quasiparticle, or low-energy limit when those limits exist.

14.12 Electron

Mechanism

Electron is a charged spinor constructor: its identity is fixed by mass, charge, spin-1/2 representation, fermionic statistics, and electromagnetic coupling.

Electron places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

The electron is not a generic object label in this tree. Its native mechanism combines a spinor state, a Schrödinger/Pauli/Dirac generator depending on regime, conserved charge, and fermionic anticommutation. The observables are charge, spin, momentum, energy, and scattering response.

Physical Construction

The state carrier is Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data. The governing operation is Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal. The calculated observables are Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.

Topic Equations

Topic-specific constructor: the equations express relativistic spinor transport, electromagnetic coupling, and fermionic field statistics.

$$\begin{aligned}(i\hbar\gamma^\mu D_\mu - mc)\psi &= 0 \\ D_\mu &= \partial_\mu + \frac{ie}{\hbar c}A_\mu \\ \{\psi_\alpha(x), \psi_\beta^\dagger(y)\} &= \delta_{\alpha\beta}\delta(x-y)\end{aligned}$$

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

The relation between prepared states, observables, and spectral probability measures. The use of eigenvalues, projectors, modes, or outcome channels to represent admissible observations. The dependence of the observable on basis, domain, potential, preparation, or measurement context. The commutator structure that limits which observables can be jointly diagonalized.

The physical carrier: particle, wave, field mode, spin, qubit, detector, or excitation. The representation: wave mechanics, matrix mechanics, density matrices, path integrals, circuits, or fields. Where time dependence is placed: on the state, on the operator, in a propagator, or in

a path weight. The implementation of preparation, boundary condition, detector, or outcome channel.

Discriminating Consequences

A transfer target provides a state space, a transformation law, and a spectral or categorical observable, with one compatibility relation experimentally unresolved. A useful validation varies the basis, domain, or measurement context and measures whether the allowed observable changes while the underlying transformation law remains identifiable. A stronger validation contains two candidate observables whose predicted commutator controls joint resolvability.

14.13 Spin foam

Mechanism

Spin foam belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Spin foam places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

Geometry supplies the realization, boundary, gauge, or dual description in which the operator construction becomes physically legible.

Physical Construction

The state carrier is a spacetime, boundary algebra, gauge orbit, spin network, bulk/boundary pair, or geometric representation of a quantum state space. The governing operation is Hamiltonian, action, constraint, boundary operator, correlation map, or dictionary between two representations. Gauge, boundary, metric, covariance, and constraint conditions decide which geometric descriptions represent the same physical content. The calculated observables are Boundary correlators, spectra, entropies, scattering data, geometric invariants, or reconstructed bulk quantities.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (superposition, wave function, or quantum state) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

Geometry supplies the realization, boundary, gauge, or dual description in which the operator construction becomes physically legible. Spin foam extends the state-operator-spectrum constructor to many modes, fields, particles, gauge structure, or scale. Particle identity is treated

as a stable excitation or representation role rather than as the starting object. Creation/annihilation, field operators, gauge constraints, and scale flow preserve operator structure across realizations.

The local title, representation, and physical realization may change while the constructor role is preserved. The carrier can be a field state, occupation-number state, gauge orbit, spin network, or effective theory. The same formal role may be displayed through particles, modes, amplitudes, correlation functions, or boundary dictionaries. Scale and geometry can change the realization while preserving operator or spectral content.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Commutation, anticommutation, gauge, and occupation rules define which many-mode states are admissible. The field or many-mode construction must reduce to the appropriate single-particle, quasiparticle, or low-energy limit when those limits exist.

14.14 AdS/CFT correspondence

Mechanism

AdS/CFT correspondence is a geometry-translation constructor: bulk gravitational data and boundary field data are treated as dual presentations of one operator structure.

AdS/CFT correspondence places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

AdS/CFT belongs at the interface where geometry becomes a representation of the quantum construction rather than the invariant root. The practical content is a dictionary between bulk fields and boundary operators.

Physical Construction

The state carrier is a spacetime, boundary algebra, gauge orbit, spin network, bulk/boundary pair, or geometric representation of a quantum state space. The governing operation is Hamiltonian, action, constraint, boundary operator, correlation map, or dictionary between two representations. Gauge, boundary, metric, covariance, and constraint conditions decide which geometric descriptions represent the same physical content. The calculated observables are Boundary correlators, spectra, entropies, scattering data, geometric invariants, or reconstructed bulk quantities.

Topic Equations

Standard constructor skeleton: boundary-source/bulk-field dictionary.

$$\begin{aligned} Z_{\text{grav}}[\phi_0] &\simeq Z_{\text{CFT}}[J = \phi_0] \\ \left\langle \exp \int J \mathcal{O} \right\rangle_{\text{CFT}} &= Z_{\text{bulk}}[\phi|_{\partial} = J] \end{aligned}$$

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system, basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

14.15 Loop quantum gravity

Mechanism

Loop quantum gravity belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Loop quantum gravity places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

Geometry supplies the realization, boundary, gauge, or dual description in which the operator construction becomes physically legible.

Physical Construction

The state carrier is a spacetime, boundary algebra, gauge orbit, spin network, bulk/boundary pair, or geometric representation of a quantum state space. The governing operation is Hamiltonian, action, constraint, boundary operator, correlation map, or dictionary between two representations. Gauge, boundary, metric, covariance, and constraint conditions decide which geometric descriptions represent the same physical content. The calculated observables are Boundary correlators, spectra, entropies, scattering data, geometric invariants, or reconstructed bulk quantities.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (spectrum, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

Geometry supplies the realization, boundary, gauge, or dual description in which the operator construction becomes physically legible. Loop quantum gravity extends the state-operator-spectrum constructor to many modes, fields, particles, gauge structure, or scale. Particle identity is treated as a stable excitation or representation role rather than as the starting object.

Creation/annihilation, field operators, gauge constraints, and scale flow preserve operator structure across realizations.

The local title, representation, and physical realization may change while the constructor role is preserved. The carrier can be a field state, occupation-number state, gauge orbit, spin network, or effective theory. The same formal role may be displayed through particles, modes, amplitudes, correlation functions, or boundary dictionaries. Scale and geometry can change the realization while preserving operator or spectral content.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Commutation, anticommutation, gauge, and occupation rules define which many-mode states are admissible. The field or many-mode construction must reduce to the appropriate single-particle, quasiparticle, or low-energy limit when those limits exist.

14.16 Spin network

Mechanism

Spin network belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Spin network places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state.

Physical Construction

The state carrier is an input state, register, channel state, error syndrome, key, or controlled experimental configuration. The governing operation is an ordered sequence of gates, channels, measurements, corrections, encodings, or conditional maps. Each step must belong to the claimed map class: unitary, completely positive, trace-preserving, projective, conditional, or corrective. The calculated observables are Output state, key, error rate, fidelity, channel capacity, algorithmic success probability, or sensor estimate.

Defining Relations

Preparation or domain terms (context, basis, or preparation context) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (spectrum, eigenstate, or eigenvalue) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

The mechanism is a controlled composition of allowed maps: a sequence that prepares, transforms, protects, transmits, or reads a quantum state. Spin network extends the state-operator-spectrum constructor to many modes, fields, particles, gauge structure, or scale. Particle identity is treated as a stable excitation or representation role rather than as the starting object. Creation/annihilation, field operators, gauge constraints, and scale flow preserve operator structure across realizations.

The local title, representation, and physical realization may change while the constructor role is preserved. The carrier can be a field state, occupation-number state, gauge orbit, spin network, or effective theory. The same formal role may be displayed through particles, modes, amplitudes, correlation functions, or boundary dictionaries. Scale and geometry can change the realization while preserving operator or spectral content.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Commutation, anticommutation, gauge, and occupation rules define which many-mode states are admissible. The field or many-mode construction must reduce to the appropriate single-particle, quasiparticle, or low-energy limit when those limits exist.

14.17 Quantum chemistry

Mechanism

Quantum chemistry belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Quantum chemistry places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

Field and many-body theories enlarge the state space to variable occupation, local fields, gauge sectors, and collective modes.

Physical Construction

The state carrier is a Fock space, field configuration space, gauge sector, many-body Hilbert space, or effective low-energy sector. The governing operation is Field, creation, annihilation, charge, Hamiltonian, constraint, or renormalization operators. Statistics, locality, gauge symmetry, domain conditions, and renormalization prescriptions determine the physical sector. The calculated observables are Correlation functions, particle spectra, charges, scattering amplitudes, effective couplings, or geometric observables.

Defining Relations

Preparation or domain terms (potential or domain condition, basis, or preparation context) determine which states are admissible. State terms (wave function, quantum state, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (observable operator, Hamiltonian, or unitary operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

Quantum chemistry extends the state-operator-spectrum constructor to many modes, fields, particles, gauge structure, or scale. Particle identity is treated as a stable excitation or representation role rather than as the starting object. Creation/annihilation, field operators, gauge constraints, and scale flow preserve operator structure across realizations.

The local title, representation, and physical realization may change while the constructor role is preserved. The carrier can be a field state, occupation-number state, gauge orbit, spin

network, or effective theory. The same formal role may be displayed through particles, modes, amplitudes, correlation functions, or boundary dictionaries. Scale and geometry can change the realization while preserving operator or spectral content.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Commutation, anticommutation, gauge, and occupation rules define which many-mode states are admissible. The field or many-mode construction must reduce to the appropriate single-particle, quasiparticle, or low-energy limit when those limits exist.

14.18 Standard Model

Mechanism

Standard Model belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Standard Model places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

The mechanism is the field/mode version of the constructor: a state space is decomposed into modes or sectors, operators create, annihilate, or constrain those modes, and observables are occupation, charge, spin, momentum, energy, or scattering response.

Physical Construction

The state carrier is Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data. The governing operation is Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal. The calculated observables are Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

The mechanism is the field/mode version of the constructor: a state space is decomposed into modes or sectors, operators create, annihilate, or constrain those modes, and observables are occupation, charge, spin, momentum, energy, or scattering response. Standard Model extends the state-operator-spectrum constructor to many modes, fields, particles, gauge structure, or scale. Particle identity is treated as a stable excitation or representation role rather than as

the starting object. Creation/annihilation, field operators, gauge constraints, and scale flow preserve operator structure across realizations.

The local title, representation, and physical realization may change while the constructor role is preserved. The carrier can be a field state, occupation-number state, gauge orbit, spin network, or effective theory. The same formal role may be displayed through particles, modes, amplitudes, correlation functions, or boundary dictionaries. Scale and geometry can change the realization while preserving operator or spectral content.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Commutation, anticommutation, gauge, and occupation rules define which many-mode states are admissible. The field or many-mode construction must reduce to the appropriate single-particle, quasiparticle, or low-energy limit when those limits exist.

14.19 Klein–Gordon equation

Mechanism

Klein–Gordon equation belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Klein–Gordon equation places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement.

Physical Construction

The state carrier is a state vector, density operator, wavefunction, field state, or register on a specified domain. The governing operation is Hamiltonian, unitary map, channel generator, action, constraint, or differential operator that transports the state. Self-adjointness, complete positivity, trace preservation, gauge constraints, and boundary/domain conditions decide whether the evolution is legal. The calculated observables are Time-dependent probabilities, spectra, transition amplitudes, conserved quantities, or response functions.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (spectrum, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (uncertainty relation, commutator, or incompatible observable) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

The Hamiltonian, Liouvillian, action, or channel determines the change of state that precedes measurement. Klein–Gordon equation extends the state-operator-spectrum constructor to many modes, fields, particles, gauge structure, or scale. Particle identity is treated as a stable excitation or representation role rather than as the starting object. Creation/annihilation, field operators, gauge constraints, and scale flow preserve operator structure across realizations.

The local title, representation, and physical realization may change while the constructor role is preserved. The carrier can be a field state, occupation-number state, gauge orbit, spin network, or effective theory. The same formal role may be displayed through particles, modes, amplitudes, correlation functions, or boundary dictionaries. Scale and geometry can change the realization while preserving operator or spectral content.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Commutation, anticommutation, gauge, and occupation rules define which many-mode states are admissible. The field or many-mode construction must reduce to the appropriate single-particle, quasiparticle, or low-energy limit when those limits exist.

14.20 Bose–Einstein statistics

Mechanism

Bose–Einstein statistics is a symmetric-state constructor: exchange statistics permits collective occupation of one quantum mode.

Bose–Einstein statistics is needed because identical bosons occupy symmetric many-particle states. Multiple particles may share one mode, which produces collective occupation effects absent from distinguishable particles and fermions.

Bose–Einstein statistics is an admissibility rule for symmetric many-particle states. Commutation permits unrestricted mode occupation, while the thermal distribution determines the mean population and the conditions for macroscopic ground-state occupation.

Physical Construction

The state carrier is a bosonic Fock space assembled from symmetric many-particle sectors or occupation-number modes. The governing operation is Creation, annihilation, and number operators obeying canonical commutation relations. Exchange symmetry permits any non-negative integer occupation of a one-particle mode. The calculated observables are Mode occupations, condensate fraction, particle density, pressure, heat capacity, and coherence observables.

Topic Equations

Topic-specific construction: symmetric mode algebra, unrestricted occupation, and equilibrium population.

$$= \delta_{ij}, \quad n_i \in \{0, 1, 2, \dots\}$$
$$\bar{n}_i = \frac{1}{e^{\beta(\epsilon_i - \mu)} - 1}$$

Physical Meaning

The Bose–Einstein distribution gives the mean occupation of each energy mode at thermal equilibrium. The chemical potential and density determine whether a macroscopic population accumulates in the lowest mode, as in Bose–Einstein condensation.

Cooling a dilute bosonic gas below its critical temperature transfers a macroscopic fraction of the atoms into one quantum mode. The condensate is not caused by an attractive binding force; it follows from the symmetric occupation rule and thermodynamic constraints.

Fock space provides one common language for bosonic and fermionic occupations, while field operators create and remove individual mode excitations.

Invariance And Realization

Exchange symmetry and canonical commutation define the bosonic many-particle sectors. Each one-particle mode admits any non-negative integer occupation. The Bose–Einstein function gives the equilibrium mean occupation once energy, temperature, and chemical potential are specified.

The dispersion relation, dimensionality, density of states, conserved particle number, and interaction approximation depend on the physical system. Photons, cold atoms, phonons, and bosonic quasiparticles realize the same exchange rule with different chemical-potential constraints. Macroscopic occupation of the lowest mode occurs only when the density of states and thermodynamic limit support condensation.

Discriminating Consequences

The theory recovers unrestricted mode occupation and Bose enhancement relative to distinguishable particles. The theory recovers the Maxwell-Boltzmann distribution in the dilute low-fugacity limit. The defining relation is particle-number or energy constraints and test whether the inferred condensate fraction follows the correct dimensional dependence.

14.21 String theory

Mechanism

String theory belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

String theory places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

Geometry supplies the realization, boundary, gauge, or dual description in which the operator construction becomes physically legible.

Physical Construction

The state carrier is a spacetime, boundary algebra, gauge orbit, spin network, bulk/boundary pair, or geometric representation of a quantum state space. The governing operation is Hamiltonian, action, constraint, boundary operator, correlation map, or dictionary between two representations. Gauge, boundary, metric, covariance, and constraint conditions decide which geometric descriptions represent the same physical content. The calculated observables are Boundary correlators, spectra, entropies, scattering data, geometric invariants, or reconstructed bulk quantities.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (Hamiltonian, observable operator, or generator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (spectrum, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

Geometry supplies the realization, boundary, gauge, or dual description in which the operator construction becomes physically legible. String theory extends the state-operator-spectrum constructor to many modes, fields, particles, gauge structure, or scale. Particle identity is treated

as a stable excitation or representation role rather than as the starting object. Creation/annihilation, field operators, gauge constraints, and scale flow preserve operator structure across realizations.

The local title, representation, and physical realization may change while the constructor role is preserved. The carrier can be a field state, occupation-number state, gauge orbit, spin network, or effective theory. The same formal role may be displayed through particles, modes, amplitudes, correlation functions, or boundary dictionaries. Scale and geometry can change the realization while preserving operator or spectral content.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Commutation, anticommutation, gauge, and occupation rules define which many-mode states are admissible. The field or many-mode construction must reduce to the appropriate single-particle, quasiparticle, or low-energy limit when those limits exist.

14.22 Quantum statistical mechanics

Mechanism

Quantum statistical mechanics belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Quantum statistical mechanics places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

Field and many-body theories enlarge the state space to variable occupation, local fields, gauge sectors, and collective modes.

Physical Construction

The state carrier is a Fock space, field configuration space, gauge sector, many-body Hilbert space, or effective low-energy sector. The governing operation is Field, creation, annihilation, charge, Hamiltonian, constraint, or renormalization operators. Statistics, locality, gauge symmetry, domain conditions, and renormalization prescriptions determine the physical sector. The calculated observables are Correlation functions, particle spectra, charges, scattering amplitudes, effective couplings, or geometric observables.

Defining Relations

Preparation or domain terms (basis, preparation context, or measurement basis) determine which states are admissible. State terms (quantum state, density matrix, or wave function) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (observable operator, matrix, or Hamiltonian) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (spectrum, eigenvalue, or eigenstate) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (commutator, uncertainty relation, or non-commuting observables) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

Quantum statistical mechanics extends the state-operator-spectrum constructor to many modes, fields, particles, gauge structure, or scale. Particle identity is treated as a stable excitation or representation role rather than as the starting object. Creation/annihilation, field operators, gauge constraints, and scale flow preserve operator structure across realizations.

The local title, representation, and physical realization may change while the constructor role is preserved. The carrier can be a field state, occupation-number state, gauge orbit, spin

network, or effective theory. The same formal role may be displayed through particles, modes, amplitudes, correlation functions, or boundary dictionaries. Scale and geometry can change the realization while preserving operator or spectral content.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Commutation, anticommutation, gauge, and occupation rules define which many-mode states are admissible. The field or many-mode construction must reduce to the appropriate single-particle, quasiparticle, or low-energy limit when those limits exist.

14.23 Quantum gravity

Mechanism

Quantum gravity asks whether geometry is a background realization, a constrained quantum carrier, or an emergent observable of a deeper quantum state.

Quantum gravity places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

The constructor cannot treat quantum gravity as an ordinary wave function on a fixed domain. A candidate theory must specify the state space of geometric and matter degrees of freedom, the constraint or evolution operators, the gauge-invariant or relational observables, and the limit in which classical spacetime is recovered.

Physical Construction

The state carrier is a state space for geometric and matter degrees of freedom, or a deeper carrier from which geometry is reconstructed. The governing operation is Constraint, evolution, or amplitude operators that do not presuppose an unexamined fixed spacetime background. Gauge and diffeomorphism constraints determine the physical state space and which operators are observable. The calculated observables are Relational observables, boundary amplitudes, geometric spectra, or semiclassical spacetime data.

Topic Equations

Schematic constructor shared by constrained approaches: quantum carrier, constraints, physical observables, and semiclassical recovery must all be specified.

$$\begin{aligned}\Psi &\in \mathcal{H}_{\text{geom}\otimes\text{matter}}, & \hat{\mathcal{C}}_a\Psi &= 0 \\ & & [\hat{\mathcal{O}}_{\text{phys}}, \hat{\mathcal{C}}_a]\Psi &= 0 \\ \langle\Psi|\hat{g}_{\mu\nu}|\Psi\rangle &\longrightarrow g_{\mu\nu}^{\text{cl}} & \text{in a controlled semiclassical regime}\end{aligned}$$

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

The rule connecting prepared states, observables, and spectral probability measures across wave, matrix, path-integral, circuit, or field notation. The operator-to-spectrum relation: admissible observations are represented through eigenvalues, projections, modes, or outcome channels. The dependence of admissible observable on measurement context or boundary condition. The non-commuting compatibility structure, which survives changes of representation.

The name of the carrier: particle, wave, field, qubit, or excitation. Where time dependence is represented: on the state, on the operator, or in a path weight. The coordinate system,

basis, or geometric picture used to display the same relation. The physical implementation of detector, boundary, preparation, or observable.

Discriminating Consequences

A concrete transfer target is a material, biological, or collective system with a state, a transformation, and a spectral or categorical observable, but without a tested incompatibility relation. The validation criterion is that varying the context changes the admissible observable while the transformation law remains identifiable; shuffled or erased contexts should weaken the effect.

14.24 Quantum geometry

Mechanism

Quantum geometry is a geometry-realization page: geometric quantities are promoted to quantum observables rather than assumed as a smooth background.

Quantum geometry places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

Quantum geometry uses a quantum state of geometry, often represented by graph or spin-network data. The operator-to-spectrum step asks for eigenvalues of geometric observables such as area or volume. This places the page near the geometry/boundary interface rather than inside a generic many-mode field layer.

Physical Construction

The state carrier is a spacetime, boundary algebra, gauge orbit, spin network, bulk/boundary pair, or geometric representation of a quantum state space. The governing operation is Hamiltonian, action, constraint, boundary operator, correlation map, or dictionary between two representations. Gauge, boundary, metric, covariance, and constraint conditions decide which geometric descriptions represent the same physical content. The calculated observables are Boundary correlators, spectra, entropies, scattering data, geometric invariants, or reconstructed bulk quantities.

Topic Equations

Topic-specific constructor: the equations express graph-based geometry states and spectral prediction of geometric observables.

$$\begin{aligned}\mathcal{H}_\Gamma &= L^2(SU(2)^E/SU(2)^V), \quad |\Gamma, j_e, \iota_v\rangle \\ \hat{A}(S)|\Gamma, j, \iota\rangle &= 8\pi\gamma\ell_P^2 \sum_{e \in S} \sqrt{j_e(j_e + 1)} |\Gamma, j, \iota\rangle \\ \text{geometry observable : } \hat{G}|g_i\rangle &= g_i|g_i\rangle\end{aligned}$$

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

The relation between prepared states, observables, and spectral probability measures. The use of eigenvalues, projectors, modes, or outcome channels to represent admissible observations. The dependence of the observable on basis, domain, potential, preparation, or measurement context. The commutator structure that limits which observables can be jointly diagonalized.

The physical carrier: particle, wave, field mode, spin, qubit, detector, or excitation. The representation: wave mechanics, matrix mechanics, density matrices, path integrals, circuits, or

fields. Where time dependence is placed: on the state, on the operator, in a propagator, or in a path weight. The implementation of preparation, boundary condition, detector, or outcome channel.

Discriminating Consequences

A transfer target provides a state space, a transformation law, and a spectral or categorical observable, with one compatibility relation experimentally unresolved. A useful validation varies the basis, domain, or measurement context and measures whether the allowed observable changes while the underlying transformation law remains identifiable. A stronger validation contains two candidate observables whose predicted commutator controls joint resolvability.

14.25 Quantum spacetime

Mechanism

Quantum spacetime belongs to the many-mode extension: the same state, generator, observable, and compatibility logic is lifted from one system to fields, particles, scaling limits, or gauge constraints.

Quantum spacetime places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

Geometry supplies the realization, boundary, gauge, or dual description in which the operator construction becomes physically legible.

Physical Construction

The state carrier is a spacetime, boundary algebra, gauge orbit, spin network, bulk/boundary pair, or geometric representation of a quantum state space. The governing operation is Hamiltonian, action, constraint, boundary operator, correlation map, or dictionary between two representations. Gauge, boundary, metric, covariance, and constraint conditions decide which geometric descriptions represent the same physical content. The calculated observables are Boundary correlators, spectra, entropies, scattering data, geometric invariants, or reconstructed bulk quantities.

Defining Relations

Preparation or domain terms (preparation condition, measurement setup, or potential or domain condition) determine which states are admissible. State terms (quantum state, wave function, or density operator) name the predictive carrier: vector, wave function, density operator, field state, or register state. Operator terms (matrix, Hamiltonian, or observable operator) name the observable, Hamiltonian, unitary, or constraint acting on the carrier. Spectral terms (eigenvalue, energy level, or measurement outcome) name the outcome labels and projectors that define outcome channels. The probability rule maps states and projectors to recorded probabilities through the Born rule, trace rule, or projection-valued measure. Compatibility terms (uncertainty relation, commutator, or incompatible observable) mark cases where observables do not admit one common sharp observable basis.

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

Geometry supplies the realization, boundary, gauge, or dual description in which the operator construction becomes physically legible. Quantum spacetime extends the state-operator-spectrum constructor to many modes, fields, particles, gauge structure, or scale. Particle identity is treated as a stable excitation or representation role rather than as the starting object.

Creation/annihilation, field operators, gauge constraints, and scale flow preserve operator structure across realizations.

The local title, representation, and physical realization may change while the constructor role is preserved. The carrier can be a field state, occupation-number state, gauge orbit, spin network, or effective theory. The same formal role may be displayed through particles, modes, amplitudes, correlation functions, or boundary dictionaries. Scale and geometry can change the realization while preserving operator or spectral content.

Discriminating Consequences

The topic is physically defined by its state carrier, operator or map, observable consequence, and compatibility condition. Commutation, anticommutation, gauge, and occupation rules define which many-mode states are admissible. The field or many-mode construction must reduce to the appropriate single-particle, quasiparticle, or low-energy limit when those limits exist.

14.26 Fock space

Mechanism

Fock space is the occupation-number state space: the construction that replaces a fixed-particle Hilbert space by a direct sum over particle number.

Fock space places quantum dynamics in a relativistic, many-body, field, gauge, geometric, or scale-dependent setting. The state space and operator domain must therefore be specified for that setting rather than inferred from a single-particle model.

Fock space changes the carrier of the quantum state. Instead of describing one system in one Hilbert space, it builds sectors with zero, one, two, and more identical quanta, then imposes the bosonic or fermionic exchange rule. Creation and annihilation operators are the native coordinates of this page because they move the state between occupation sectors.

Physical Construction

The state carrier is Fock space, field configuration space, or a sector selected by charge, spin, momentum, statistics, or gauge data. The governing operation is Creation, annihilation, field, charge, spin, Hamiltonian, or scattering operators acting on the admissible sector. Statistics, gauge constraints, commutation or anticommutation rules, domain conditions, and sector labels decide which states are legal. The calculated observables are Occupation number, charge, spin, momentum, energy, correlation function, cross-section, or scattering amplitude.

Topic Equations

Topic-specific constructor: the equations express variable particle number, exchange symmetry, and occupation-number observable.

$$\begin{aligned}\mathcal{F}_{\pm}(\mathcal{H}) &= \bigoplus_{n=0}^{\infty} \mathcal{S}_{\pm} \mathcal{H}^{\otimes n} \\ [a_i, a_j^{\dagger}]_{\mp} &= \delta_{ij}, \quad [a_i, a_j]_{\mp} = 0 \\ N &= \sum_i a_i^{\dagger} a_i, \quad N|n_1, n_2, \dots\rangle = \left(\sum_i n_i \right) |n_1, n_2, \dots\rangle\end{aligned}$$

Physical Meaning

Different topics in this branch use different carriers: spinor wave functions, Fock spaces, many-body states, gauge sectors, geometric states, or effective low-energy sectors. Their physical content is fixed by the associated field equation or Hamiltonian, its constraints and domain, and the amplitudes, spectra, charges, or correlation functions it predicts.

Field and many-body mechanisms become experimentally useful when assembled into an ordered intervention. The protocol chapter shows how preparation, controlled evolution, measurement, and correction compose into one executable map.

Invariance And Realization

The relation between prepared states, observables, and spectral probability measures. The use of eigenvalues, projectors, modes, or outcome channels to represent admissible observations. The dependence of the observable on basis, domain, potential, preparation, or measurement context. The commutator structure that limits which observables can be jointly diagonalized.

The physical carrier: particle, wave, field mode, spin, qubit, detector, or excitation. The representation: wave mechanics, matrix mechanics, density matrices, path integrals, circuits, or fields. Where time dependence is placed: on the state, on the operator, in a propagator, or in a path weight. The implementation of preparation, boundary condition, detector, or outcome channel.

Discriminating Consequences

A transfer target provides a state space, a transformation law, and a spectral or categorical observable, with one compatibility relation experimentally unresolved. A useful validation varies the basis, domain, or measurement context and measures whether the allowed observable changes while the underlying transformation law remains identifiable. A stronger validation contains two candidate observables whose predicted commutator controls joint resolvability.

Part VI

Historical And Interpretive
Annotations

Chapter 15

Annotations: history, interpretations, and popular frames

Some pages help readers navigate the subject but do not form steps in the mechanism. They are kept as annotations so books, historical figures, interpretations, and popular frames do not distort the constructive tree.

Historical and interpretive sources explain how the formalism developed and how its probabilities are read. They remain linked to the physical relations they discuss.

15.1 Topics

Page
Introduction to Quantum Mechanics (book) (<i>annotation</i>)
Quantum mind (<i>annotation</i>)
Erwin Schrödinger (<i>annotation</i>)
History of quantum mechanics (<i>annotation</i>)
Introduction to quantum mechanics (<i>annotation</i>)
History of quantum field theory (<i>annotation</i>)
David Hilbert (<i>annotation</i>)
Quantum Computing: A Gentle Introduction (<i>annotation</i>)
Quantum mysticism (<i>annotation</i>)
Interpretations of quantum mechanics (<i>annotation</i>)
Quantum Theory: Concepts and Methods (<i>annotation</i>)
QBism (<i>annotation</i>)
Relational quantum mechanics (<i>annotation</i>)
Modern Quantum Mechanics (<i>annotation</i>)
Applications of quantum mechanics (<i>annotation</i>)
Werner Heisenberg (<i>annotation</i>)
The Quantum Universe (<i>annotation</i>)

15.2 Introduction to Quantum Mechanics (book)

Historical or interpretive annotation.

Historical Role

Introduction to Quantum Mechanics (book) records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

The source discusses states, operators, probability rules, or protocols whose mathematical consequences are developed in the physical chapters. Its interpretation changes the theory only when it changes an equation, admissibility condition, probability law, or experimental prediction.

15.3 Quantum mind

Historical or interpretive annotation.

Historical Role

Quantum mind records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

The source discusses states, operators, probability rules, or protocols whose mathematical consequences are developed in the physical chapters. Its interpretation changes the theory only when it changes an equation, admissibility condition, probability law, or experimental prediction.

15.4 Erwin Schrödinger

Historical or interpretive annotation.

Historical Role

Erwin Schrödinger records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

The source discusses states, operators, probability rules, or protocols whose mathematical consequences are developed in the physical chapters. Its interpretation changes the theory only when it changes an equation, admissibility condition, probability law, or experimental prediction.

15.5 History of quantum mechanics

Historical or interpretive annotation.

Historical Role

History of quantum mechanics records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

The source discusses states, operators, probability rules, or protocols whose mathematical consequences are developed in the physical chapters. Its interpretation changes the theory only when it changes an equation, admissibility condition, probability law, or experimental prediction.

15.6 Introduction to quantum mechanics

Historical or interpretive annotation.

Historical Role

Introduction to quantum mechanics records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

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15.7 History of quantum field theory

Historical or interpretive annotation.

Historical Role

History of quantum field theory records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

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15.8 David Hilbert

Historical or interpretive annotation.

Historical Role

David Hilbert records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

The source discusses states, operators, probability rules, or protocols whose mathematical consequences are developed in the physical chapters. Its interpretation changes the theory only when it changes an equation, admissibility condition, probability law, or experimental prediction.

15.9 Quantum Computing: A Gentle Introduction

Historical or interpretive annotation.

Historical Role

Quantum Computing: A Gentle Introduction records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

The source discusses states, operators, probability rules, or protocols whose mathematical consequences are developed in the physical chapters. Its interpretation changes the theory only when it changes an equation, admissibility condition, probability law, or experimental prediction.

15.10 Quantum mysticism

Historical or interpretive annotation.

Historical Role

Quantum mysticism records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

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15.11 Interpretations of quantum mechanics

Historical or interpretive annotation.

Historical Role

Interpretations of quantum mechanics records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

The source discusses states, operators, probability rules, or protocols whose mathematical consequences are developed in the physical chapters. Its interpretation changes the theory only when it changes an equation, admissibility condition, probability law, or experimental prediction.

15.12 Quantum Theory: Concepts and Methods

Historical or interpretive annotation.

Historical Role

Quantum Theory: Concepts and Methods records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

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15.13 QBism

Historical or interpretive annotation.

Historical Role

QBism records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

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15.14 Relational quantum mechanics

Historical or interpretive annotation.

Historical Role

Relational quantum mechanics records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

The source discusses states, operators, probability rules, or protocols whose mathematical consequences are developed in the physical chapters. Its interpretation changes the theory only when it changes an equation, admissibility condition, probability law, or experimental prediction.

15.15 Modern Quantum Mechanics

Historical or interpretive annotation.

Historical Role

Modern Quantum Mechanics records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

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15.16 Applications of quantum mechanics

Historical or interpretive annotation.

Historical Role

Applications of quantum mechanics records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

The source discusses states, operators, probability rules, or protocols whose mathematical consequences are developed in the physical chapters. Its interpretation changes the theory only when it changes an equation, admissibility condition, probability law, or experimental prediction.

15.17 Werner Heisenberg

Historical or interpretive annotation.

Historical Role

Werner Heisenberg records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

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15.18 The Quantum Universe

Historical or interpretive annotation.

Historical Role

The Quantum Universe records the history, pedagogy, or interpretation of quantum theory. It does not define an additional state space, equation of motion, observable, or probability law.

Relation To The Formal Theory

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